# BASIC WORKSHOP PRACTICE

## INTRODUCTION:-

During first year of Engineering, all the students irrespective of the trade or discipline in which they have go admission, have to study & practice Basic workshop processes such as Fitting, Carpentry, Welding & Sheet metal.

It is a very well known fact that machine shop terms not only an important but an indispensable part of a modern workshop. It is no only inches a heavy investment but, at The same time, calls for a fairly high skill of workmanship. If carried out successfully, the operations performed in the shop are capable of producing a large number of jobs of different shapes & sized having a fine finish within very close limits of dimension,. A machine shop man should, therefore, have a thorough knowledge of the different machines he has to work on. He should also have a comprehensive knowledge of the type of tools he has to use & their effective application. In addition to this, he should be fully acquainted with the various operation performed on these machines, their working
Principles, the attachments which are used in conjunction with these machines, different Measuring & testing devices used for checking the finished products & the material sued For helping in cutting of metals & maintenance of the cutting tools & the machines.

Workshop practice imports basic knowledge of various tools & their use in different sections of manufacture such as Fitting, Blacksmith, Welding, Sheet metal, Carpentry, etc. It also helps engineers, when they occupy managerial positions, in understanding the activities & practical difficulties so that they can take appropriate decisions.

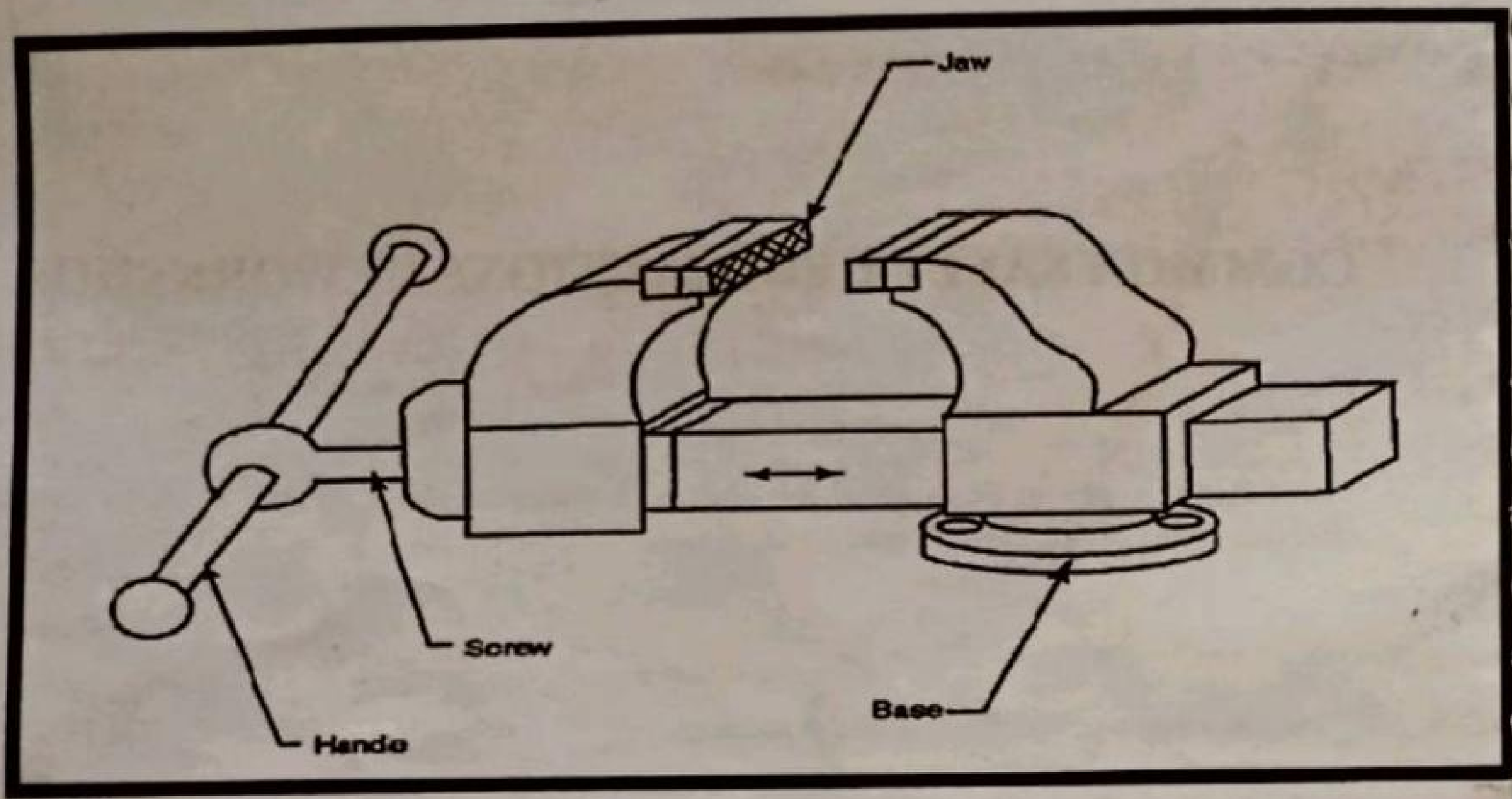
Even when a large amount of mechanical equipment is available for producing or repairing parts, there are still some elements of work which have to be performed by hand methods. This practice is intended to instruct the correct use of hand tools. Finally the engineers must also be familiar with first-aid practices.

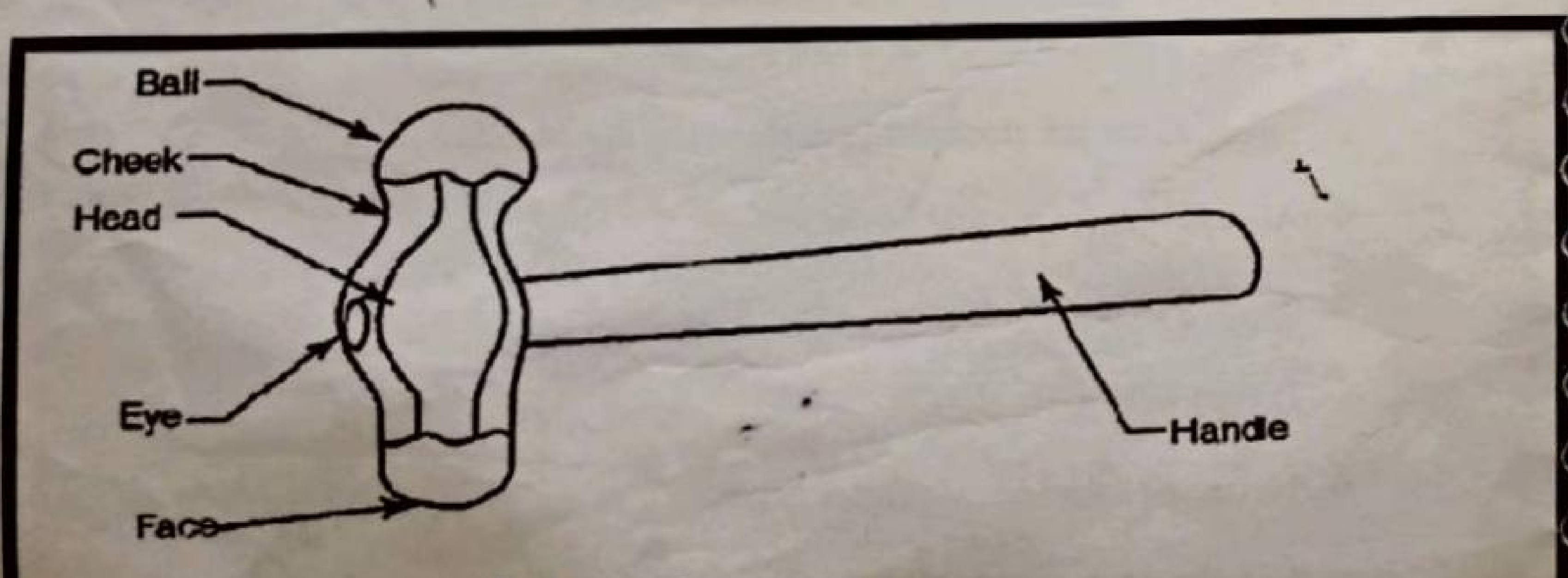
# COMMON SAFETY PRECAUTIONS IN WORKSHOP:-

**The following precautions must be observed while working in the shop to achieve effective & accidental free work.**

- Always wear uniform in the workshop.
- Never walk bare footed inside the workshop. Use of rubbershoe is recommended.
- Never touch moving parts, belts or rotating tools, etc.
- Defective equipments & tools should not be used for any work.
- Never touch any switch, knob or lever of the machine without knowing it.
- Do not touch any live wire inside the machine.
- Incase of fire the electric supply should be disconnected first.
- Always keep your mind on the job.
- Shop floor must be kept clean free from debris, scrap, oil & grease.
- Do not wear loose cloth.
- Never use hammers with loose heads.
- Use always the right tool for right job.

## VICE:-



Ball peen hammer





## STUDY OF TOOLS IN FITTING.

### AIM:

To make a detailed study of tools in Fitting

### INTRODUCTION:

The term fitting is used with bench work in assembling of parts by removing excess metal to obtain the necessary fit. Various tools are used in fitting shop to finish the job to required shape and size. The following are the commonly used tools in the fitting shop.

### BENCH VICE:

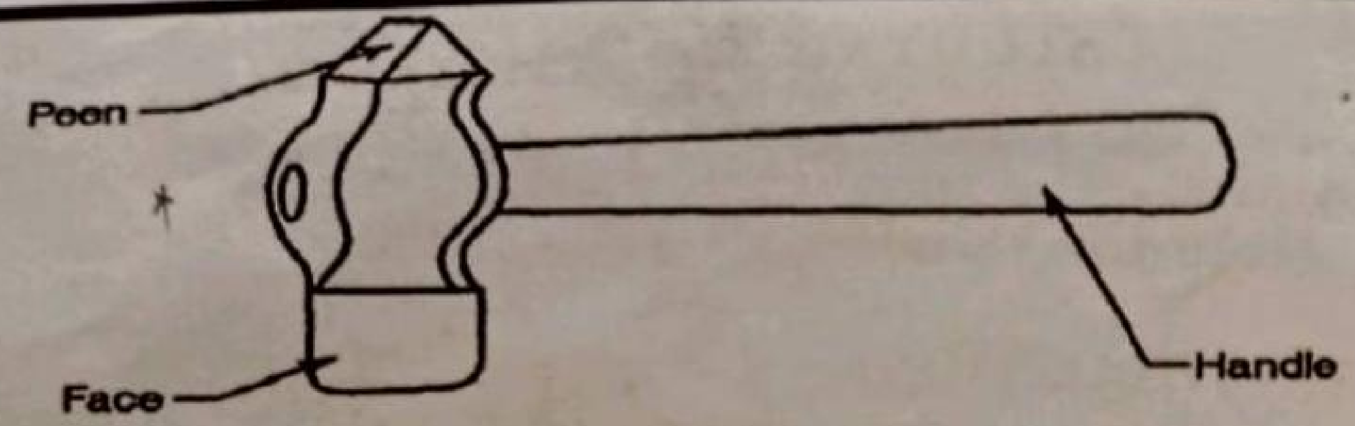
The most commonly used vice is the Engineer's parallel-jaw bench vice, sometimes called Fitter's vice. The bench vice is used for holding the work-piece in the fitting operations. The body of the bench vice is made up of cast iron and consists of two jaws, one fixed and the other movable. The movable jaw slides inside the fixed one by a screw and box nut arrangement. The vice has two hardened steel jaws with serrated faces used to hold the work firmly. These jaws are fixed with screws. The vice size is known by the width of its jaws and is varying from 80 to 140mm.

### HAMMERS:

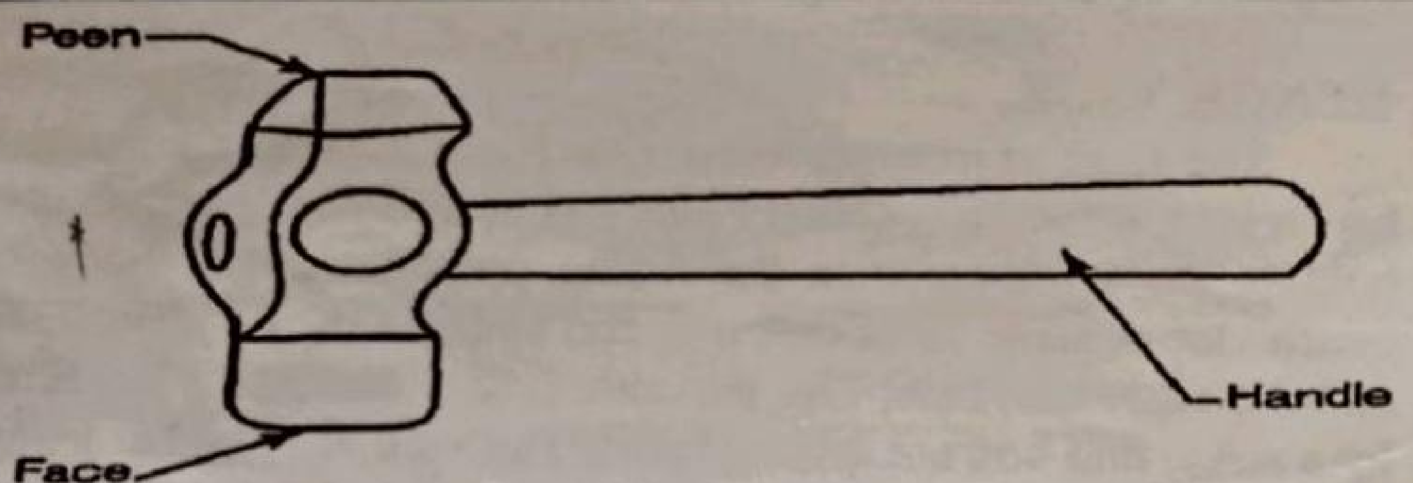
Hammers are used to strike a job or a tool. They are made of Forged steel of various sizes (weights) and shapes to suit various purposes. A hammer consists of four parts namely peen, head, eye and face. Hammers are classified, according to the shape of the peen, as ball peen, cross peen and straight peen hammers.

### BALL PEEN HAMMER:

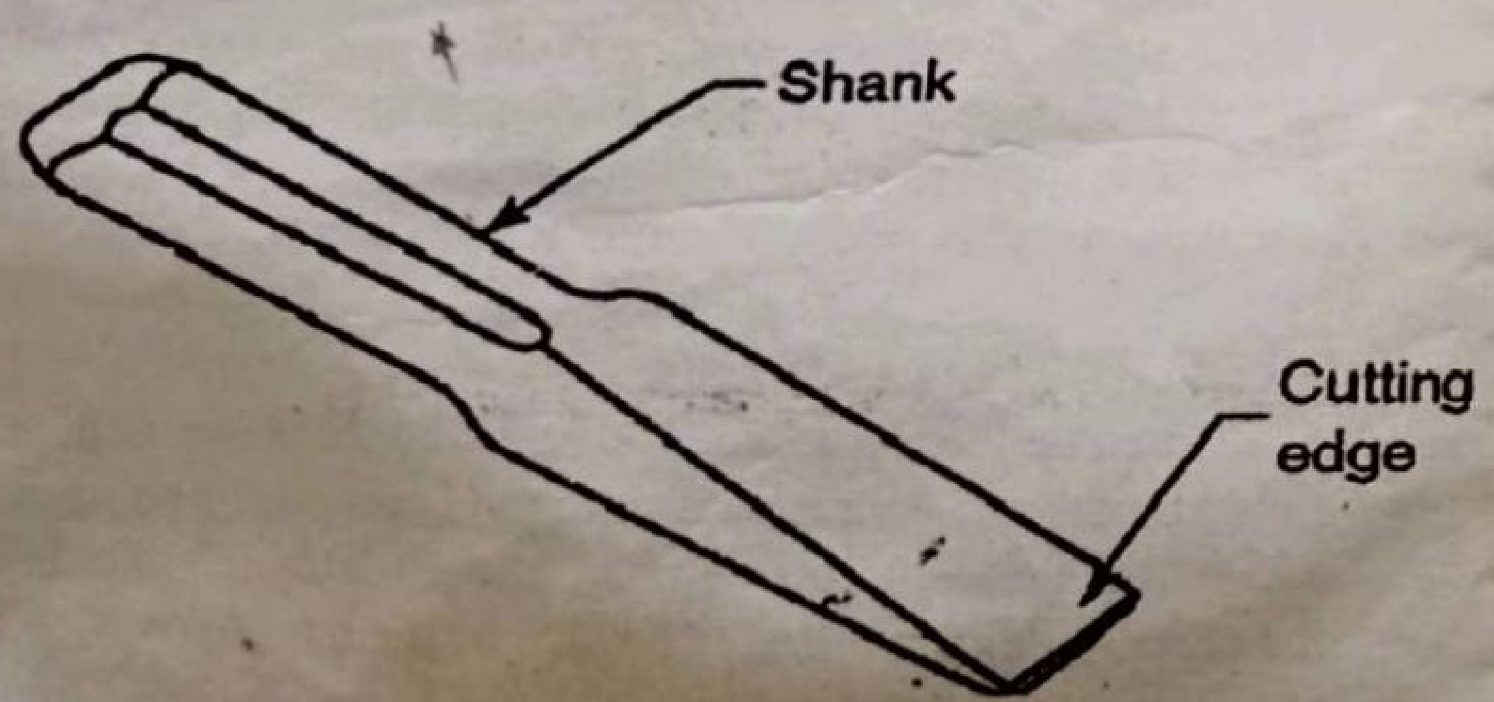
This is the most commonly used hammer and is sometimes called engineer's hammer, or chipping hammer. The peen has a shape of a ball which is hardened and polished. This hammer is chiefly used for chipping and riveting. The size of the hammer varies from 0.11kg to 0.91kg.



Cross peen hammer



Straight peen hammer



Flat or cold chisel



### CROSS PEEN HAMMER:

This is similar to ball peen hammer in shape and size except the peen which is cross the shaft (or Handle). This is mainly used for bending, stretching, hammering into shoulders, inside curves, etc. The size varies from 0.22 to 0.91kg.

### STRAIGHT PEEN HAMMER:

This hammer has a peen straight with the shaft i.e, parallel to the axis of the shaft. This is used for stretching or peening the metal. The size varies from 0.11 to 0.91kg.

### FLAT OR COLD CHISEL :

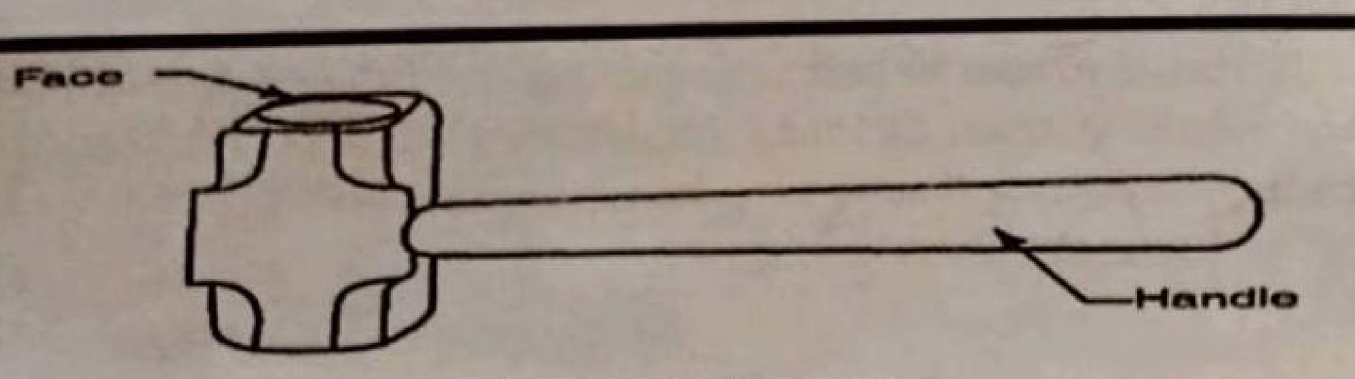
This is used for metal cutting and chipping. It is made up of carbon steel. Its shank is rectangular, hexagonal or octagonal in cross-section. The length of a flat chisel varies from 100 to 400mm and width of the cutting edge varies from 16 to 32mm.

### FILES:

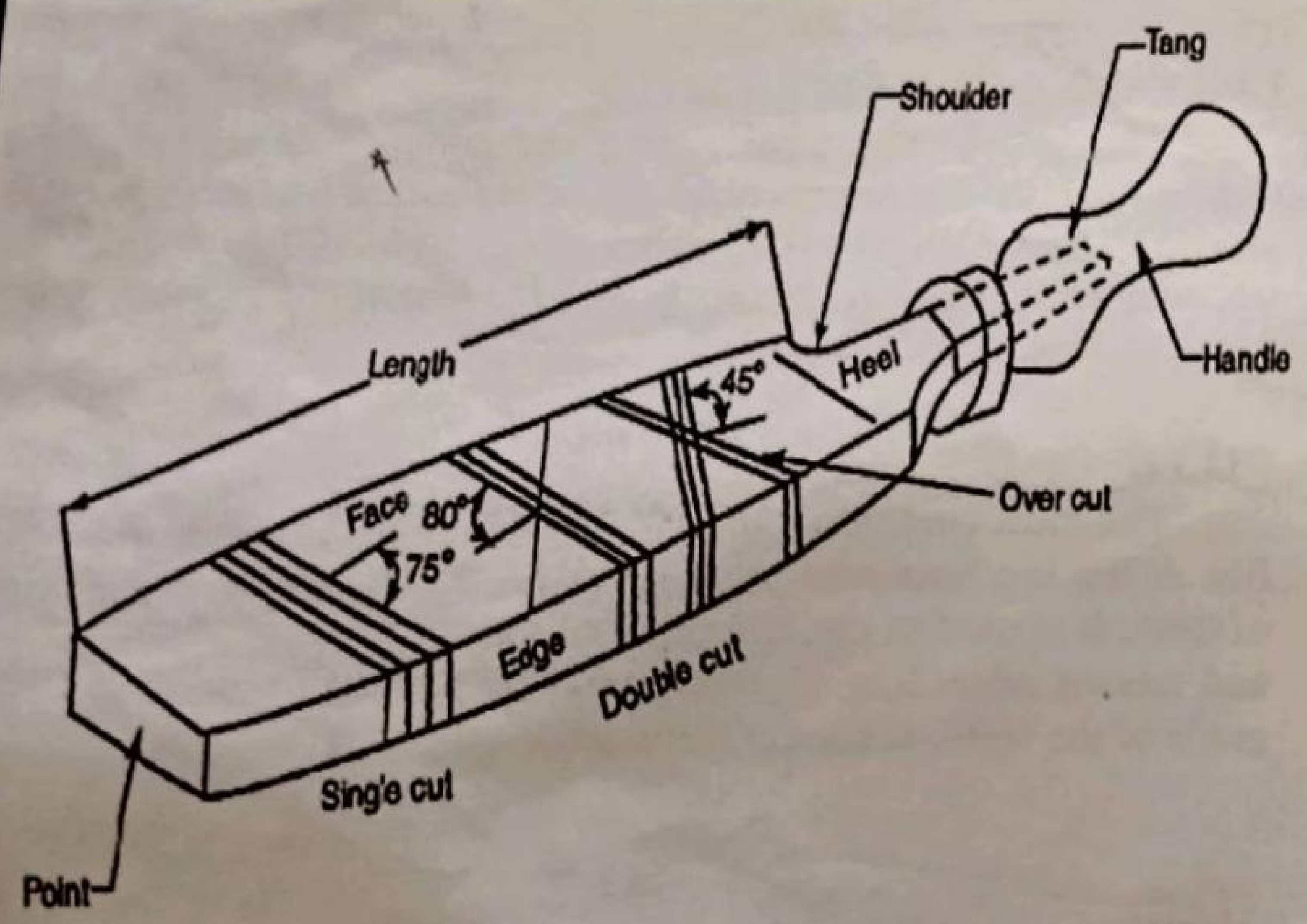
The most commonly used hand tool to be found in fitting shop is file. A file is a hardened piece of high grade steel with slanting rows of teeth. It is used to cut, smooth, or fit metal parts. Files are classified and named according to four principal factors - size, cut of teeth, grade of the teeth, sectional form.

### SIZE:

The size of a file is its length. This is the distance from the *point* to *heel*, without the tang. Files for fine work are usually from 100 to 200mm and those for heavier work from200to450mminlength.



Sledge hammer



Parts of file





**CUT OF TEETH:**

Cuts of teeth are divided into two groups (1) single-cut (2) double-cut.

**Single-cut:**

On single-cut files the teeth are cut parallel to other across the file at an angle of about 60° to the center line of the file. This type of file is used for finishing surface and sharpening edges. It is used for filing soft metals.

**Double-cut:**

Double-cut files have two sets of teeth, the over-cut teeth being cut at about 60° and up-cut at 75° to 80° to the center line. The double-cut file can do filing faster than single-cut file.

**GRADES OF FILE TEETH:**

**Rough-cut File:**

This file has very coarse spaced teeth. It is used for rough work. It removes more material.

**Bastard File:**

This file has less coarsely spaced teeth than rough-cut file. It is also used for rough work.

**Second-cut File:**

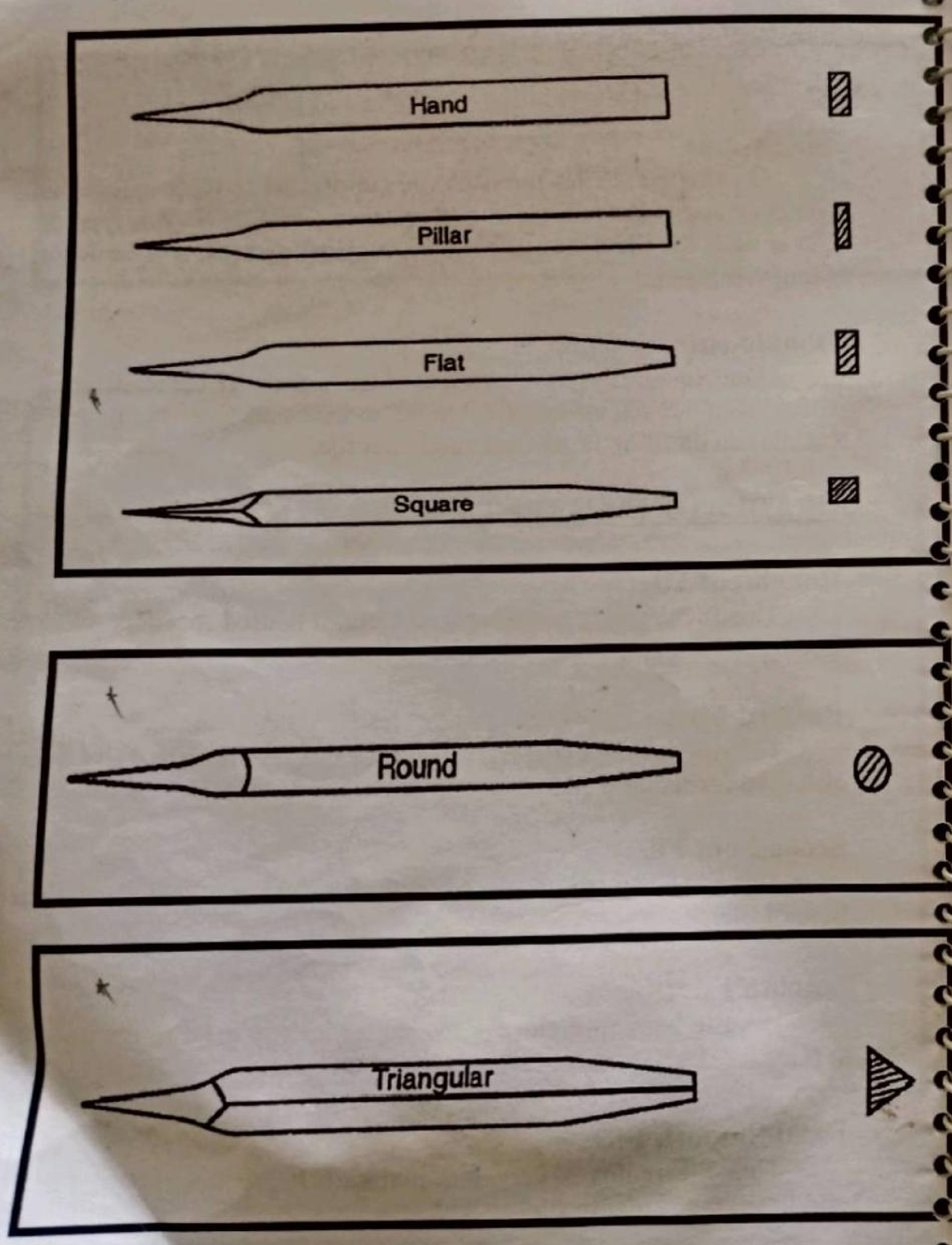
This file has medium spaced teeth. It produces better finish than bastard file.

**Smooth File:**

His file has more closely spaced teeth. It is used to get smooth surface.

**Dead Smooth File:**

This file removes very less material. It produces better polish and finish than smooth file.








**SHAPES OF FILES:**

The shape of a file is its general outline and *cross-section.*

**Flat File:**

This is tapered in width and thickness, and one of the most commonly used files for general work. They are always double-cut on the faces and single-cut on the edges.

**Hand File:**

This is parallel in its width, and tapered in thickness. A hand file is used for finishing flat surfaces. It has one safe edge (i.e, it is left uncut) and therefore is useful where the flat file cannot be used. They are always double-cut.

**Square File:**

This is square in cross-section double-cut tapered towards the point. This is used for filing square corners, enlarging square or rectangular openings as splines and keyways.

**Pillar File:**

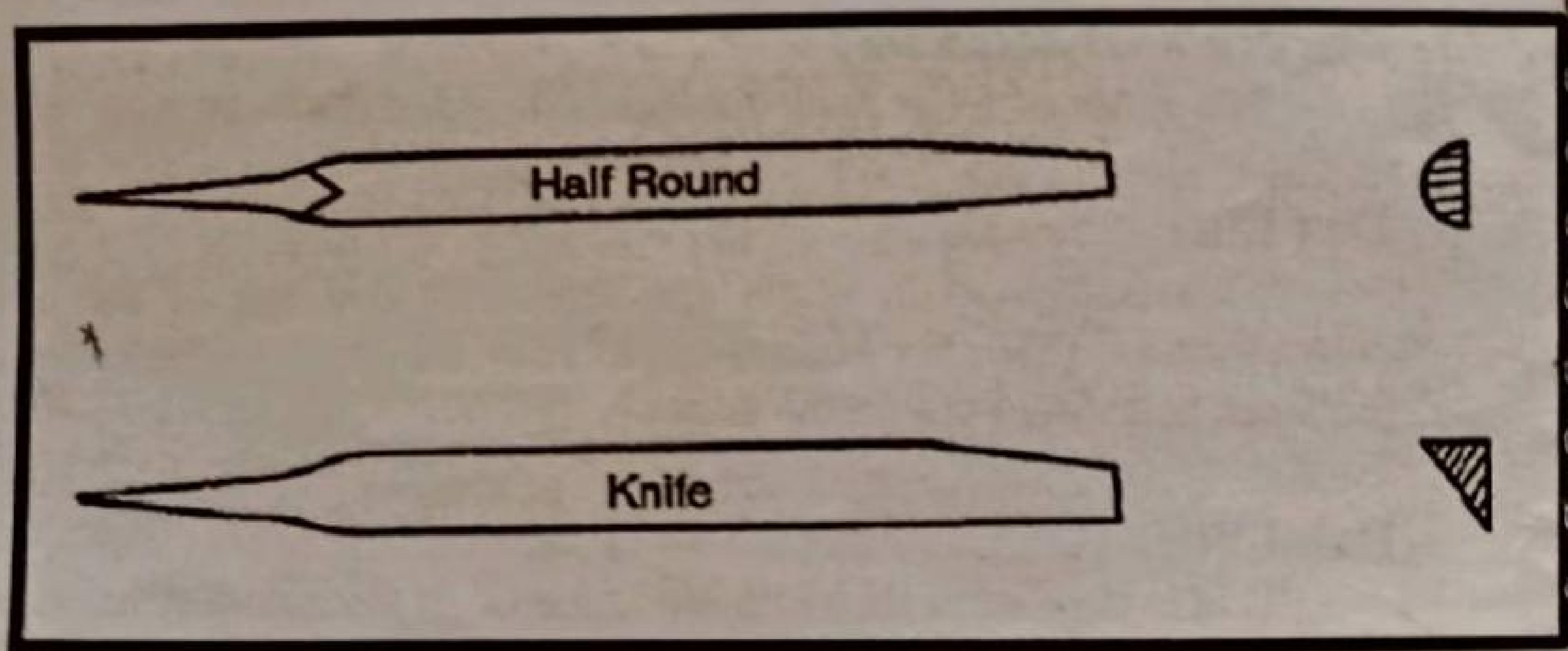
Pillar files are double-cut, narrow and of rectangular section. It has one safe edge, and is used for narrow work, such as keyways, slots and grooves.

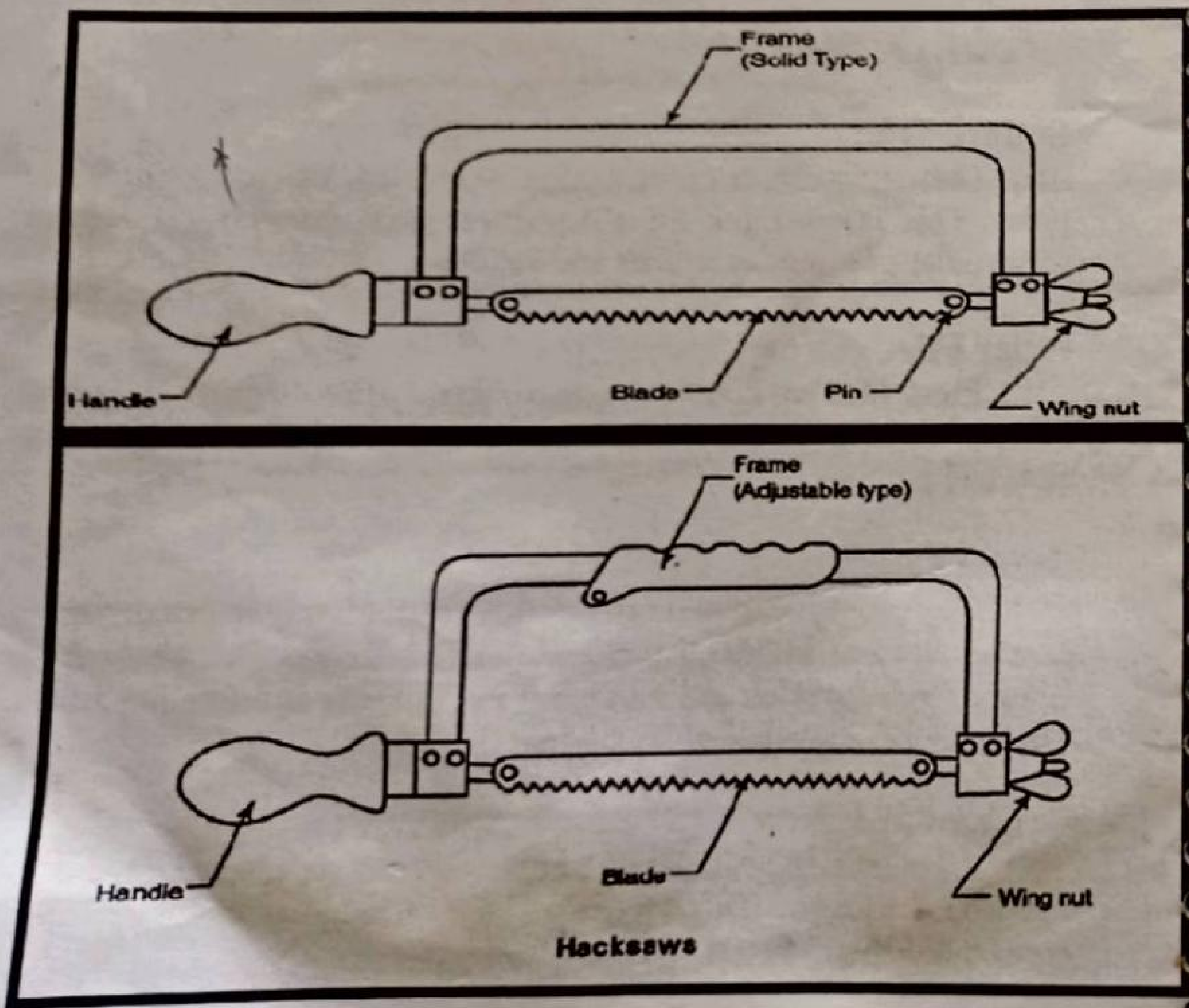
**Round File:**

They are round in cross-section and usually tapered. When parallel they are described as parallel round. Round files are used for filing curved surfaces and enlarging round holes and forming fillets. They may be single-cut or double-cut.

**Triangular File:**

Triangular file is tapered, double-cut, and the shape is that of an equilateral triangle. They are used for rectangular cuts and filing corners less than 90°.








### Half-round File:

This is tapered double-cut and its cross-section is not a half-circle but only about one-third circle. This file is used for round cuts and filing curves surfaces.

### Knife edge File:

This is shaped like a knife, tapered in width, thickness and double-cut. They are used for filing narrow slots, notches and grooves.

### ***Specification of a File:***

1. Length , say, 100mm
2. Single or Double cut
3. Shape , say , flat
4. grade, say, bastard.

### HACKSAW:

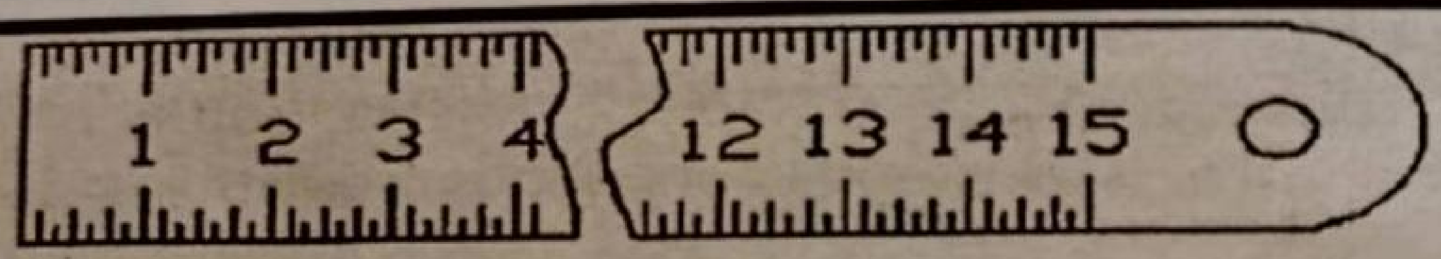
Hacksaw is used for cutting rods, strips, bars and pipes into desired length. It consists of a fixed or adjustable frame, a wing nut and blade. The hacksaw blade is made up of hard, tempered alloy steel. There are three types of blades:

(a) All hard (b) Semi-flexible (c) Flexible back.

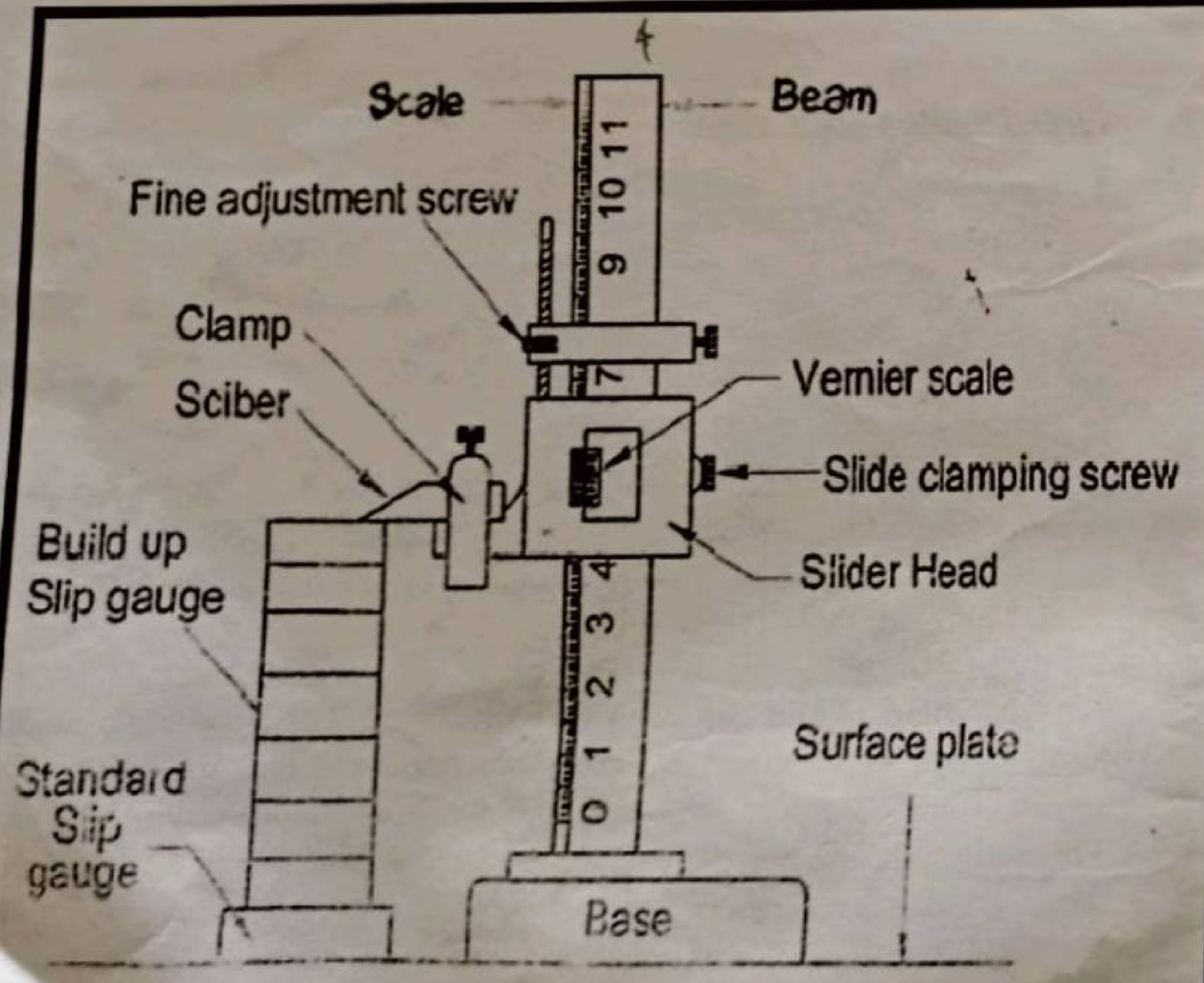
All hard blades are used for cutting the harder metals such as alloy steels, while flexible blades are good for use by unskilled or semi-skilled operators or where the work is inconveniently placed. These flexible blades are less liable to break and are used for general work. Blades are measured by the:

(1) length (2) width (3) Thickness (4) pitch of the teeth.

The points of the teeth are bent to cut a wide groove and provide clearance for the blade to prevent binding. This binding of the teeth to the sides is called the *set* of the teeth.




Steel rule




Vernier height gauge



**MEASURING TOOLS:**

**Steel Rule:**

The steel rule is one of the most useful tool in the shop for taking linear measurements of blanks and articles to an accuracy of from 1.0mm to 0.5mm. It consists of a strip of hardened steel having line graduations etched or engraved at interval of fraction of a standard unit of length. This is usually marked in both inches and centimeters. Steel rules are made 150,300,500 and 1000mm long.

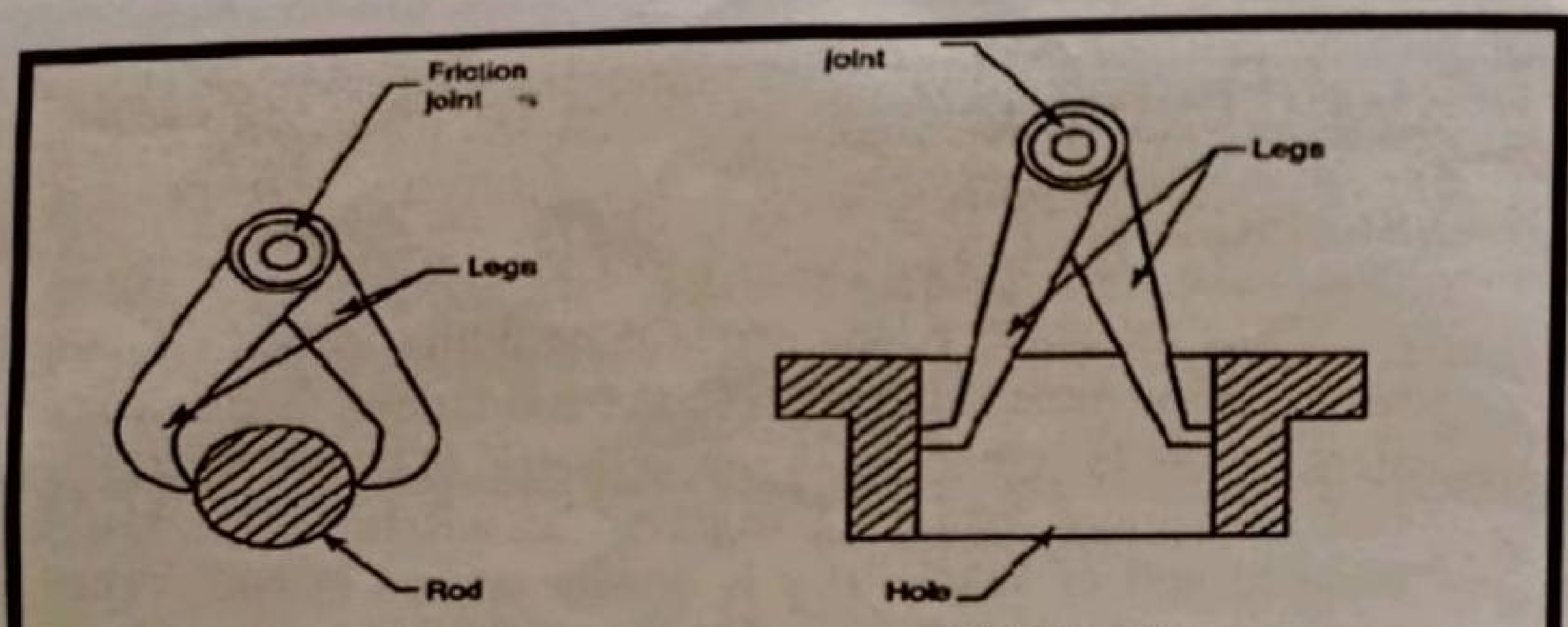
**Vernier Height Gauge:**

The vernier height gauge is used to measure the height of parts to an accuracy of 0.02mm in metric measurement and 0.001 in in English measurement. This is also used for precision layout work. The essential parts of the instrument are
(1) Instrument base with a lapped undersurface (2) Graduated beam or main scale
(3) Sliding head with vernier (4) Sliding jaw holding the scriber (5) Vernier clamp which moves with the sliding head (6) Fine adjustment screw in the vernier clamp
(7) Two knurled screws which lock the vernier head and clamp to the rule at any desired setting.

The vernier height gauges are available for the following lower and upper limits of measurement : 0 to 200 mm, 20 to 250mm, 30 to 300mm, 40 to 500mm, 60 to 800mm, 60 to 1000mm. For marking out, the scriber is set for the specified height and then the lines are scribed by moving the scriber along the work-piece.

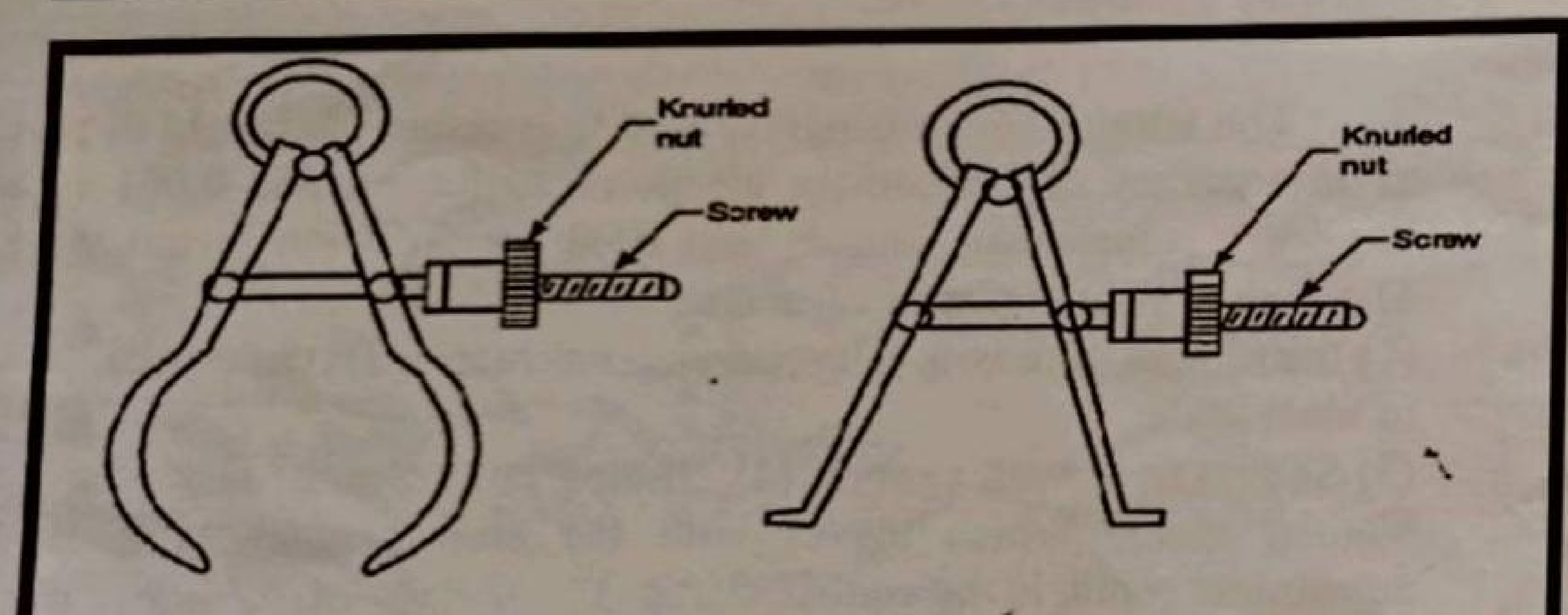
**Vernier Caliper:**

The vernier caliper is primarily intended for measuring both inside and outside diameters of shafts, thickness of parts, etc. to an accuracy of 0.02mm by a vernier scale attached to the caliper. A vernier scale is the name given to any scale making use of the difference between two scales which are nearly, but not quite alike, for obtaining small differences



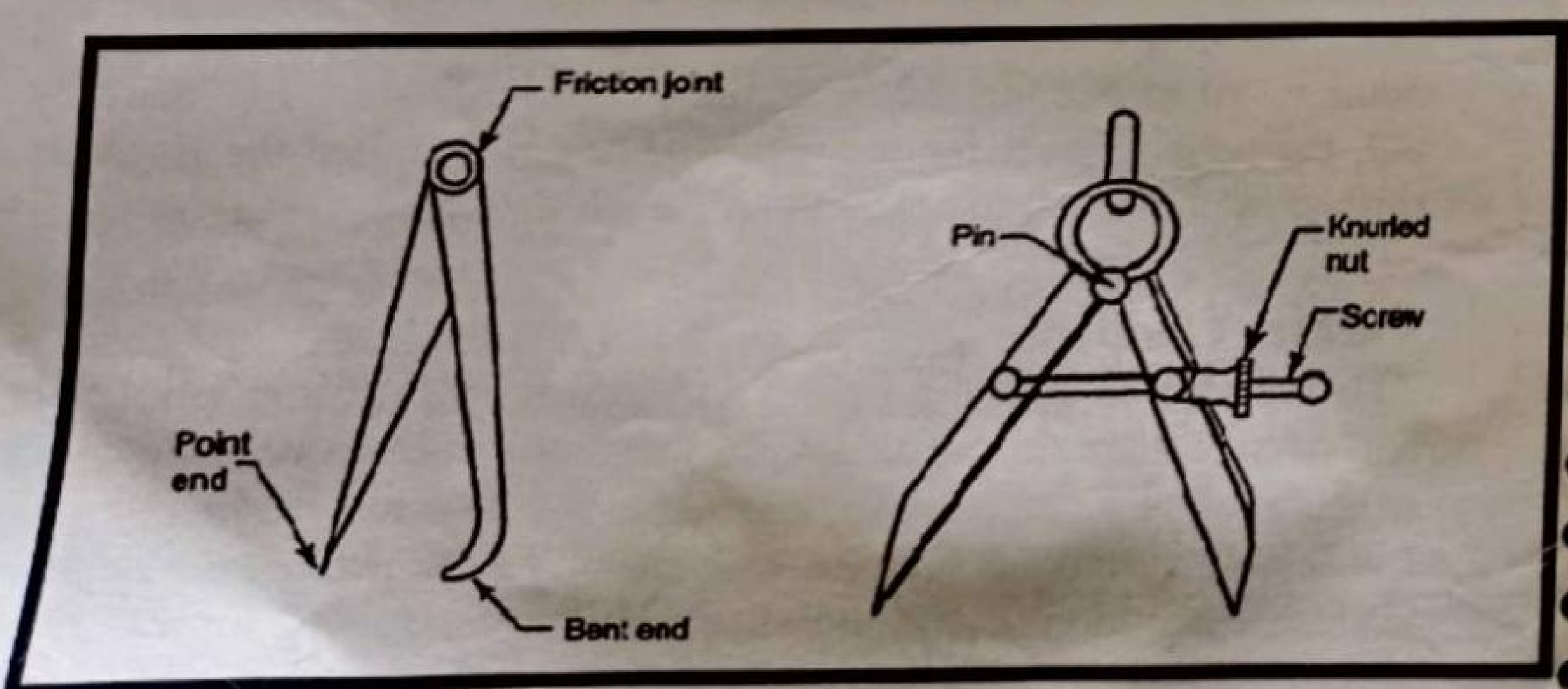

a) Outside caliper (b) Inside caliper

Firm joint calipers




a) Outside spring caliper (b) Inside spring caliper

Spring calipers






The instrument comprises a beam or main scale which carries the fixed graduations, two measuring jaws, a vernier head having a vernier scale engraved on it, and an auxiliary head of a vernier clamp which is used for a specified dimension by a micrometer screw. The vernier head and the auxiliary head can be locked to the main scale by the knurled screw attached to each head.

### Calipers:

A caliper is used to transfer and compare a dimension from one object to another or from a part to a scale or micrometer where the measurement can not be made directly. An *Outside Caliper* is used for measuring or comparing thicknesses, diameters, and other outside dimensions. An *Inside Caliper* is used for comparing or measuring hole diameters, distances between shoulders, or other parallel surfaces of any inside dimensions.

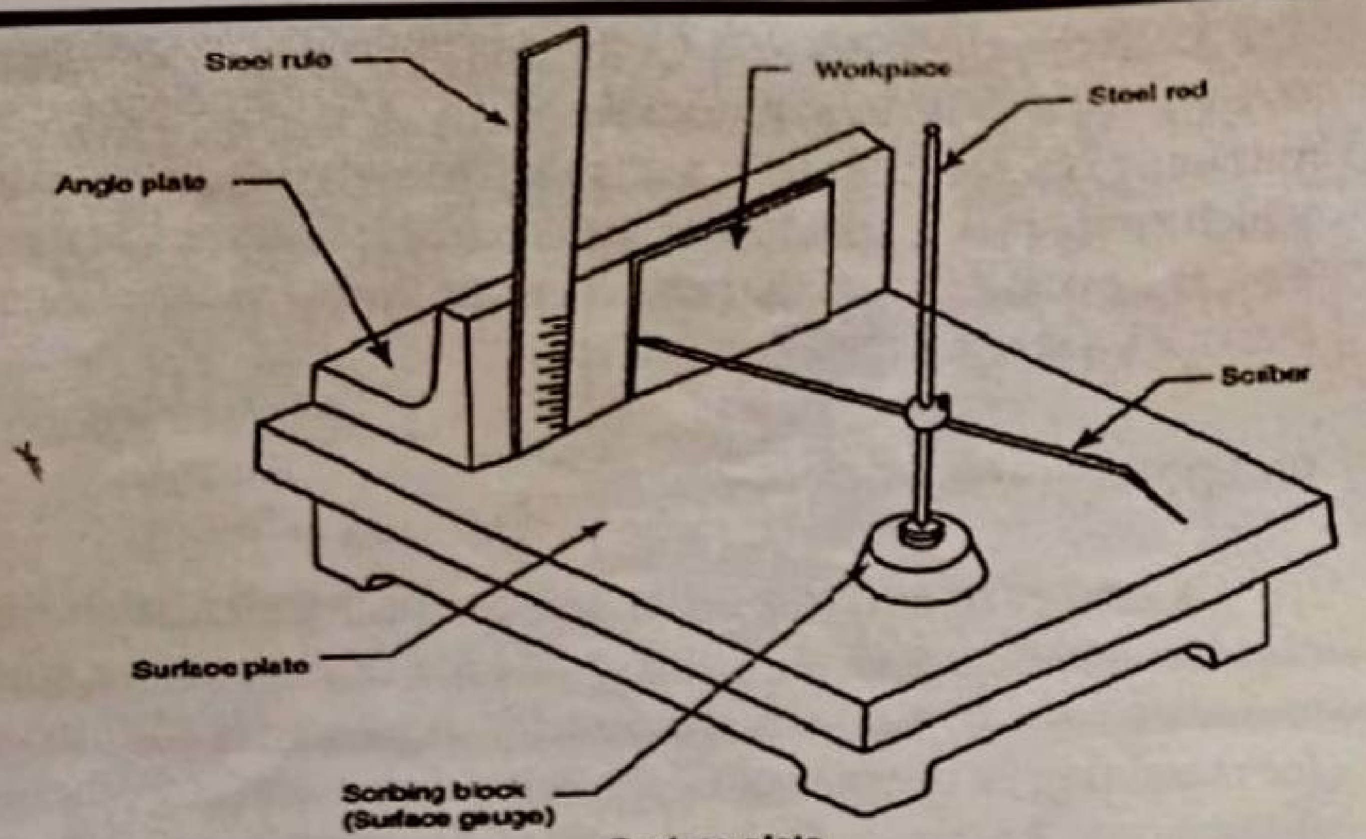
### Dividers:

A divider is similar in construction to a caliper except that both legs are straight with sharp hardened points at the end. This tool is used for transferring dimensions, scribing circles, and doing general layout work.

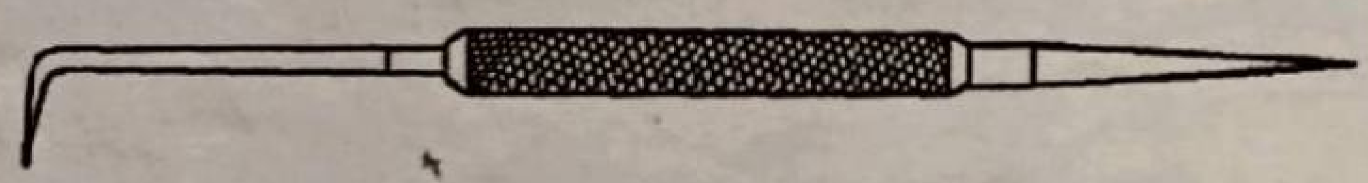
## MARKING TOOLS:

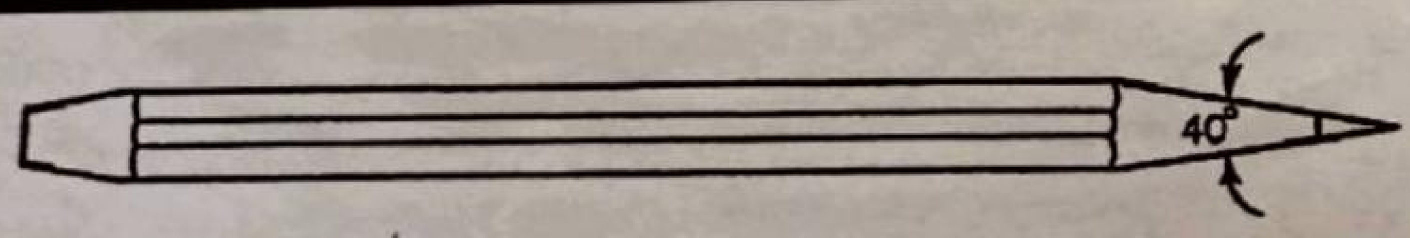
### Surface Plate:

The surface plate is used for testing the flatness of work itself and is also used for marking-out work. Generally, surface plate is used for small pieces of work and the *marking-out table* is used for larger jobs. Surface plates are made of gray cast iron and of solid design or with ribs. They should be well and reflection-free illuminated and rest horizontally on a firm support, the working height being about 800mm from the floor. These plates may be of following types 1.5 * 5 m, 1.5 * 3m, 2 * 2 m, 2 * 4m.

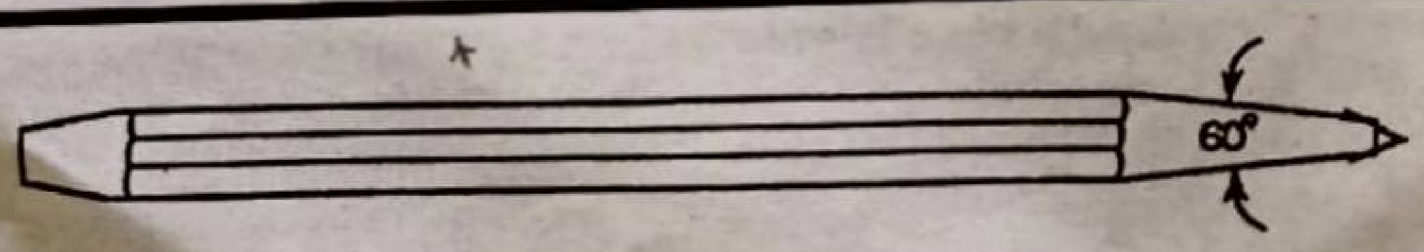


Surface plate


Scriber



Prick punch



Centre punch



## Angle Plate:

The angle plate which is made of grey cast iron has two plane surfaces at right angles to each other. This is used in conjunction with the surface plate for supporting work in the perpendicular position.

## Scriber:

The scriber is a piece of hardened steel about 150 to 300mm long and 3 to 5 mm in diameter pointed one or both ends like a needle. It is held like a pencil to scratch or scribe lines on metal. The bent end is used to scratch line in places where the straight end cannot reach. The ends are sharpened on an oilstone when necessary.

## Punch:

A punch is used in a bench work for marking out work, locating centers, etc. in a more permanent manner. Two types of punches are used: (1) prick punch (2) center punch.

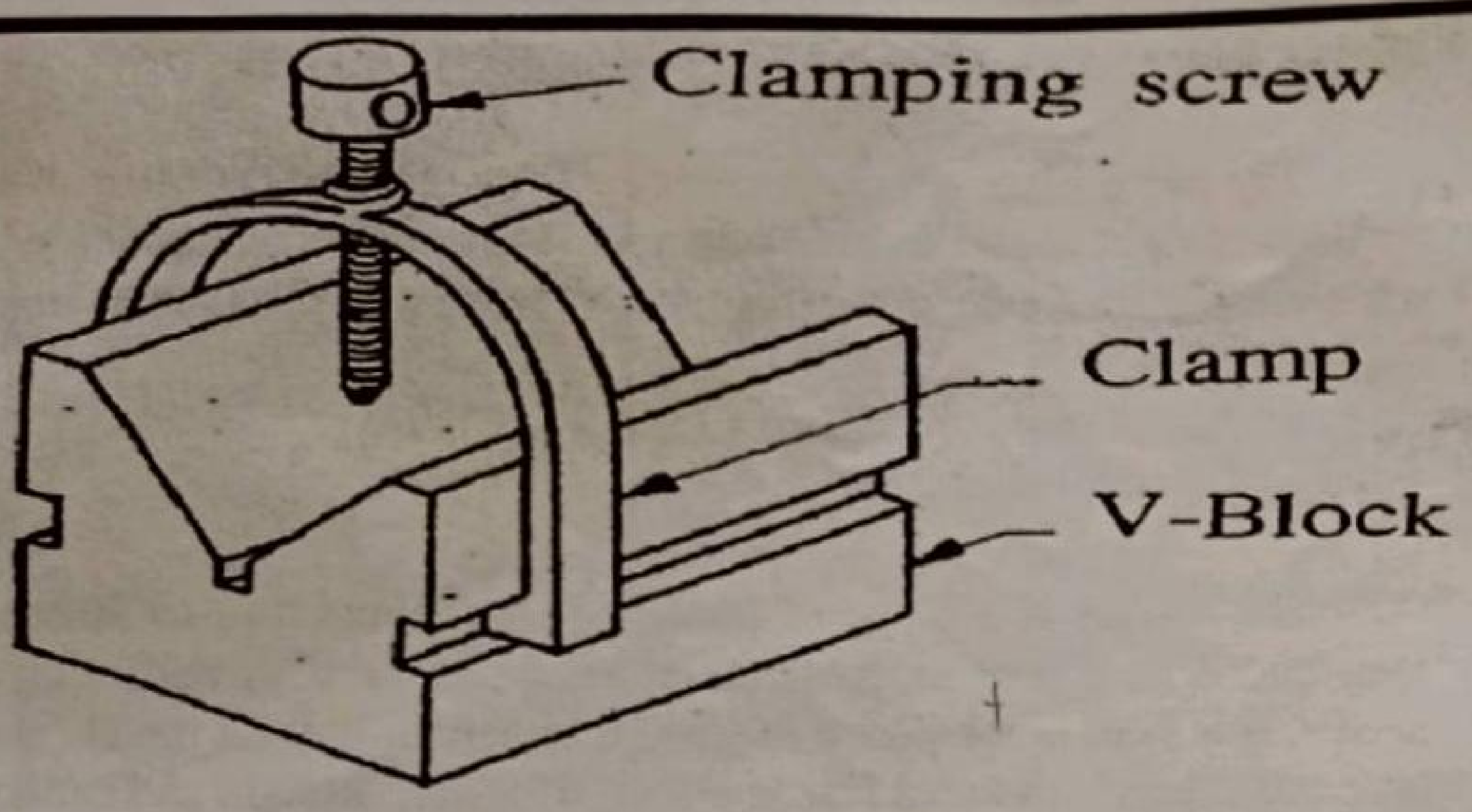
## The prick punch:-

It is a sharply tool. The tapered point of the punch has an angle of usually 40°. it is used to make small punch marks on layout lines in order to make them last longer.

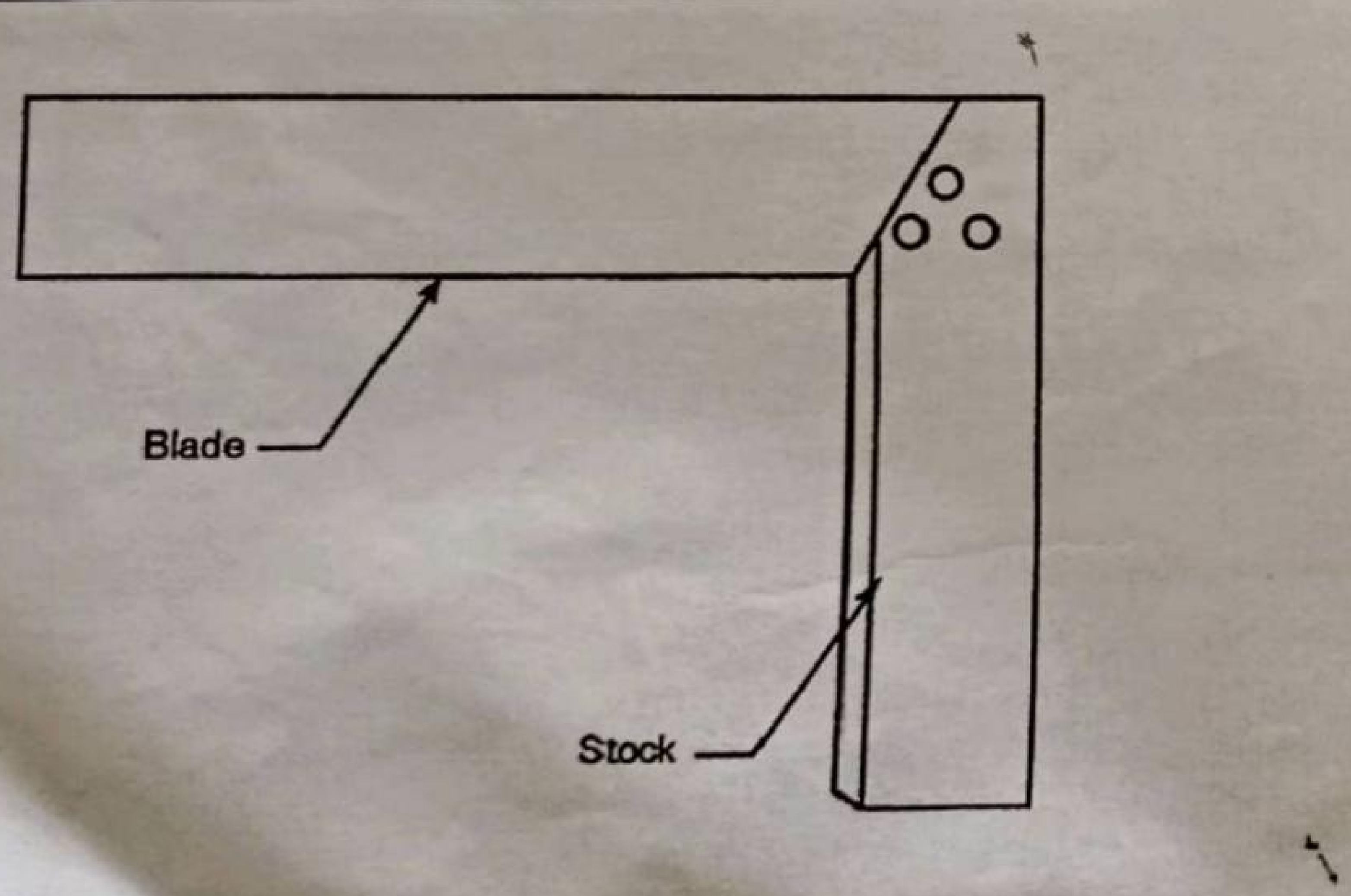
## The center punch:-

It looks like a prick punch. Its point has an angle more obtuse than that of the prick punch point, this angle being 60°. The center punch is used only to make the prick punch marks larger at the centers of holes that are to be drilled, hence the name center punch. A strong blow of the hammer is needed to mark the point.

In its body portion the punch is a steel rod 90 to 150mm long and 8 to 13mm in diameter.




V-block with clamp




Try-square



**V- Block:**

The V-Block is a block of steel with V-shaped grooves. Roundly shaped work pieces which are to be marked or drilled are placed on V-supports. V-blocks of the following sizes are found to be most useful: length from 50 to 250mm, width and height 50 to 100mm. For long cylindrical work, several blocks of the same size are used as set.

**Try-square:**

The try-square is made in one piece, both blade and beam. This is used when it is necessary to get another edge or surface exactly at right angles to an already trued edge or surface and also for laying out work. The squareness of any square may be tested by placing the beam of the square against a straight edge with the blade resting on a smoother surface. While in this position a line may be scribed along the edge of the blade.

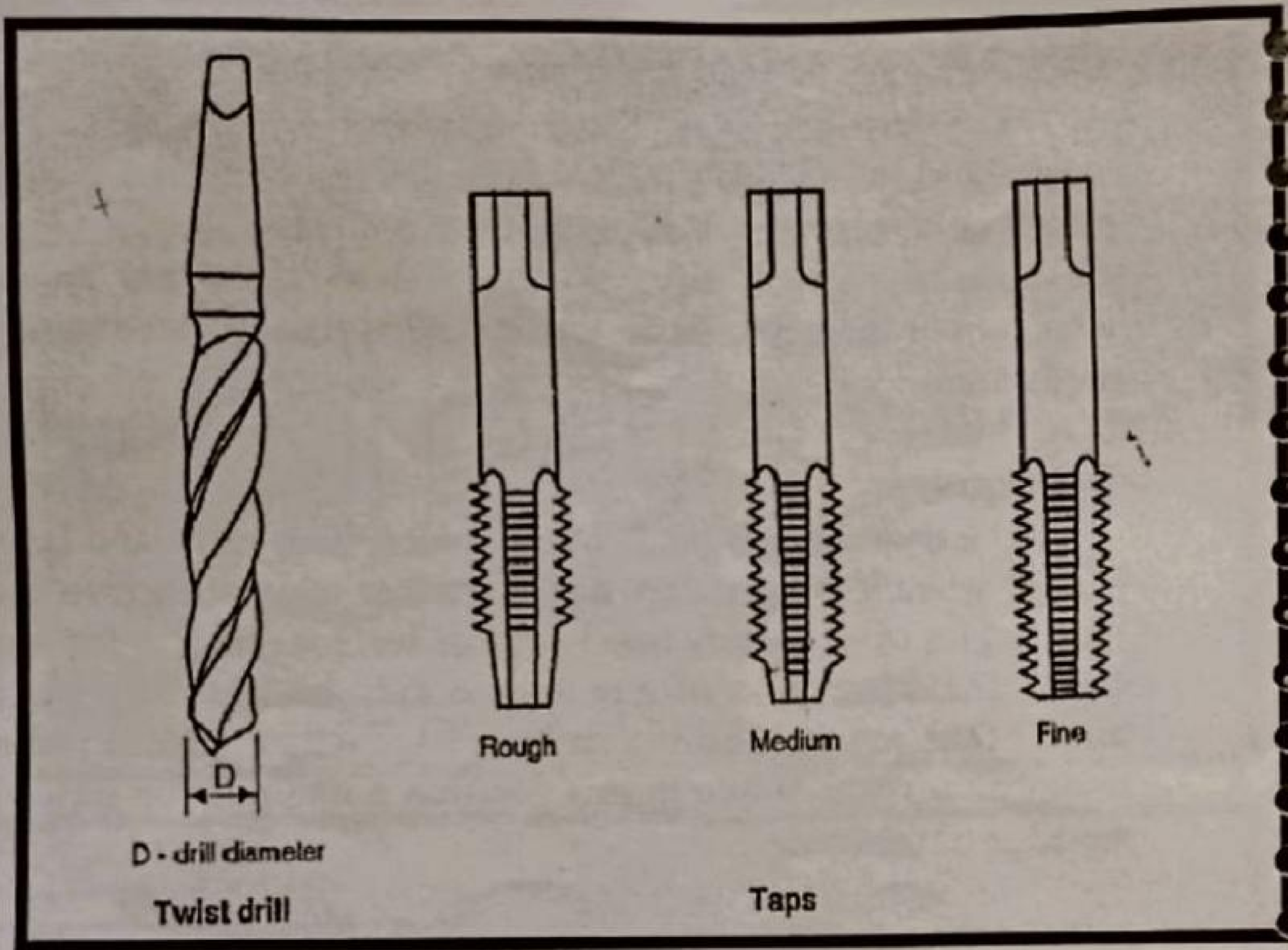
**DRILL :**

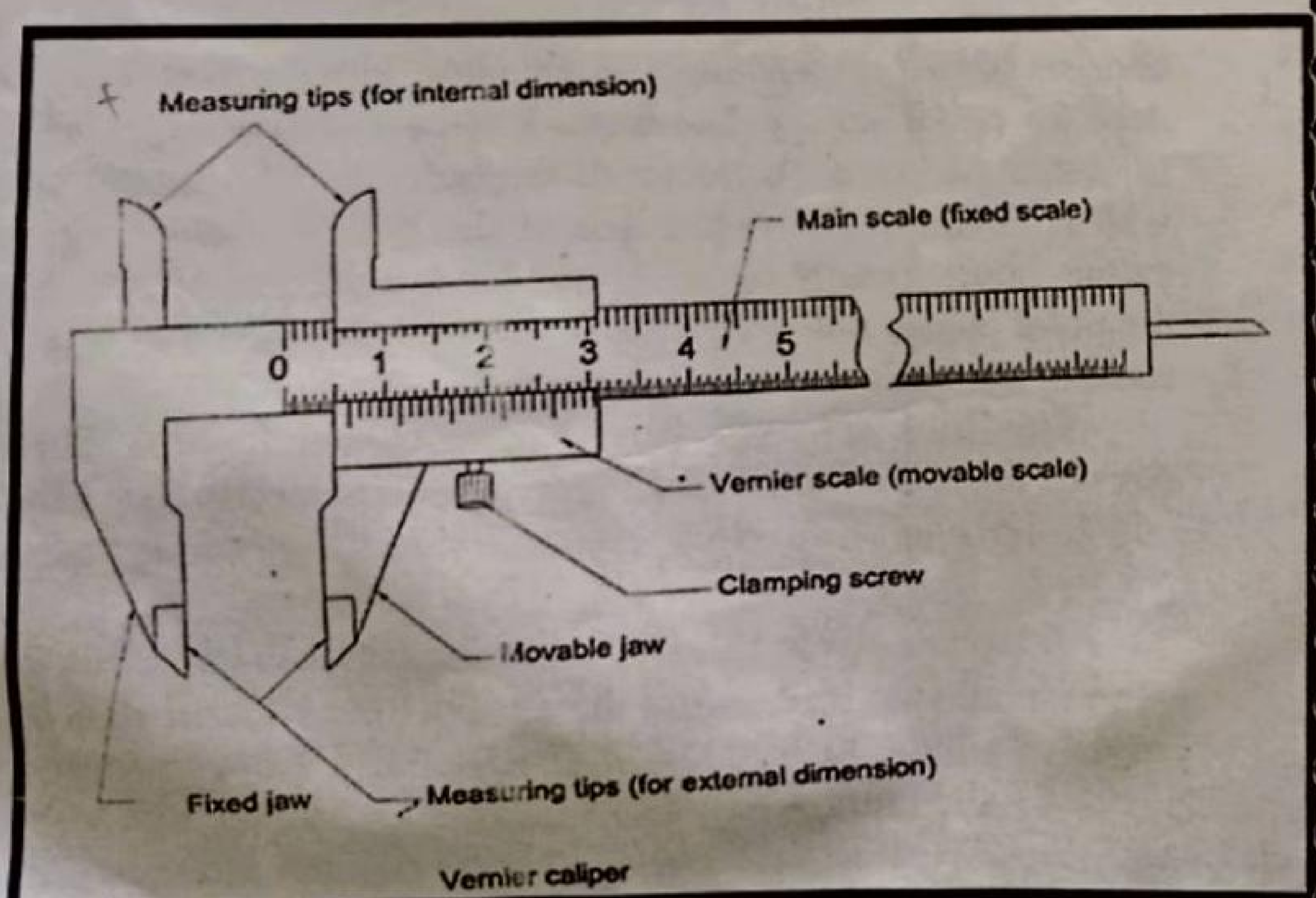
A drill is a tool for making holes in metal piece. It usually consists of two cutting edges set an angle with the axis. Twist drill are used for rapid and accurate work. The best cutting angle is 118° and to obtain the correct diameter of the hole, the drill should be ground with both lips at 59° to the axis of the drill, with the lengths of the cutting edges exactly equal. A drill having unequal cutting edges and the point angles not symmetrical will probably result in a larger hole running out of line.

The drills are made from 3.0mm to 100mm size. The smaller sizes of drills are not usually marked and the size is found by the use of a drill gauge.

**TAP:**

A tap is a screw-like tool which has threads like a bolt and three or four flutes cut across the thread. It is used to cut threads on the inside of a hole, as in a nut. The edges of the thread formed by the flutes are the cutting edges. The lower part of the tap is somewhat tapered so that it can well attack the walls of the drill hole. The upper part of the tap consists of a shank ending in a square






for holding the tap by tap-wrench. This is a two-handled wrench, and it may be either fixed or adjustable.

Taps are made from carbon steel or high speed steel and are hardened and tempered. Hand taps are usually made in sets of thrice. (1) Taper tap (2) Second tap, called Plug tap (3) bottoming tap. According to IS specification they are called Rougher, Intermediate and Finisher respectively.

The end of the *Rougher* has about six threads tapered. This is used to start the thread so that the threads are formed gradually as the tap is tuned into the hole.

The *Intermediate* is tapered back from the edge about three or four threads. This is used after the taper tap has been used to cut the thread as far as possible.

The *Finisher* has full threads for the whole length. This is used to finish the work prepared by the other two taps.

**Size of Tap Drill:**

Tap drill size is derived from the following formula.

$$D = T - 2d$$

Where D = diameter of the tap drill size.
T = diameter of the tap and
d = depth of the thread.

**Result:**

Thus the tools used in Fitting has been studied

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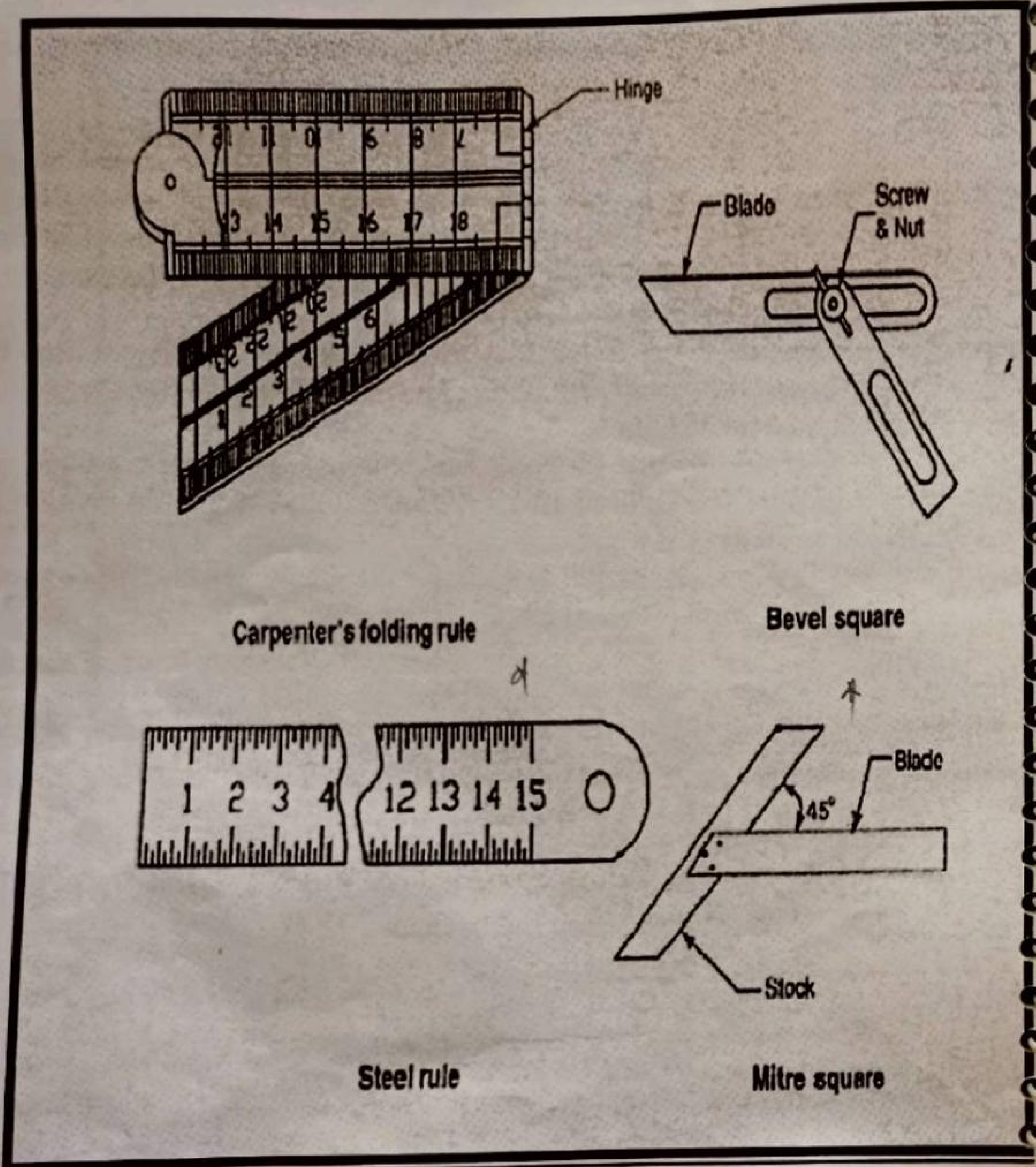



Carpenter's folding rule

Bevel square

Steel rule

Mitre square



# STUDY OF TOOLS IN CARPENTRY

## AIM:

To make a detailed study of the tools used in carpentry.

## INTRODUCTION:

The term carpentry is used with wood working. Wood works such as making wooden partition, roofing, flooring, etc. are carpentry works. The term joinery is used with making of doors, windows, wooden frames, etc. Teak, Mahogany, Sal, mango, Deodar, Padak, etc. are commonly used in carpentry.

Seasoning of wood is the process whereby the surplus moisture is drawn from the green wood by evaporation, thus allowing the natural juices to harden. The trees are felled when reaching maturity and the logs roughly squared into balks. The circulation of air is provided all round them.

**Wood (Timber) for commercial purposes are divided into two classes** :

(1) Soft Wood (2) Hard Wood. Soft wood cannot resist any kind of stress developed across their fibres and the timber gets splitted easily. Hard woods are nearly equally strong both along and across the fibres and can resist axial stress as well as transverse strain, shock and vibration quite satisfactorily.

### Marking and measuring tools:

### Rules:

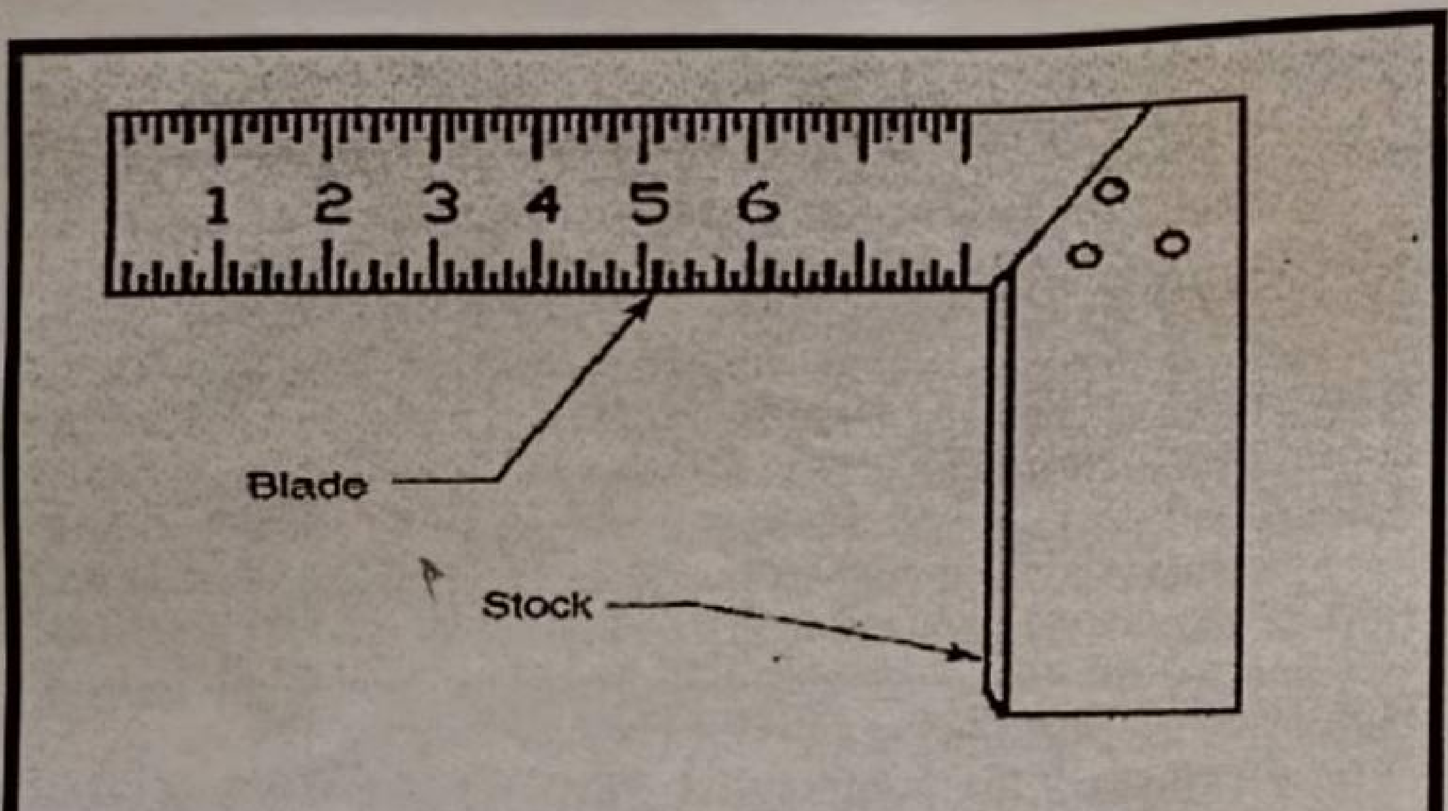
Rules of various sizes and designs are used by wood workers for measuring and setting out dimensions, but they usually work with a *four-fold box-wood rule* ranging from 0 to 60 cm. All the four pieces are joined with each other by means of hinged joints which make the scale folding.

For larger measurements carpenters use a *flexible measuring rule or tape*. Such rules are very useful for measuring curved and angular surfaces. When not in use, the blade is coiled into a small, compact, watch-size, case.

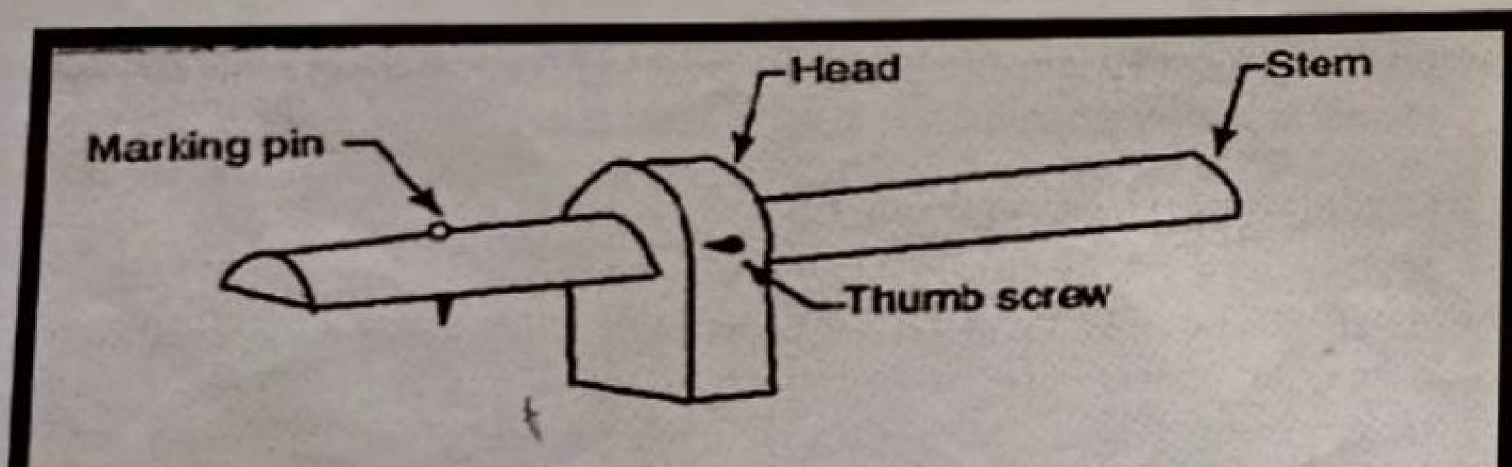
Steel rule is also used for measuring and marking dimensions. It is made of steel. It is graduated on both sides in millimeters, centimeters and inches. It is available in 150mm ( 6 inches) and 300mm ( 12 inches) length.

### Try square:

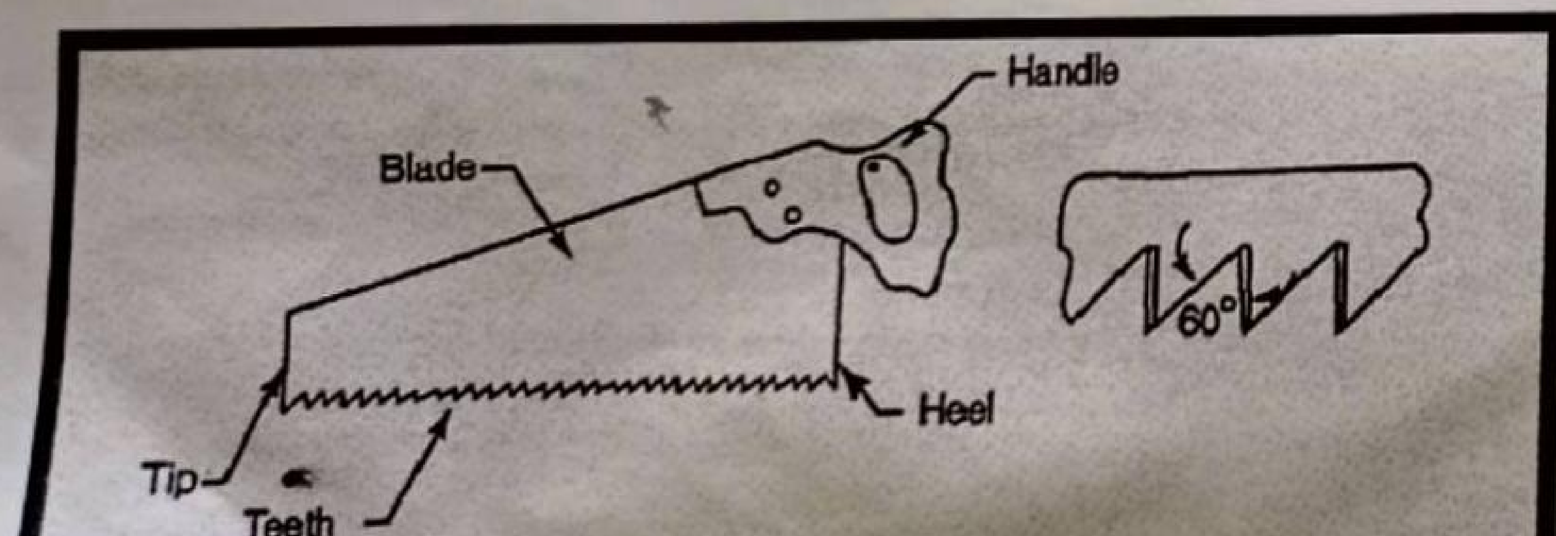
It is used for marking and checking right angles. It consists of a steel blade riveted at right angles to the edge of machined stock. The stock when in use acts as a fence and blade as guide for marking. The length of blade varies from 150 to 300mm.



Try square



Marking gauge



Rip saw





**Mitre square:**
It is similar to a try square in construction , but its blade is fitted to the stock at an angle of 45°.

**Bevel Square:**
It is similar to the try square but has a blade that can be swivelled to any angle from 0 to 180 degrees.

**Marking knife:**
Marking knives are used for converting the pencil lines onto cut lines. They are made of steel having one end pointed and the other end formed into a sharp cutting edge.

**Gauges:**
Gauges are used to mark lines parallel to the edge of a piece of wood. It consists of a small stem sliding in a stock. The stem carries one or more steel marking points or a cutting knife. The stock is set to the desired distance from the steel point and fixed by the thumbscrew. The gauge is then held firmly against the edge of the wood and pushed along the sharp steel point marking the line.

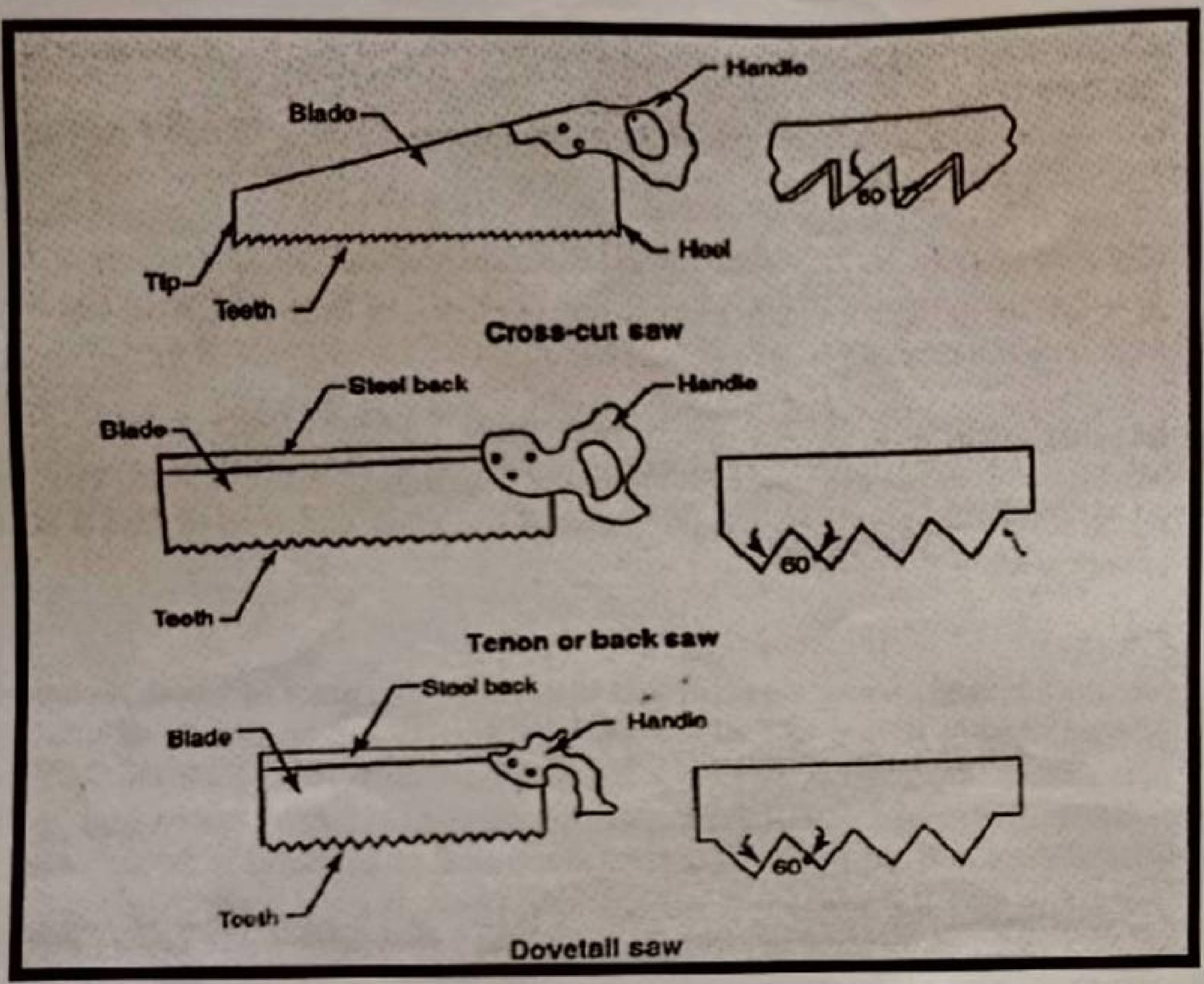
The *marking gauge* has one marking point. It gives an accurate cut line parallel to a true edge. The mortise gauge has two marking points – one fixed near to the end of the stem and the other attached to a brass sliding bar. These two teeth cut two parallel lines, called mortise lines.

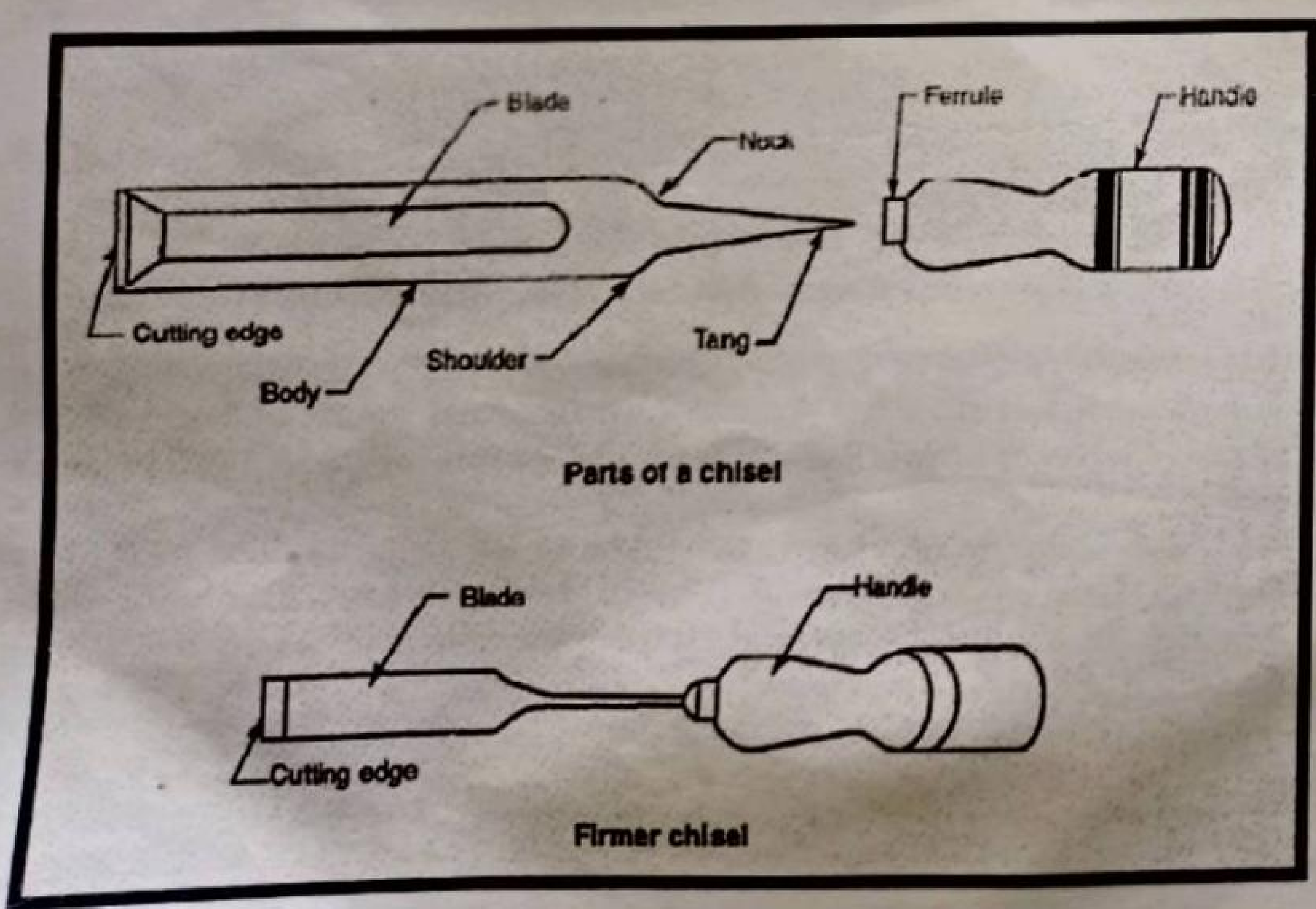
## CUTTING TOOLS:

Cutting tools include saws, chisels and gouges.

**Rip saw:**
Rip saws are used for cutting *along* the grain in thick wood. The blade is made of high grade tool steel. It is fitted in a wooden handle made of hard wood by means of rivets or screws. Rip saws are about 700mm long with 3 to 5 points or teeth per 25mm.

The teeth are bent alternatively, one to the right, the next to the left. Bending the teeth in this manner is called "setting in saw". The set of a saw provides clearance to prevent the blade from binding during the sawing operation.








**Cross-cut saw:**

Cross-cut saws or "Hand saws" as they are sometimes called, are used for cutting *across* the grain in thick wood. They are 600 to 650mm long with 8 to 10 teeth per 25mm.

**Panel saw:**

A panel saw is about 500mm long with 10 to 12 teeth per 25mm and is very much like the cross-cut saw. It has a finer blade and is used for fine work, mostly on the bench.

**Tenon or Back saw:**

This saw is mostly used for cross cutting when a finer and more accurate finish is required. The blade, being very thin, is reinforced with a rigid steel back. Tenon saw blades are from 250 to 400mm in length and generally have 13 teeth per 25mm.

**Dovetail saw:**

A smaller version of the tenon, this saw is used where the greatest accuracy is needed and fine shallow cuts are to be made. The number of teeth may be from 12 to 18 per 25mm, while the length may vary from 200 to 350mm.

**Firmer chisel:**

The firmer chisel is the most useful for general purposes and may be used by hand pressure or mallet. It has a flat blade about 125mm long. The width of the blade varies from 1.5 to 50mm.

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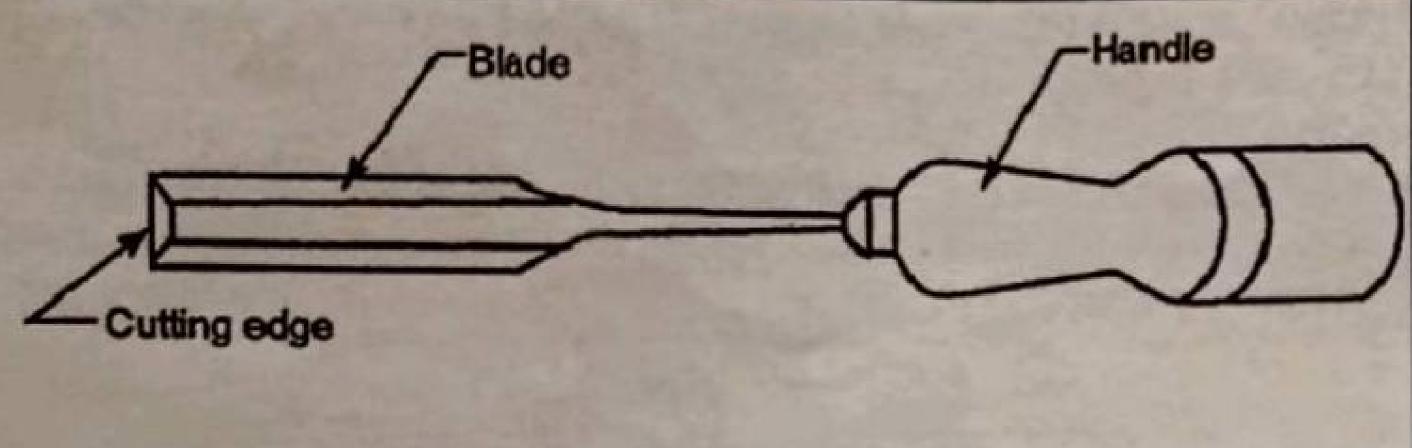
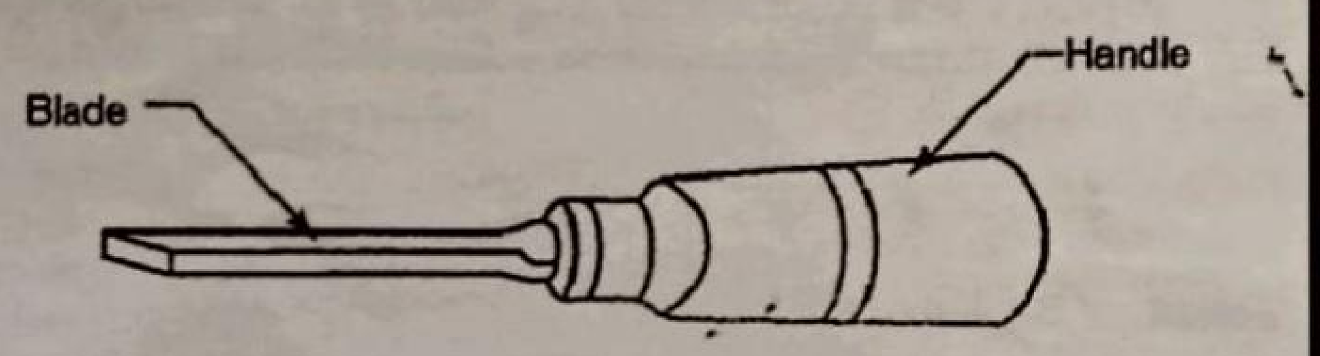
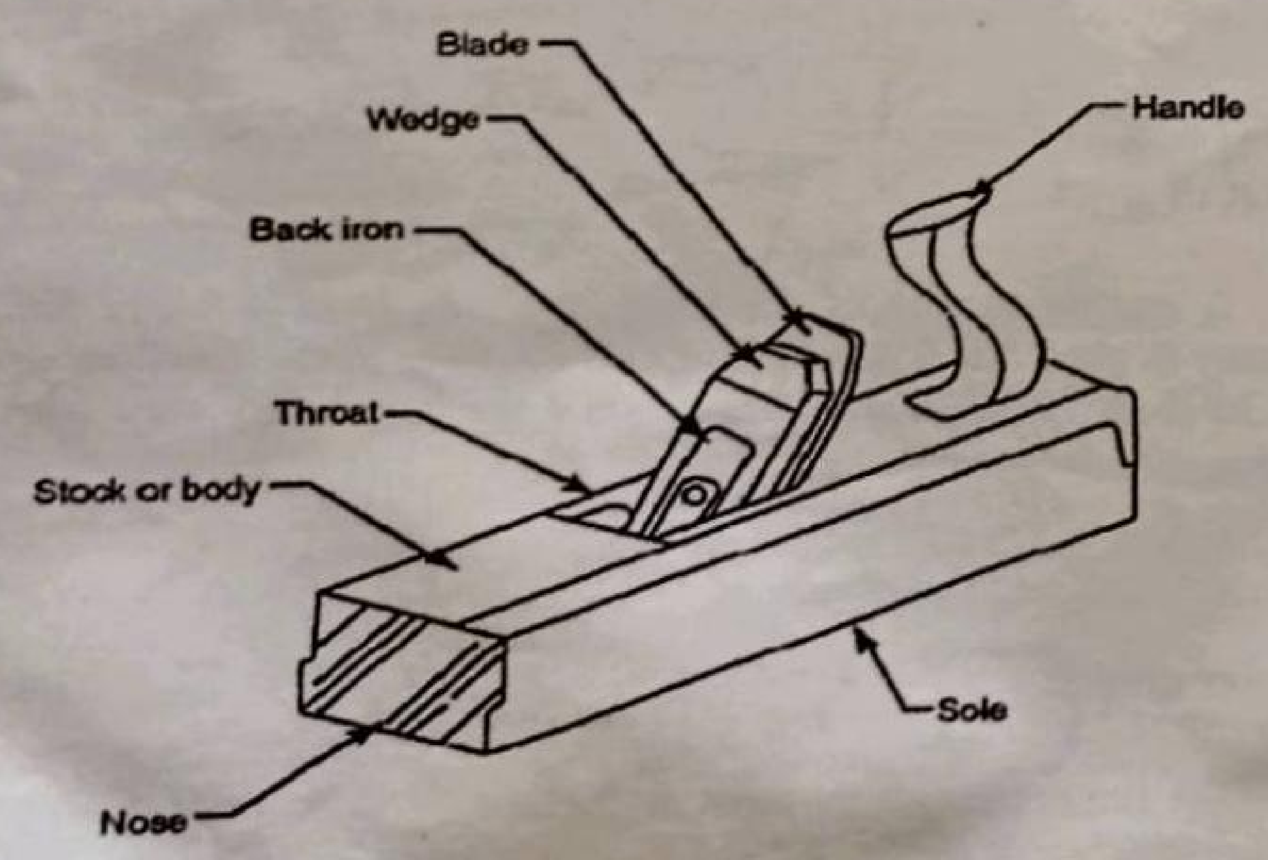



Bevelled edge firmer chisel




Mortise chisel




Wooden jack plane



**Beveled edge firmer chisel:**

The beveled edge firmer chisel is used for more delicate or fine work. They are useful for getting into corners where the ordinary firmer chisel would not be used.

**Mortise chisel:**

The mortise chisel is used for chopping out mortises. These chisels are designed to withstand heavy work. They are made with a heavy deep blade with a generous shoulder or collar to withstand the force of the mallet blows on the oval-sectioned handle. Blades vary in width from 3 to 16mm.

**Gauges:**

Gouges are chisels with curved sections and may be inside or outside ground. Inside ground gouges are used in exactly the same way for inside curved edges as a chisel would be for straight one. Outside ground gouges are used for curving hollows. Gouges are made to a large number of different curves for different work, and the size ranges from 6mm, with intermediate sizes to a maximum of 40mm wide.

**Planes:**

The plane can be likened to the chisel fastened into a block of metal or wood, and its blade acts exactly like a wide chisel. These are used for smoothening plane surfaces.

**Wooden Jack Plane:**

A wooden jack plane is the commonest and is used for the first truing-up of a piece of wood. It consists of a block of wood into which the blade is fixed by a wooden wedge. The blade is set at an angle of 45° to the sole. On the cutting blade another blade is fixed called cap iron or back iron. This does not cut, but stiffens the blade near its cutting edge to prevent chattering. It is the back iron which causes the shavings to be curled when they come out of the plane. Some types of planes do not have a cap iron. Jack planes are obtainable from 350 to 425mm in length and with blades 50 to 75mm wide.

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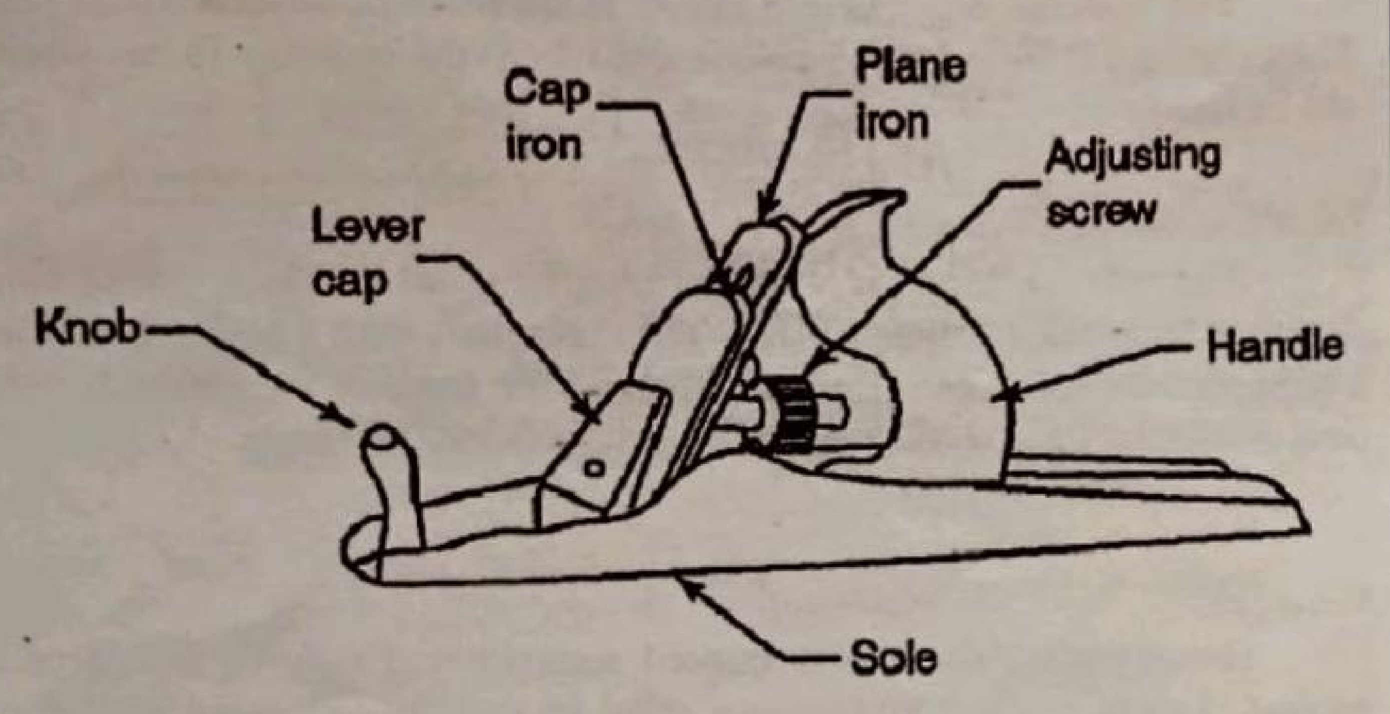
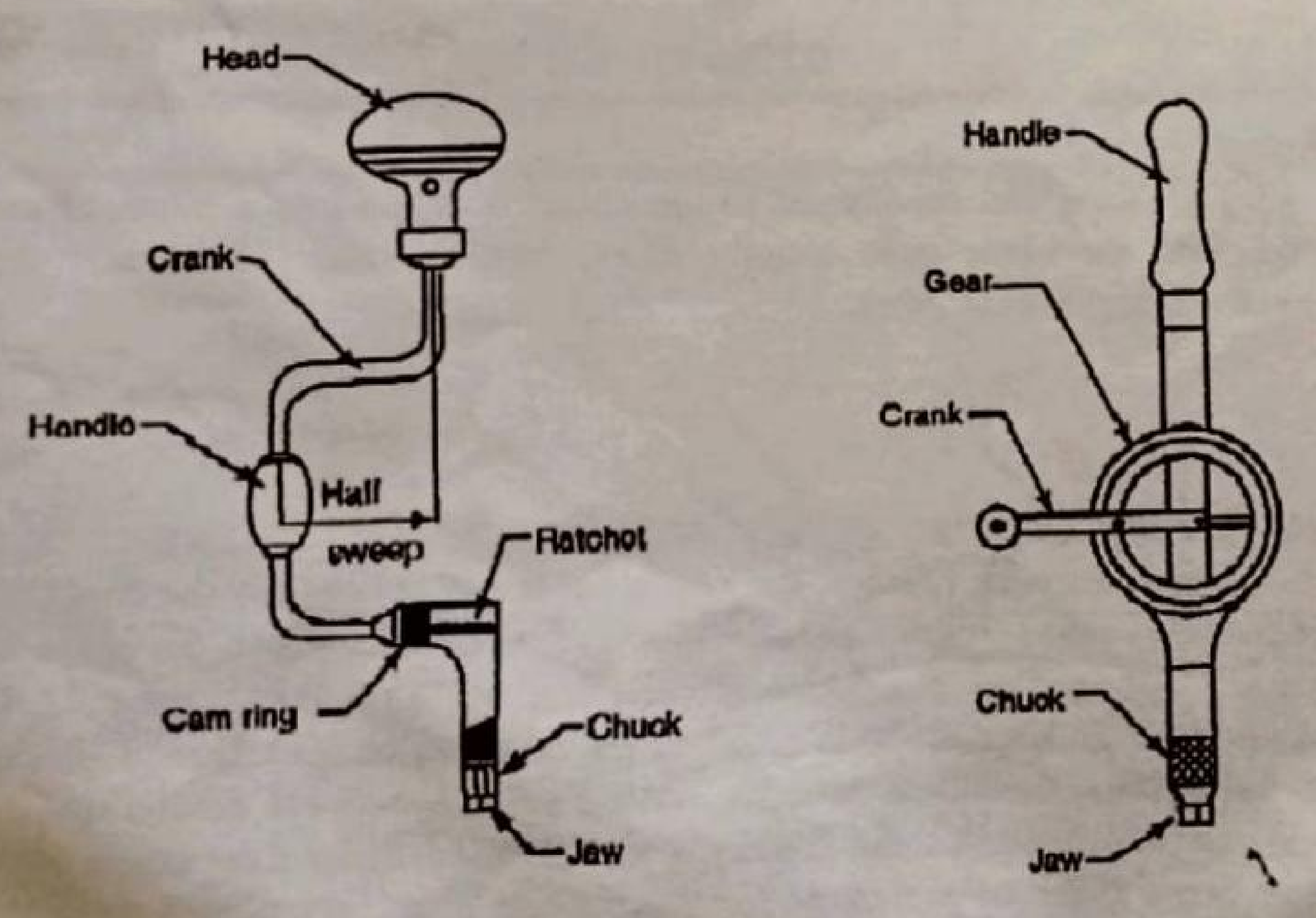


Metal jack plane



Ratchet brace

Wheel brace



### Metal Jack Plane:

Metal planes serve the same purpose as the wooden planes but facilitate a smoother operation and better finish. The body of a metal plane is made from a grey iron casting, with the side and sole machined and ground to a bright finish. The thickness of shavings are controlled by a fine screw adjustment.

### Boring Tools:

Boring tools are frequently necessary to make round holes in wood, and they are selected according to the type and purpose of the hole.

### Bradawl and Gimlet:

The bradawl and gimlet are hand-operated tools, and are used to bore small holes, such as for starting a screw or large nail.

### Brace:

The brace is a tool used for holding and turning a bit for boring tools. It has two jaws, which grip the specially shaped end of the bit. The *Ratchet Brace* is most useful for turning bits and drills of all kinds, being adaptable (a) for working in confined situations, and (b) for when the cut is particularly heavy and it is desirable to pull the handle through a quarter-turn only. The *Wheel Brace* is used to hold round and parallel-shanked drills. This tool is invaluable for cutting smooth hole accurately and quickly.

### Drills and Reamers:

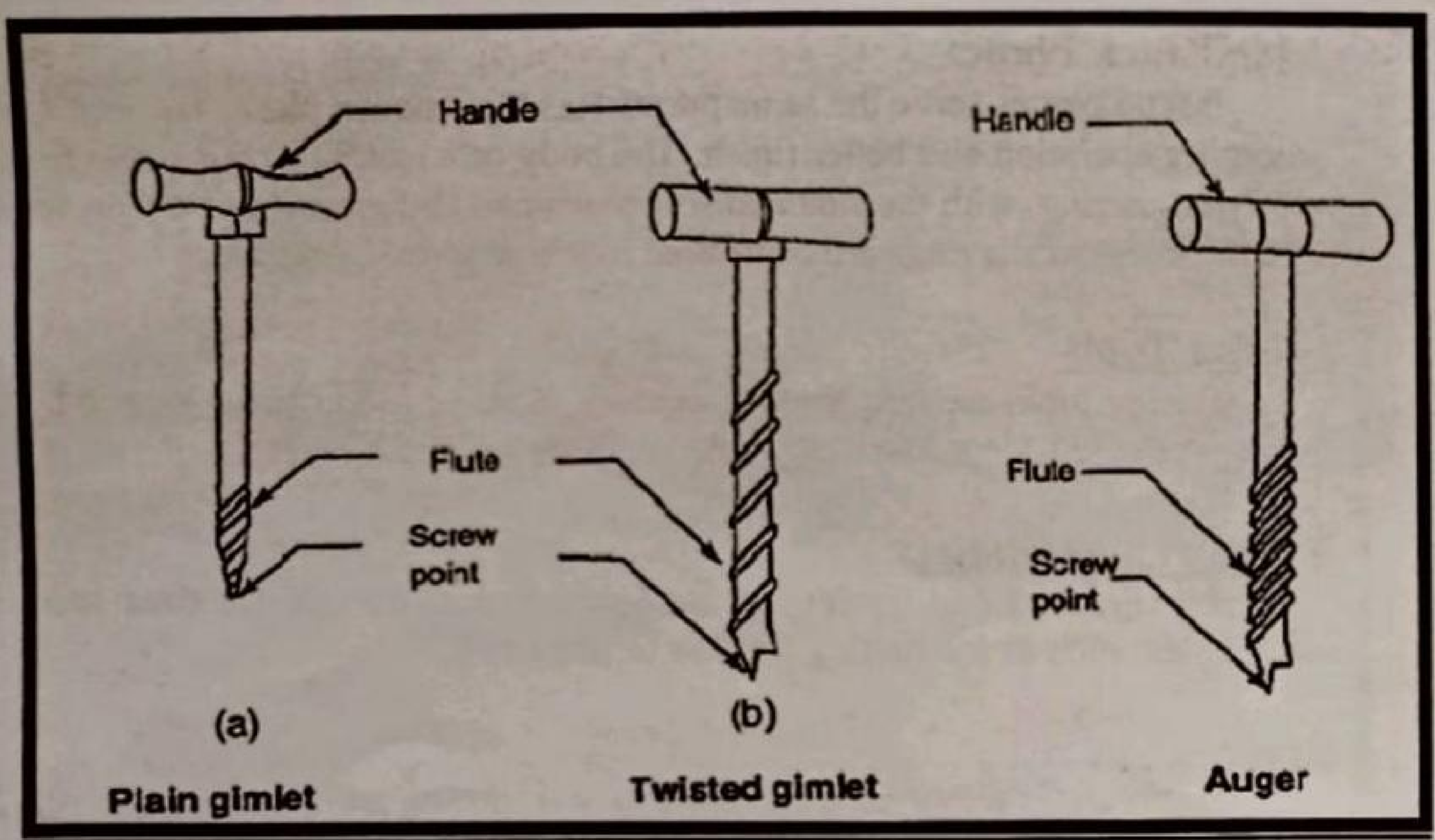
Morse drills are very convenient for making screw-holes, especially when with a wheel brace. This is adapted for drilling holes when wood working bits would be spoiled. Reamers are tapered bits shaped like shell bits and used for enlarging holes.

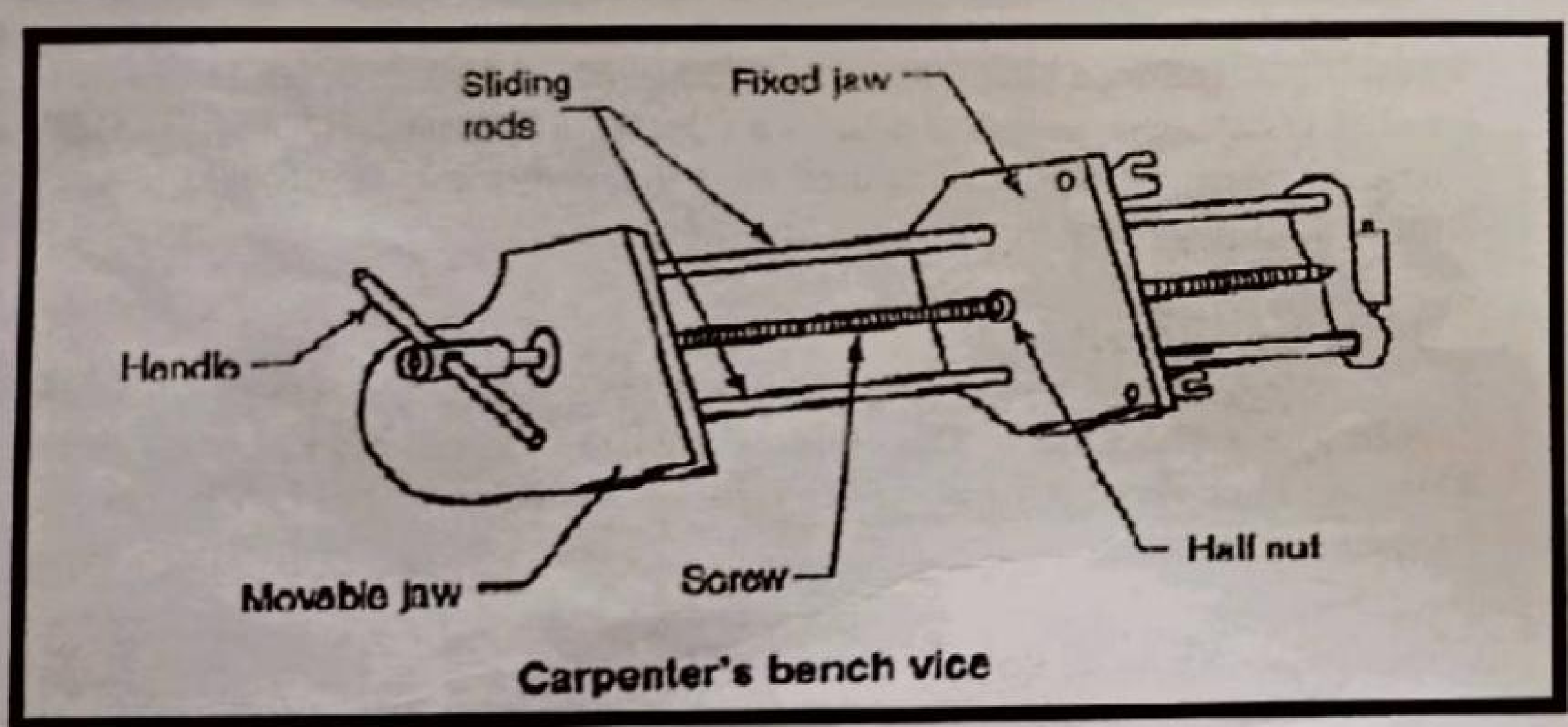
### Striking Tools:

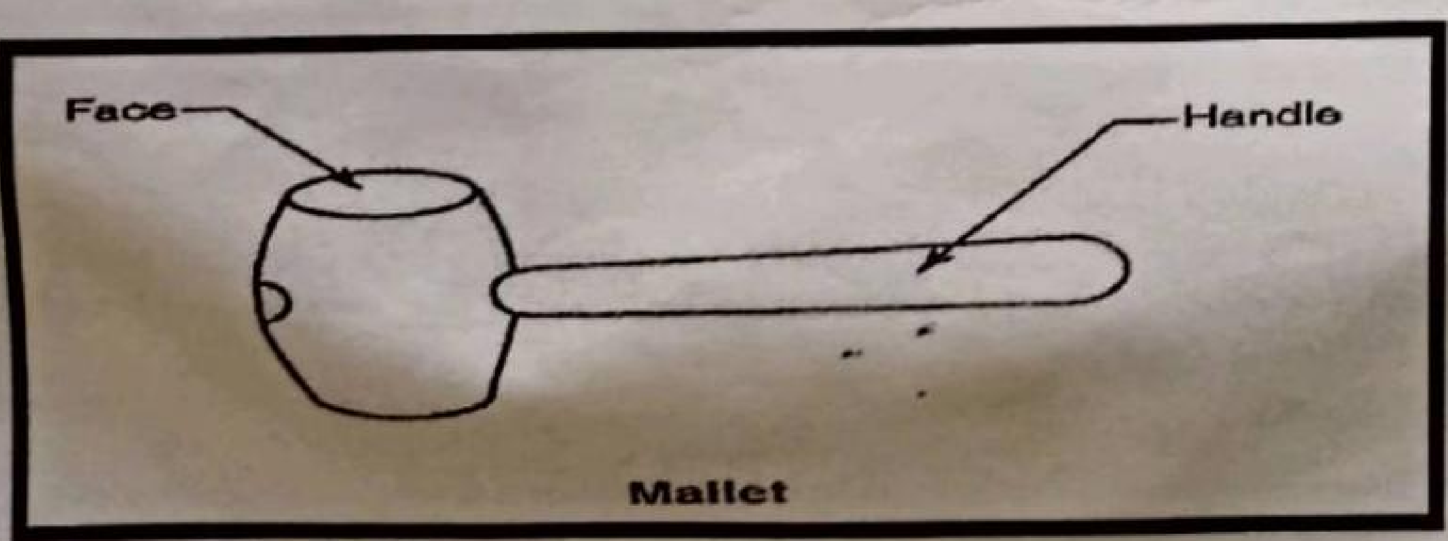
Tools like chisels, nails, etc. require pressure from the top to drive them into the wood. This pressure is obtained by striking operations using hammers and mallets.

### Warrington Hammer:

It is made of forged high carbon steel. It has a face, peen and handle. The handle is made of bamboo and fitted tightly in the eye with help of a wedge to avoid any slip out when in use.



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### Mallet:

The mallet is a wooden-headed hammer of round or rectangular cross-section. The striking face is made flat to the work. A mallet is used to give light blows to the cutting tools having wooden handle such as chisels and gouges.

### Holding Tools:

To enable the wood worker to cut his wood accurately, it must be held steady. There are a number of tools and devices to hold wood having its own purpose according to the kind of cutting to be done.

### Carpenter's Vice:

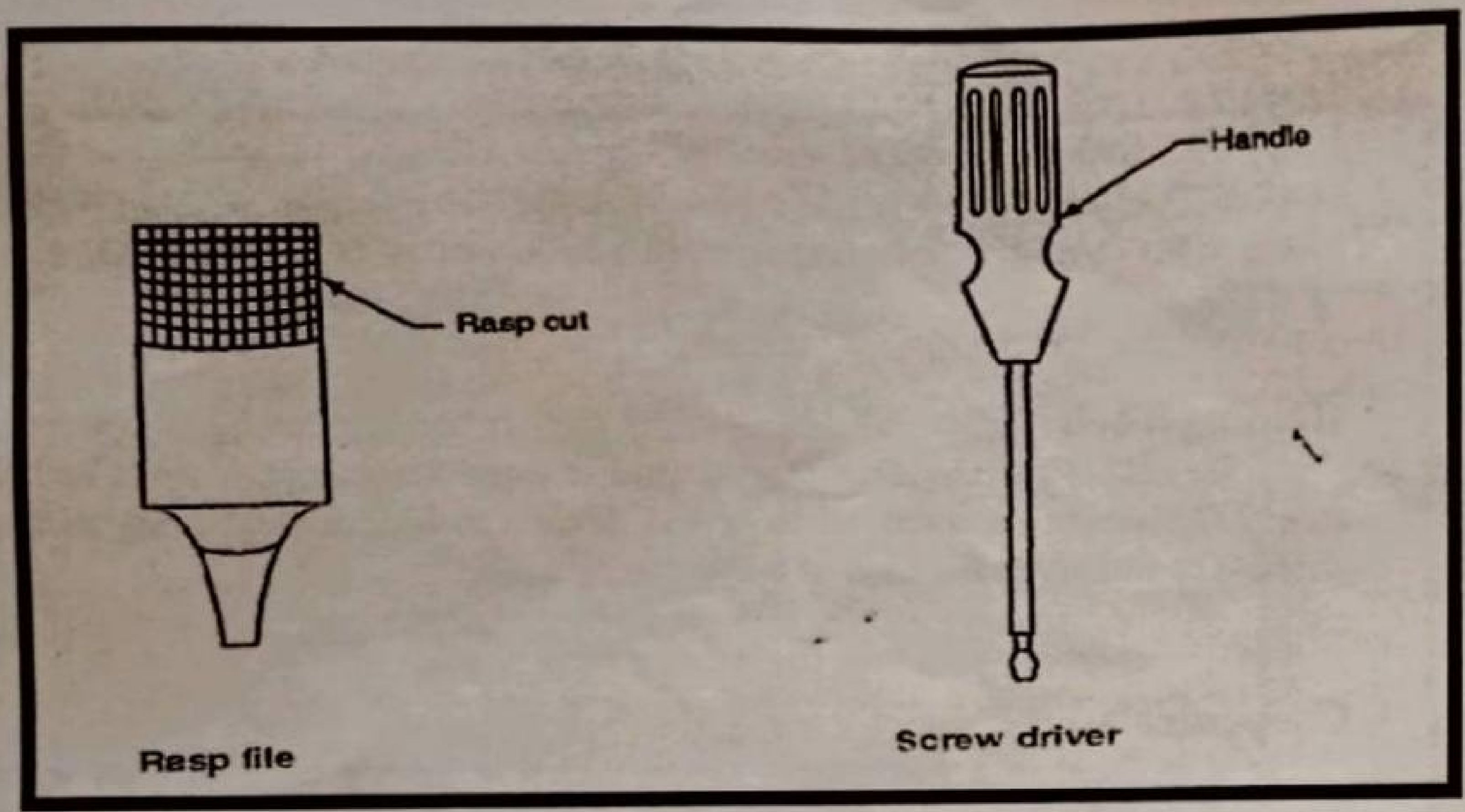
It is fixed in a carpenter's bench which is having table of hard wood usually 3 to 3.5m long, nearly 70cm wide and 70cm high. It consists of a fixed jaw and a movable jaw. The fixed jaw is rigidly fitted on the bench and the movable jaw is mounted on a screw and two sliding rods. The screw head is projected outside the movable jaw in order to carry a handle. The screw works inside a fixed half-nut that is engaged to move the movable jaw.

### Sash Cramp:

The sash cramp or bar cramp is made up of a steel bar of rectangular section, with malleable iron fittings and a steel screw. This is used for holding wide work such as frames or tops.

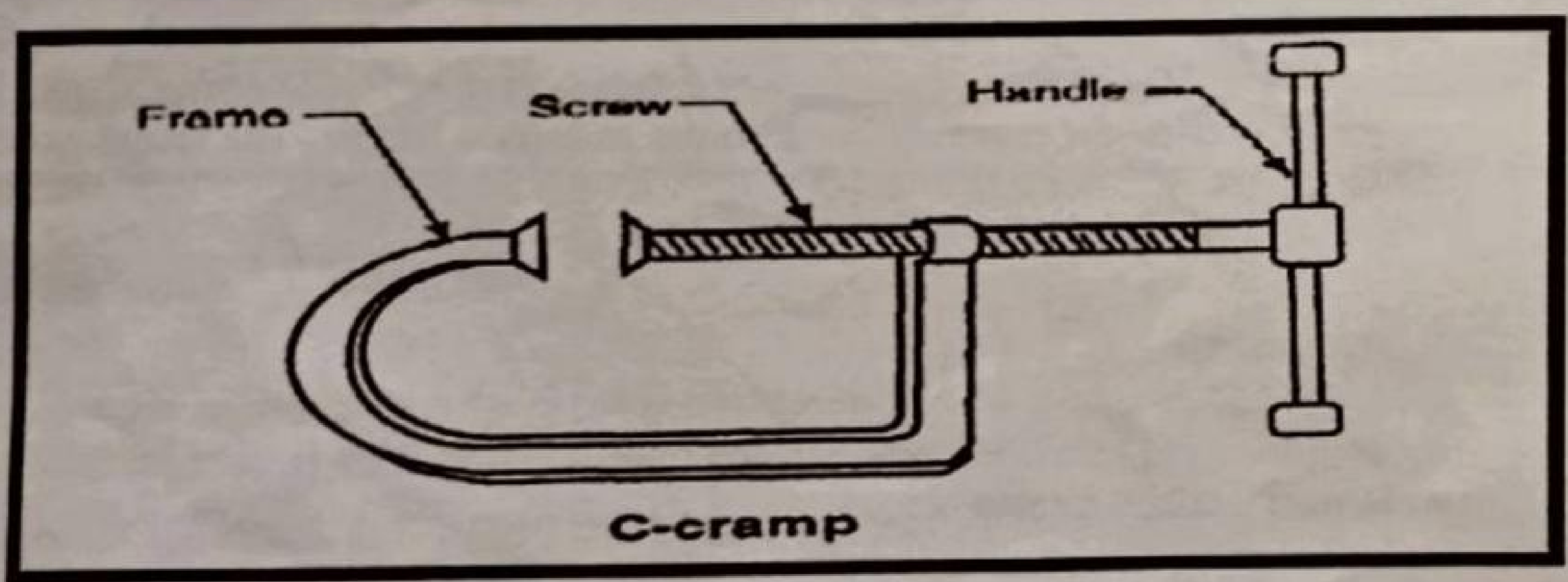
### Miscellaneous Tools:

Rasp and File: These are used for cleaning up some interior curbed surfaces where other tools can not be used.

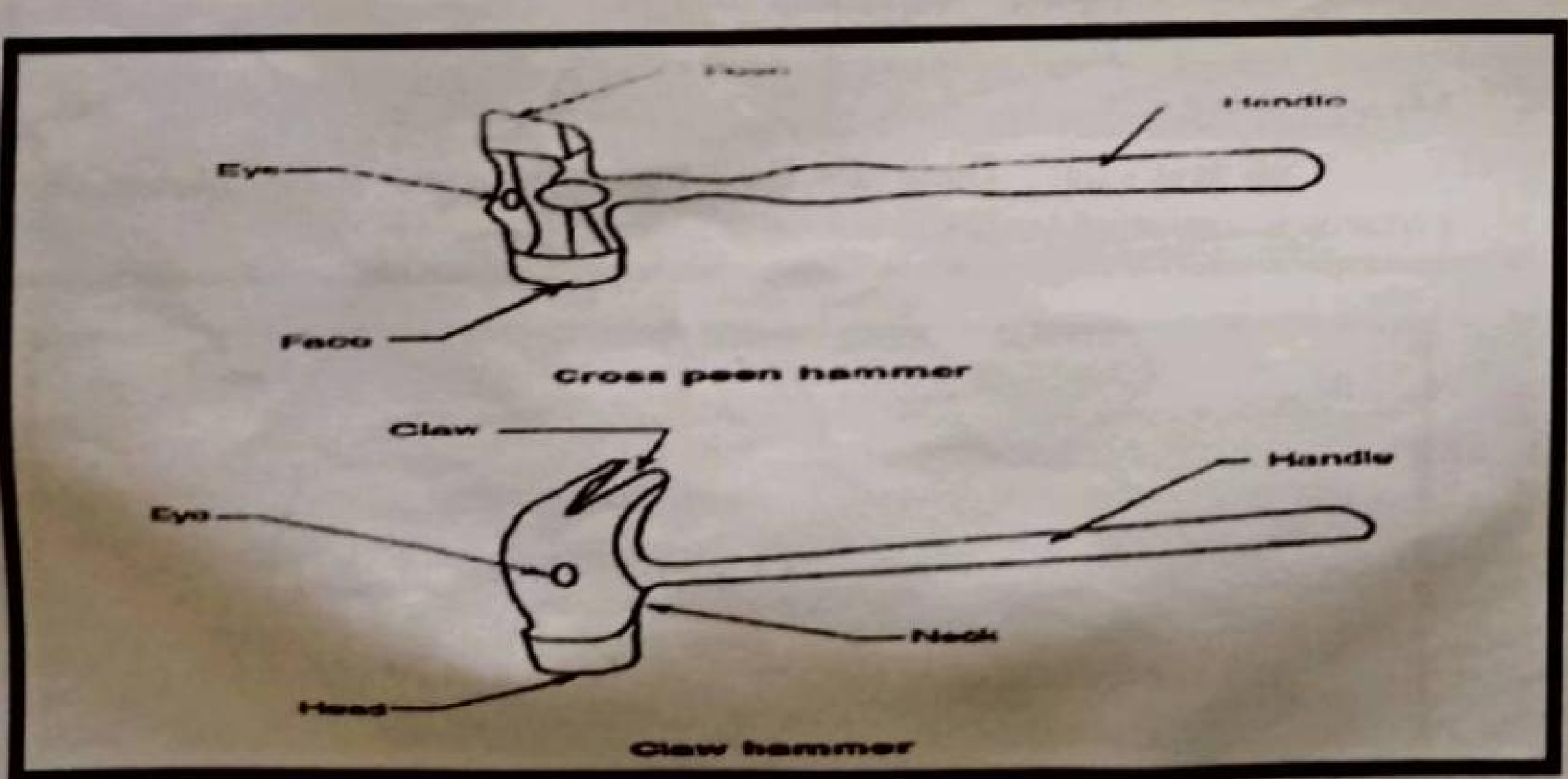



Rasp file

Screw driver




C-cramp




Cross peen hammer

Claw hammer



**Claw Hammer:**

This serves both the purpose of a hammer and a pair of pincers. The claw is used for pulling out nails. These hammers are classified according to their weight. The weight of its head ranges from 250 to 675 gms.

**G-cramp:**

The G-cramp is used for smaller work. It consists of a malleable iron frame that can be swiveled and a steel screw to which is fitted a thumbscrew.

**Oilstone:**

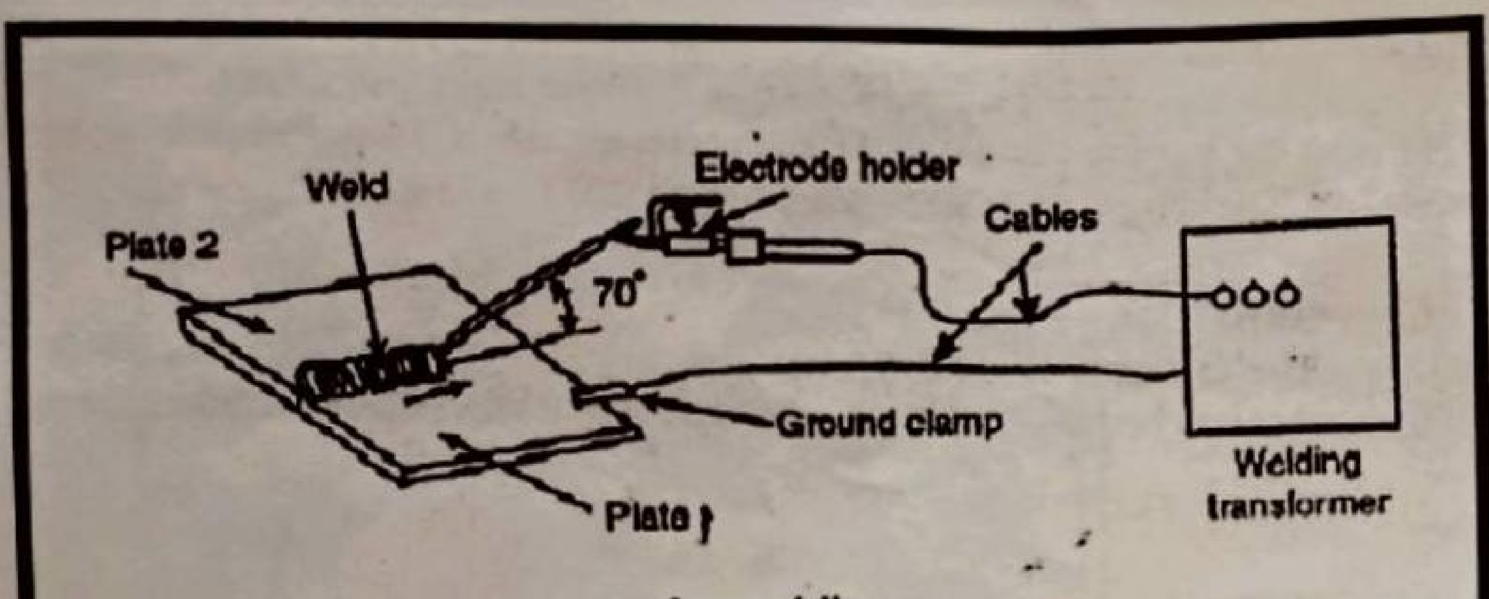
An oilstone is an essential part of a carpenter's kit of tools. Oilstones may be natural or artificial. The best-known varieties of stones are carborundum, Washita, Turkey. These may be obtained in various grades known as coarse, medium or fine.

**Pincer:**

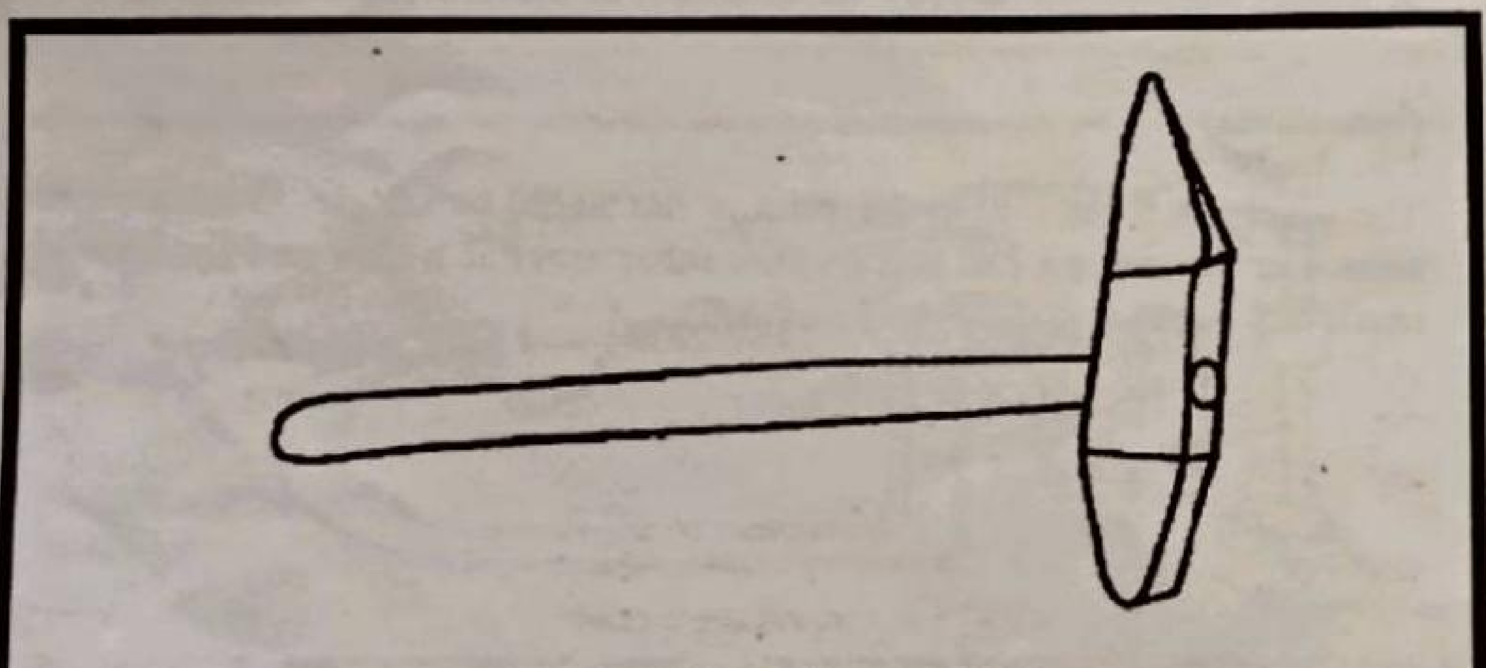
The pincer is mainly used for pulling out nails, tacks, etc. It consists of two arms – one arm has a ball end and the other arm has a claw end for levering out small tacks.

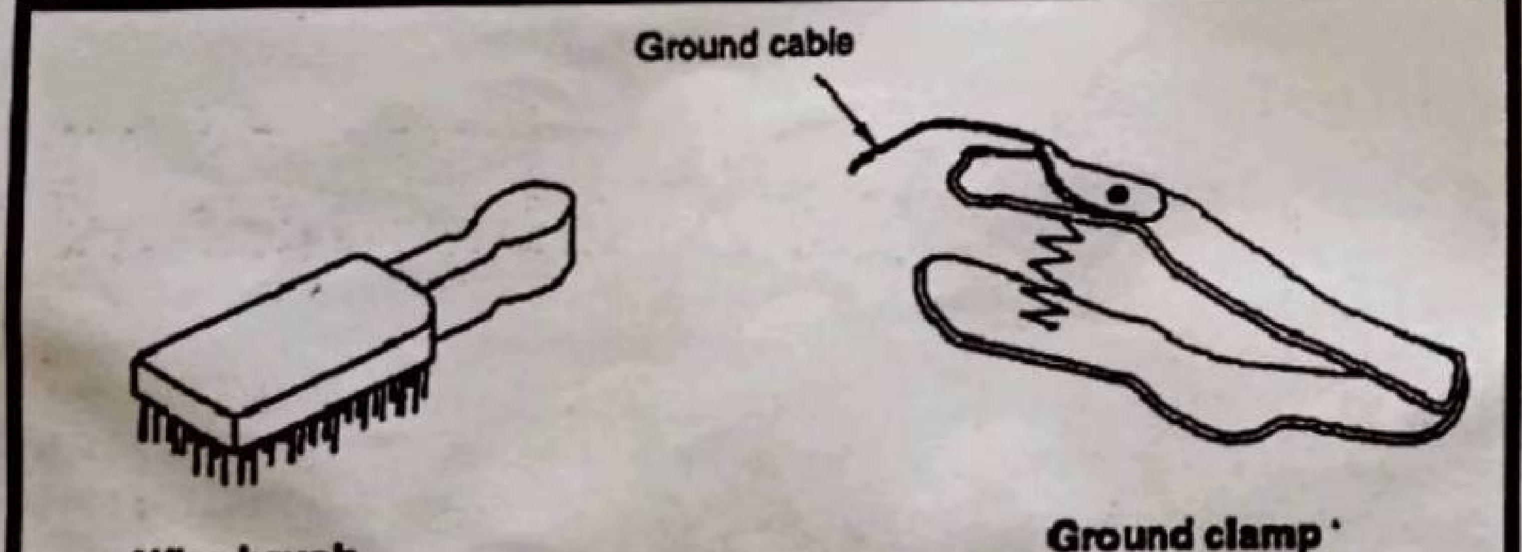
**Result :**

Thus the tools used in Carpentry has been studied.



Arc welding


Chipping hammer



Wire brush

Ground clamp



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# STUDY OF TOOLS IN WELDING

**AIM:**

To make a detail study of tools in welding.

**INTRODUCTION:**

Welding is the process of joining similar metals by application of heat with or without application of pressure and addition of filler material. The result is a continuity of homogeneous material, of the composition and characteristics of two parts which are being joined together.

Weldability has been defined as the capacity of being welded into inseparable joints having specified properties such as definite weld strength, proper structure. Weldability depends on one or more of five major factors: (1) melting point (2) Thermal conductivity (3) thermal expansion (4) surface condition and (5) change in micro-structure.

**Arc Welding:**

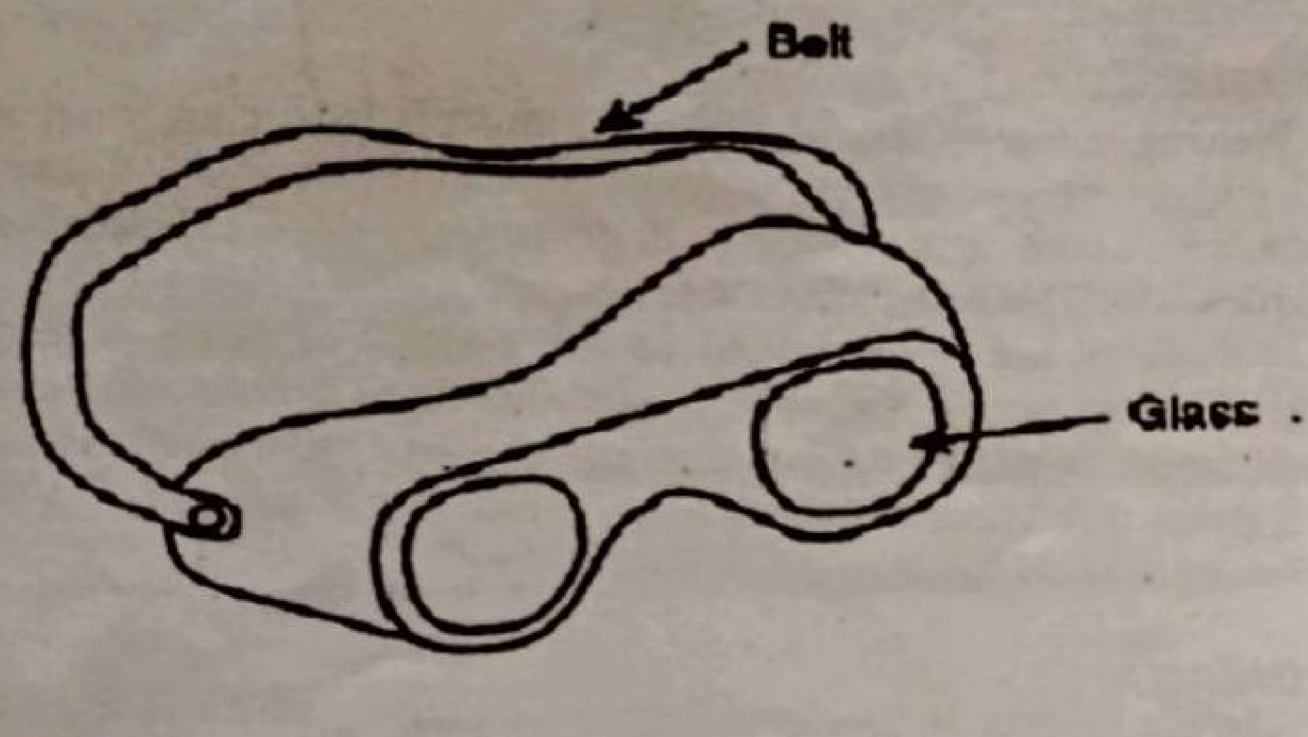
The arc column is generated between an anode, which is the positive pole of dc (direct current) power supply, and the cathode, the negative pole. When these two conductors of an electric circuit are brought together and separated for a small distance (2 to 4mm) such that the current continues to flow through a path of ionized particles (gaseous medium), called *plasma*, an electric arc is formed. This ionized gas column acts as a high-resistance conductor that enables more ions to flow from the anode to the cathode. Heat is generated as the ions strike the cathode. The temperature at the center of the arc is about 6000 to 7000° C.

The heat of the arc raises the temperature of the parent metal which is melted forming a pool of molten metal. The electrode metal is melted and transferred into the metal in the form of globules of molten metal. The deposited metal serves to fill and bond the joint or to fuse and build up the parent metal surface. The arc is extinguished by widening the arc sufficiently.

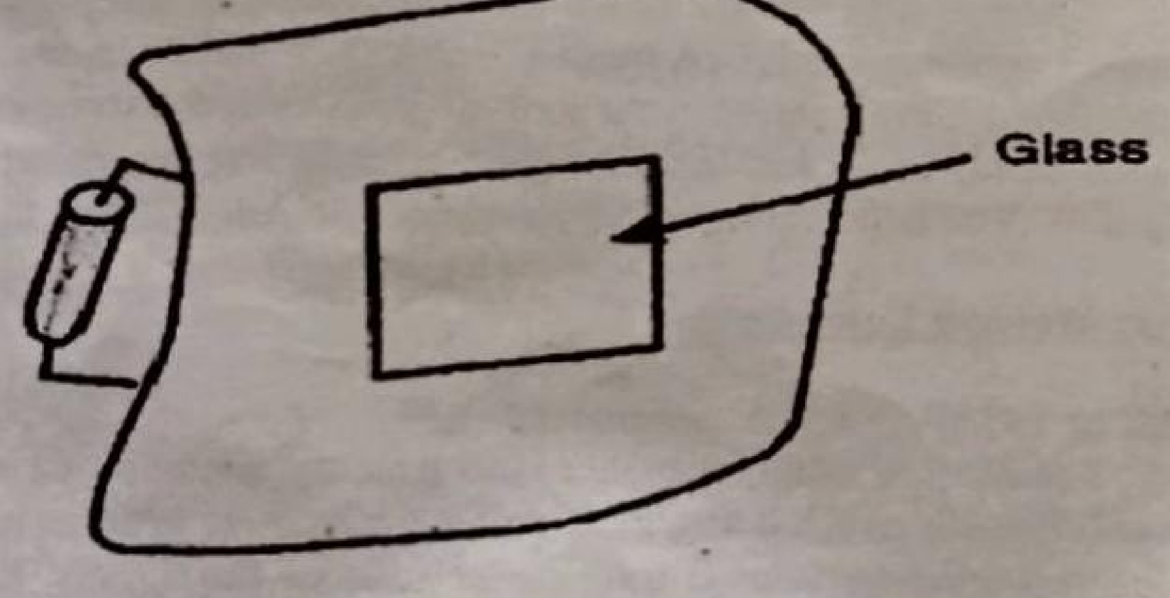
**<u>Arc Welding Equipments:</u>**

**AC or DC Welding Transformer:**

It consists of a rectangular steel tank mounted on three-tyred wheels, the front wheel swiveling and steer able by means of a draw bar. An oil-cooled, double-wound step down transformer reduces the supply mains voltage to a welding voltage of 80. All windings are totally enclosed in the steel tank. The output of the transformer can be varied by rotating a hand wheel which alters the air gap in the core of the choke resulting in step less regulation of the current between 50 and 400amps.



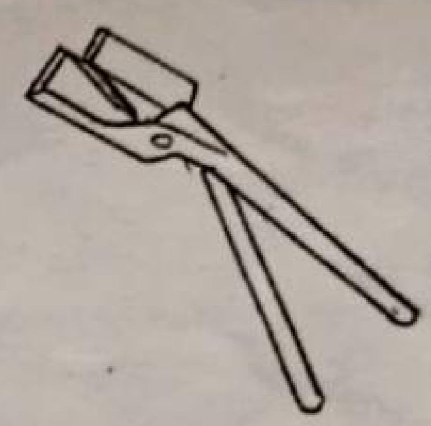

Goggle



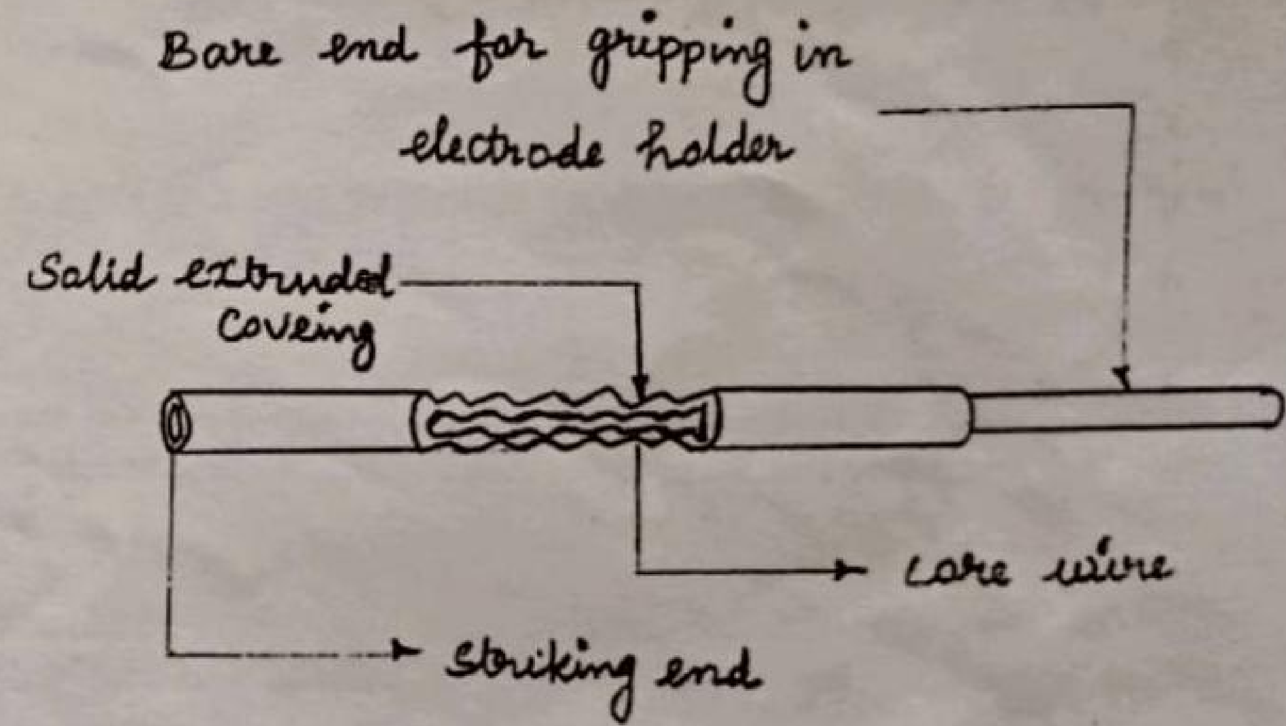

Face shield



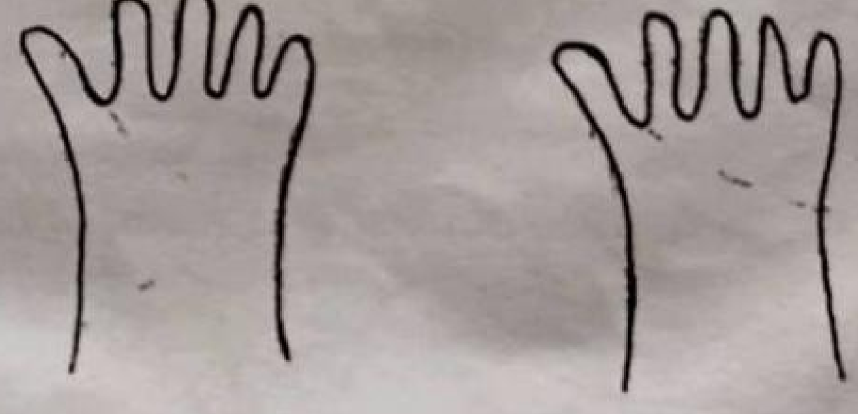




TONG




ELECTRODE



HAND GLOVES

The welding current setting can be directly read at the window on the top cover. The set can be connected to two lines of 400/440 volts, 3-phase 50 cycles supply.

**Electrodes:**

Both non-consumable and consumable electrodes are used for arc welding. *Non-consumable electrodes* may be made of carbon, graphite or tungsten which do not consume during the welding operation. Consumable electrodes may be made of various metals depending upon their purpose and the chemical composition of the metal to be welded. These consumable electrodes may be classified into *bare* and *coated.*

**Face Shield:**

A face shield is also used to protect the eyes of the welder from the light sparks produced during welding process. It is held in hand.

**Goggles:**

Goggles are used to protect eyes from slag chips during chipping process. During chipping operation, chipping hammer is used to remove the slag from the weld surface.

**Chipping Hammer:**

A chipping hammer is used to remove slag which is formed during welding.

**Ground Clamp:**

It is connected to the end of the ground cable. It is normally clamped to the welding table or the job itself to complete the electric circuit.

**Wire Brush:**

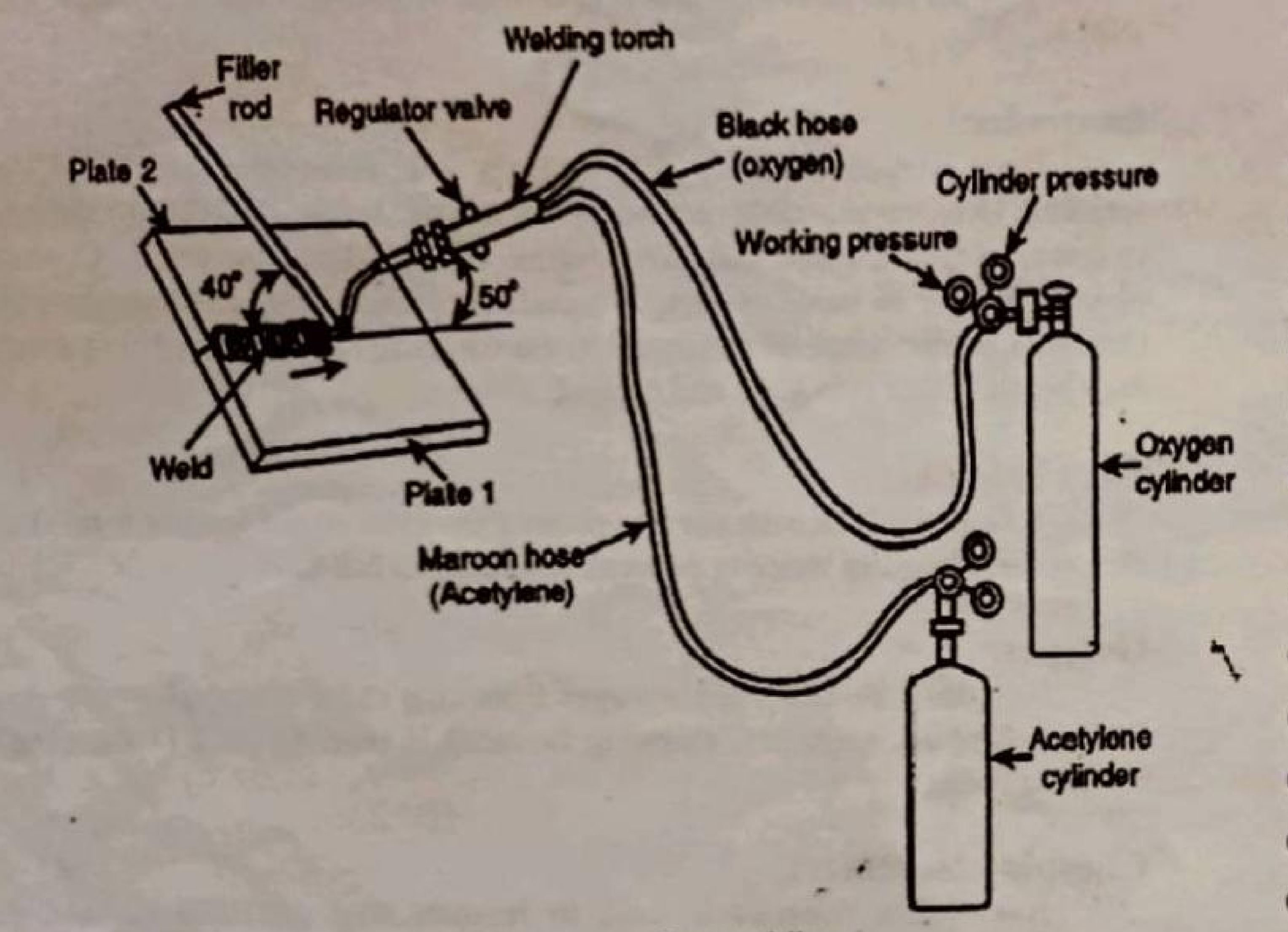
The wire brush is used to clean the welded surface so that slag is totally removed from weld surface.

**Hand gloves, Apron, Sleeves:**

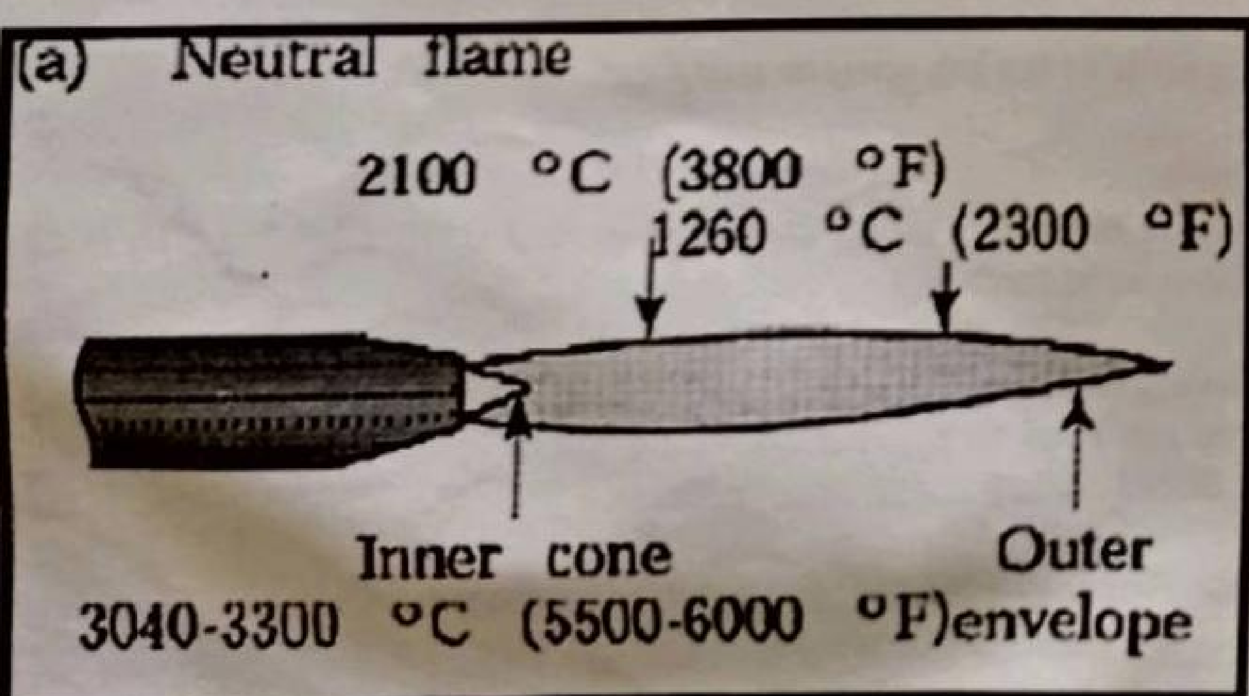
These are used to protect body and hands from light sparks which are getting produced during arc welding process.

**Electrode Holder, Cables, Cable connectors:**

These are used for connecting the electrode to the welding transformer so that the current passes through the electrode during the arc welding process.




Gas welding








# GAS WELDING

**AIM:**

To study about gas welding and equipment used in gas welding

**INTRODUCTION:**

Gas welding is also used widely used to join metal plates. The filler and parent metal plates are melted by the heat of the flame produced using oxygen and fuel gas. The most commonly gas welding is Oxy-acetylene gas welding.

**Oxy-acetylene Welding:**

**Welding Equipments :**

Acetylene supply in cylinders (painted maroon ), oxygen supply in cylinders (painted black), acetylene pressure regulators, oxygen pressure regulators, blow pipe with a range of nozzles, red colored hose for acetylene, blue colored hose for oxygen.

**Acetylene – C2H2 :**

Acetylene is a fuel gas. It is a combustible gas with high calorific value and a high flame propagation velocity. This gas is widely used in gas welding and in gas cutting. When lime stone and coal are heated together at 1900° to 2300°C in electric furnace, they fuse and form calcium carbide. Acetylene is obtained by the chemical reaction of calcium carbide and water. When calcium carbide is allowed the react with water , the carbon of the calcium carbide combines with the hydrogen of water and forms acetylene.

CaCl + H2o → C2H2 + Ca OH

Acetylene Cylinders: C2H2 cylinders are filled with a porous mass. This porous material can dissolve considerable amount of C2H2. A completely filled standard acetylene cylinder at a temperature of 15° C is under a pressure of 18bar and contains 13liters of acetone. The high pressure of the gas in acetylene cylinder is reduced to the required pressure by the pressure reducing valve.

**Oxygen :**

Oxygen is necessary to burn the fuel. Air contains 21% O2. O2 is supplied in cylinders in liquid form. Oxygen cylinders have 40liters capacity. Oxygen is compressed to a pressure of 150 bars.



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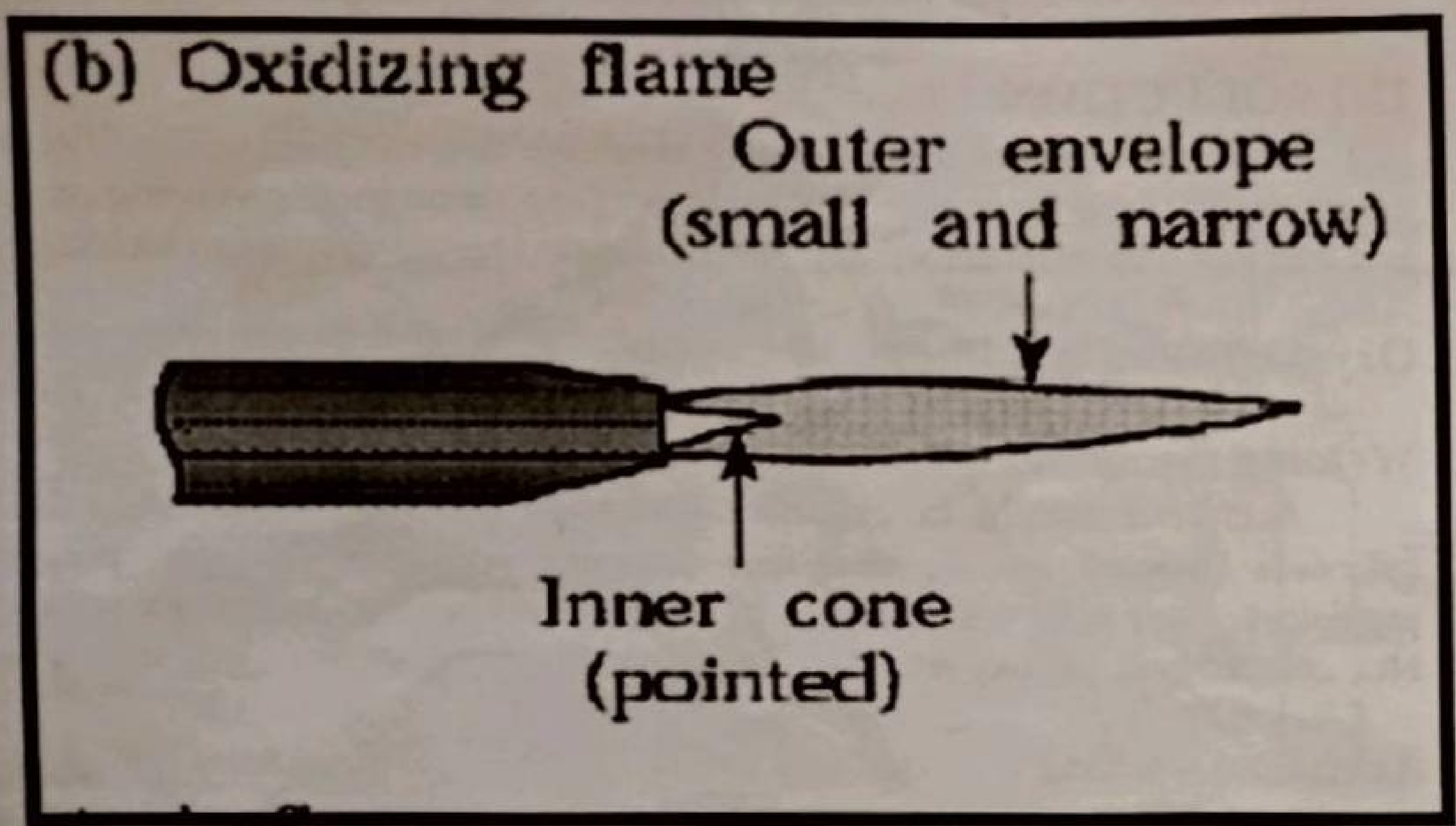
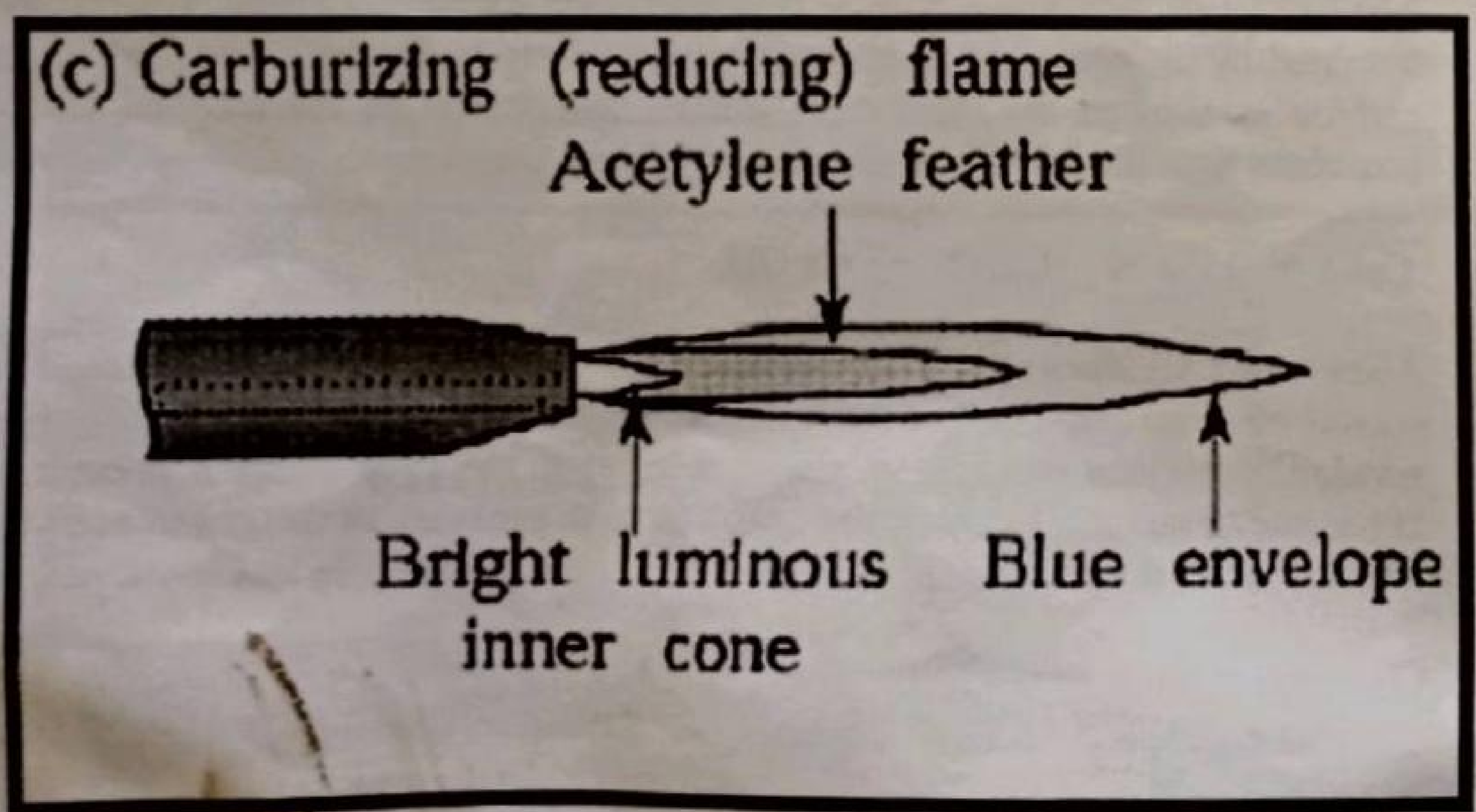







Temperature ranges for different Oxy – Fuel gas combinations:

| | | |
|---|---|---|
| Oxy – Acetylene | : | 3100° - 3200° C |
| Oxy - Butane | : | 2900° C |
| Oxy - Propane | : | 2600° C |
| Oxy – Hydrogen | : | 2300° - 2400° C |
| Oxy - Coal gas | : | 2000° - 2200° C |
| Air - Acetylene | : | 1800° - 2100° |

**Types of Gas Flames :**

Depending on the ratio of oxygen and acetylene three types of flames are obtained.

**Neutral Flame :**

It has equal quantity of oxygen and acetylene and is used for welding steel, stainless steel, cast iron, copper, etc. This flame is also used for cutting metal plates.

**Oxidizing Flame:**

It has more quantity of Oxygen. The acetylene to oxygen ratio is 1: 1.2 to 1.5. This flame is used for welding copper, brass, and bronze.

**Carburizing Flame:**

It has more quantity of acetylene. The acetylene to oxygen ratio is 1 : 0.9. This flame is used for welding ally steel and aluminium.

**Gas Welding Procedure:**

1. The surface to be welded is cleaned.
2. Open the acetylene and oxygen cylinder valve slowly, then open the acetylene valve in the torch. Keep the tip of the torch away from body and light it using lighter. Open oxygen and acetylene valves in torch slowly to get the required flame for welding.
3. Maintain a gap of 3mm between the plate to be welded and the inner cone of the flame. The torch and filler rod are moved backwards along the line to be welded.

**Advantages of Gas Welding** :

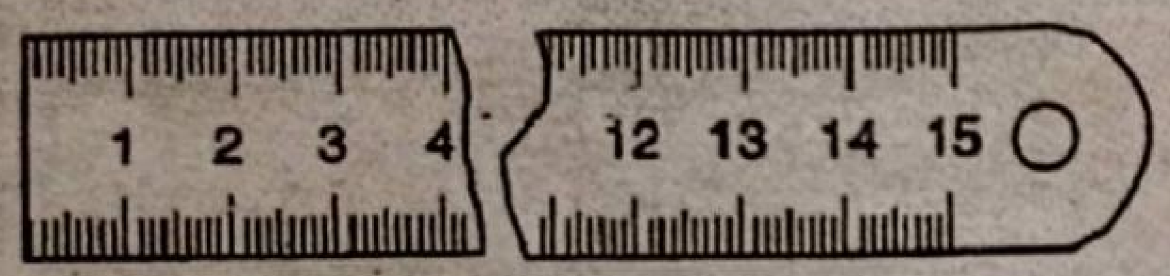
1. The equipment is inexpensive, uncomplicated.
2. It is portable.
3. Useful for welding light metals and for repair jobs.
4. gas welding can be used with all common metals.
5. A large variety of materials can be welded.
6. Welds can be produced at reasonable cost.

**Dis-advantages of Gas Welding** :

1. Acetylene is explosive.
2. Gas welding is slower than electric arc welding
3. Heated area is larger and causes more distortion
4. An excessive amount of expansion and contraction
5. Highly skilled operators are required to produce a good weld.
6. If electric arc welding is available, gas welding is seldom used for welding above 3.2mm thick
7. The gas welding is not satisfactory for heavy sections.

**Result**:

Thus Gas and Arc welding and different types of welding have been studied.

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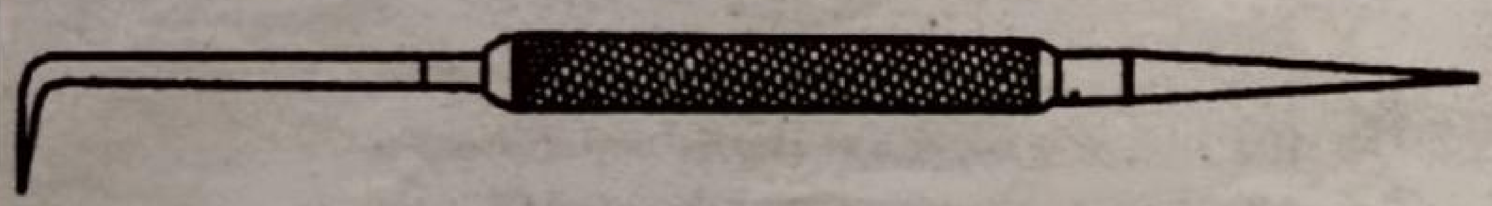
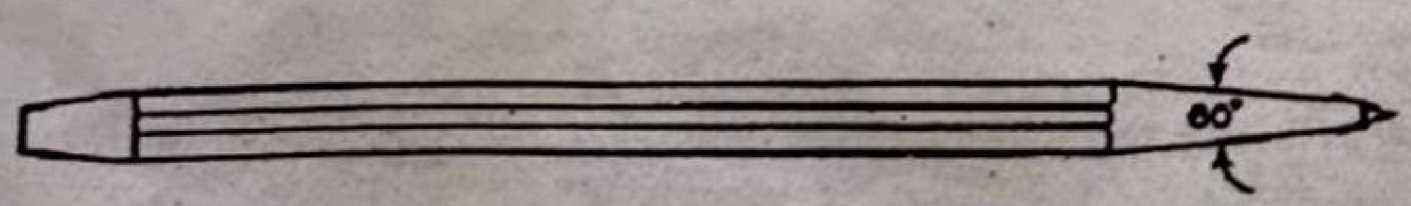
Steel rule



Scriber

Divider




Centre punch



2b

## STUDY OF TOOLS IN SHEET METAL WORK

### AIM:

To study the various tools used in the sheet metal work.

### MARKING AND MEASURING TOOLS:

Marking and measuring tools are used to produce objects to an exact shape and size. The commonly used tools are given below:

### STEEL RULE:

It is used to measure and mark dimensions. It is graduated on both sides in millimeters and centimeters or inches. Its length varies fom15 cm (6 inches) to 30 cm (12 inches)

### SCRIBER:

It is used to mark or scribe lines on the metal sheet. Its length varies from 150mm to 300mm and has diameter which varies from 3 to 5mm.

### DIVIDER:

It is used to mark the dimensions from scale to the work piece. It is also used to scribe arcs and circles on the sheets. It is similar to the compass and is used for more accurate marking.

### CENTRE PUNCH:

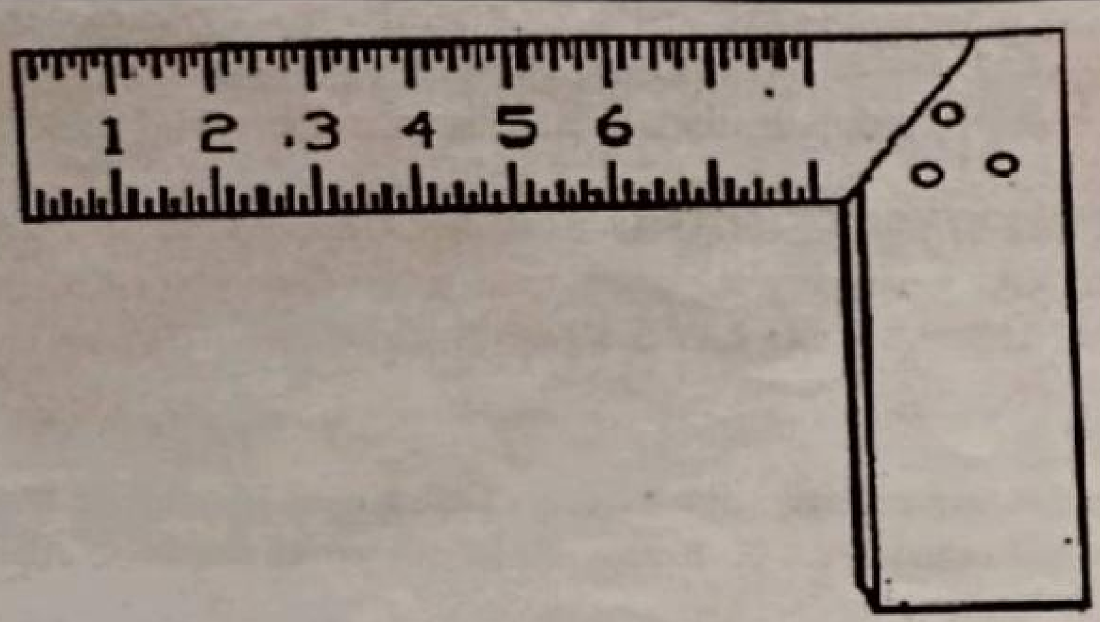
It is used to mark a centre for a hole to be drilled. It can also be used for marking points along a line for cutting. Its tapered point end has an angle of 60 degrees.

### TRY SQUARE:

It is used for marking and checking right angles. It consists of a steel blade riveted at right angles. It is used to mark square corners.

### TRAMMEL POINTS:

It is used to draw large circle and arcs. The trammel has too removable pointed legs and mounted on separate holders .the holders slide on a beam which can easily adjusted for the required length

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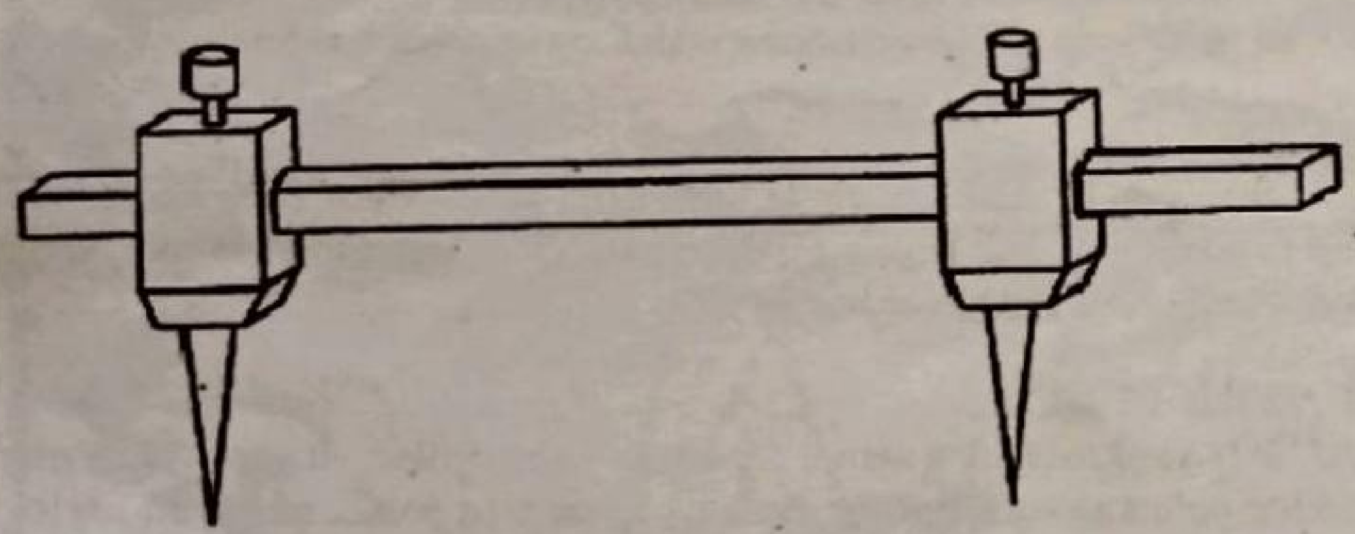
Try-square



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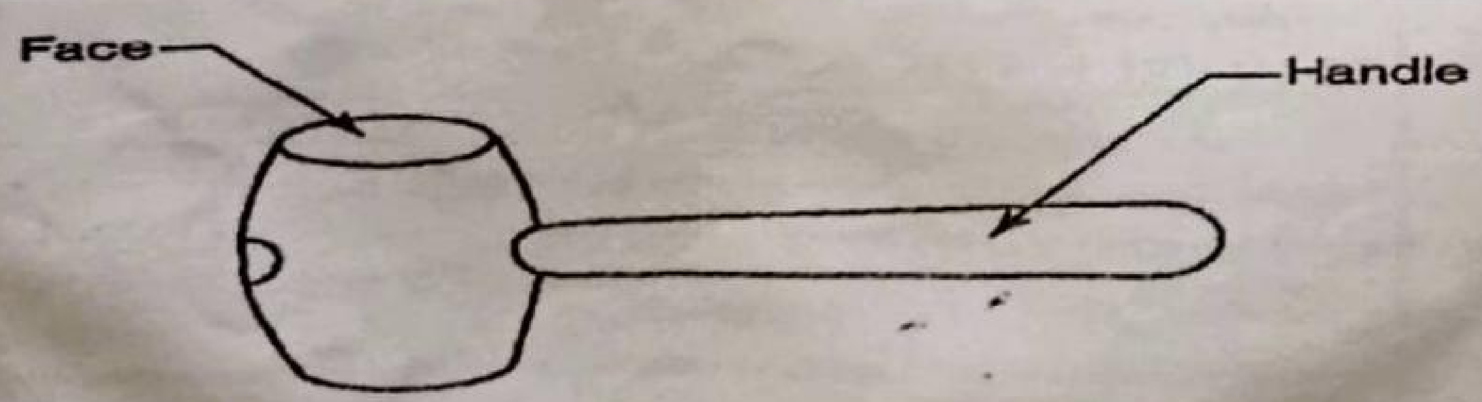
Trammel points




Mallet





### CUTTING TOOLS:

Snips are used to share or cut the metal sheets to the required size and shape. Snips are used to cut thin sheet metal .the following snips are used in sheet metal work.

### STRAIGHT SNIP:

It is used to cut or trim along a straight line. The blades in the snip are straight.

### BENT SNIP:

It is used to trim or cut along inside curves. The blades in this snip are curved back from the cutting edge, which permits the sheet to slide over top blades while cutting.

### SHEARING MACHINE:

The simple shearing machine is used to shear the sheet metal using compound lever. The shearing machine has a fixed bottom blade and a top shearing blade which is operated by a lever. It is used to shear metal plates which can't be cut or trim by snips.

### STRIKING TOOLS

### HAMMERS:

The hammer is a striking tool. It has a handle and a head. Hammers classified into four types.

(a) Ball peen hammer

(b) Cross peen hammer

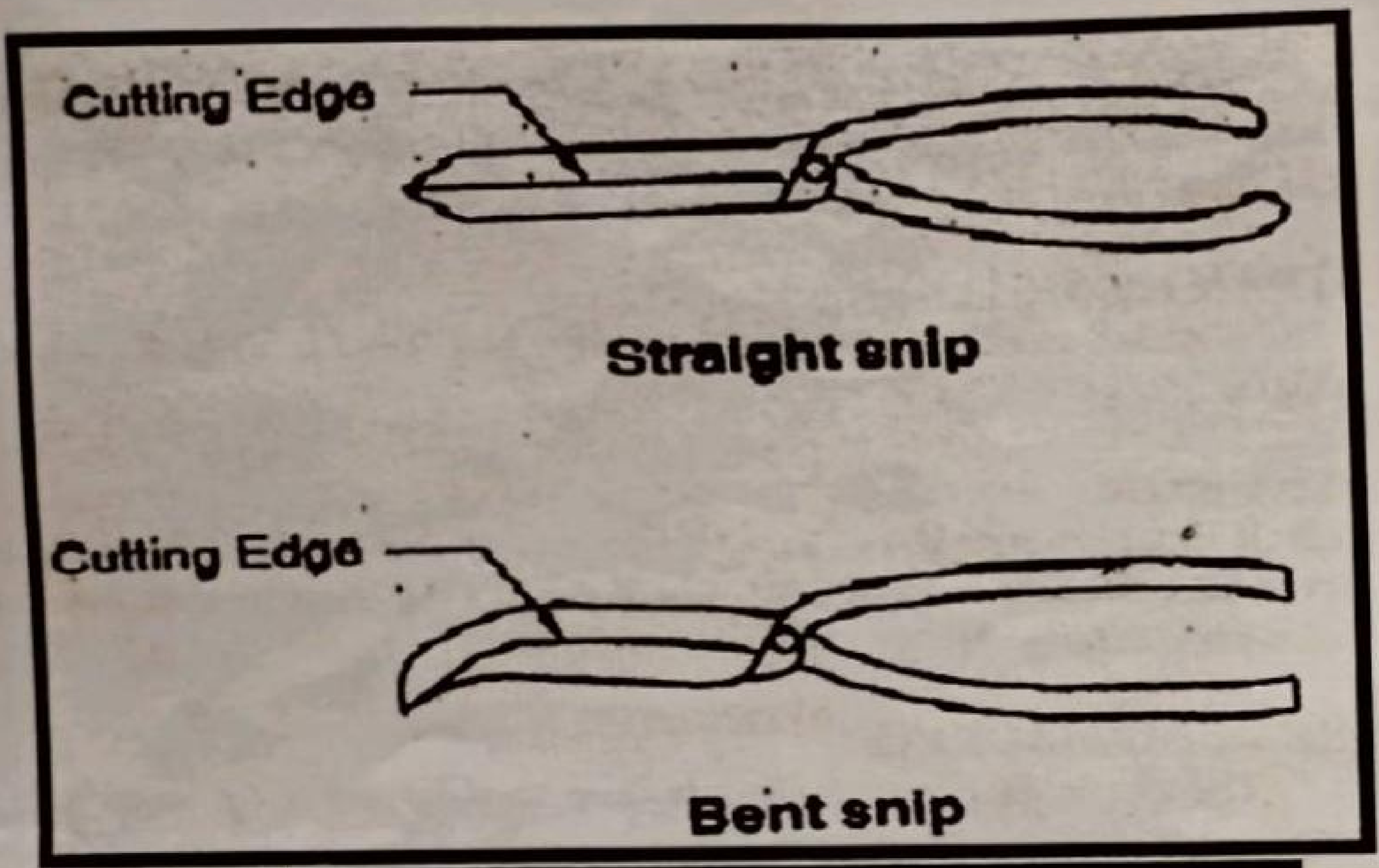
(c) Straight peen hammer

(d) Riveting hammer

The ball peen hammer is a general purpose hammer which has a slightly curved face and a round head. The cross peen and straight peen hammers are used for folding the sheet and to work in the corners of the object.

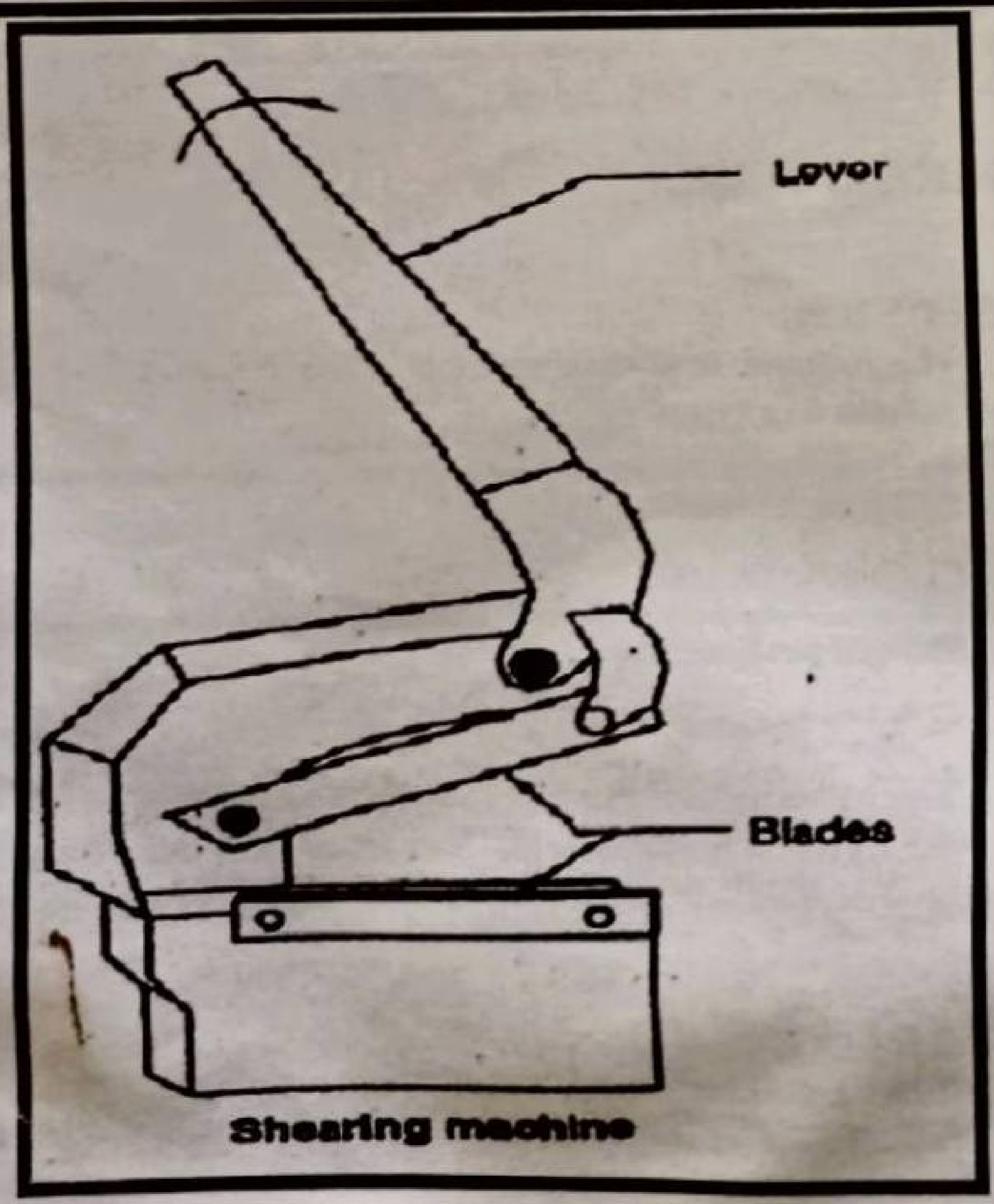
### MALLET:

It has a striking face which is used to give light blows to sheets. It is used to smoothen/flatten the surface of the sheet without spoiling it. Mallet is made of hard wood and is round or rectangular in cross-section.




Straight snip

Bent snip




Shearing machine





## SUPPORTING TOOLS

### STAKES

The stakes are the supporting tools in sheet metal work to form of shapes. It supports the sheet while bending, riveting, punching, etc. There are many types of stakes to carryout different operations. Following stakes are commonly used in sheet metal work.

### HAND STAKE:

It is used for pressing the inner sides of straight joint in the sheet. It has a flat surface with two straight edges, a concave edge and a convex edge.

### HALF MOON OR HALF ROUND STAKE :

It is used to form a round seam Joint on the inner side of the job.

### HORSE STAKE:

It has two square holes for holding one or two stakes to carry out different operations on the job.

### TAPER STAKE:

It is used to form a conical or tapering job.

## SHEET METAL JOINTS

Metal sheets are joined many ways. Some of them are given below, the joint type is selected depends on the requirement.

### LAP JOINT:

In this joint metal sheets are placed over lapping. It s commonly used to join metal sheets by riveting or soldering.

### SEAM JOINT:

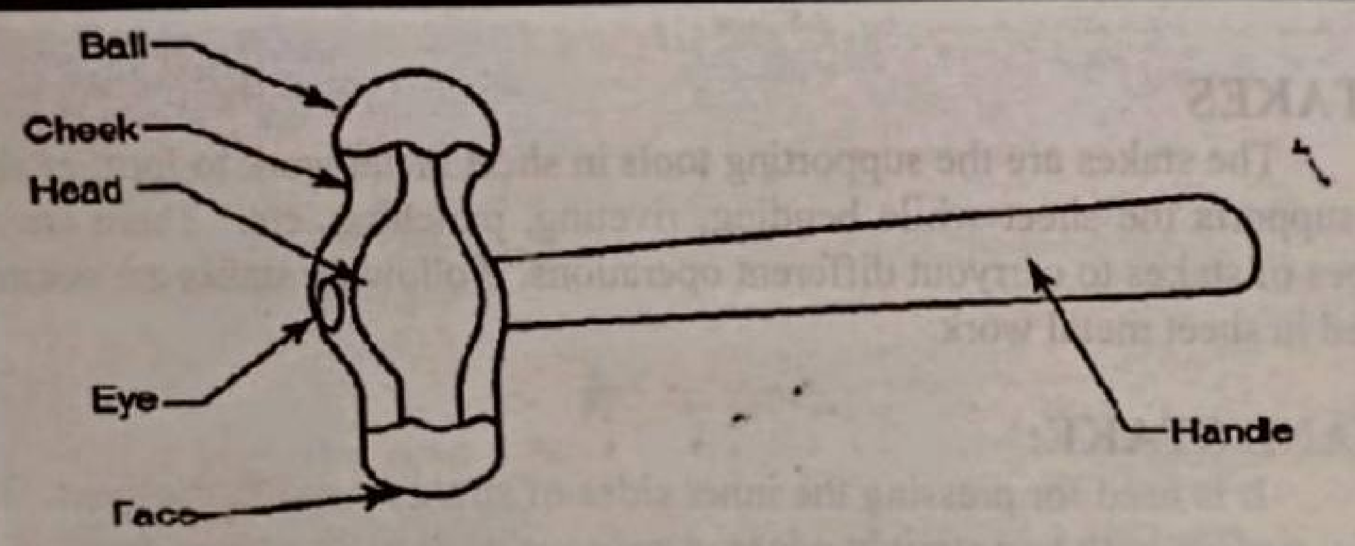
In this joint metal sheet edges are folded to lock themselves. It is also commonly used to join metal sheets.

### LOCKED SEAM JOINT:

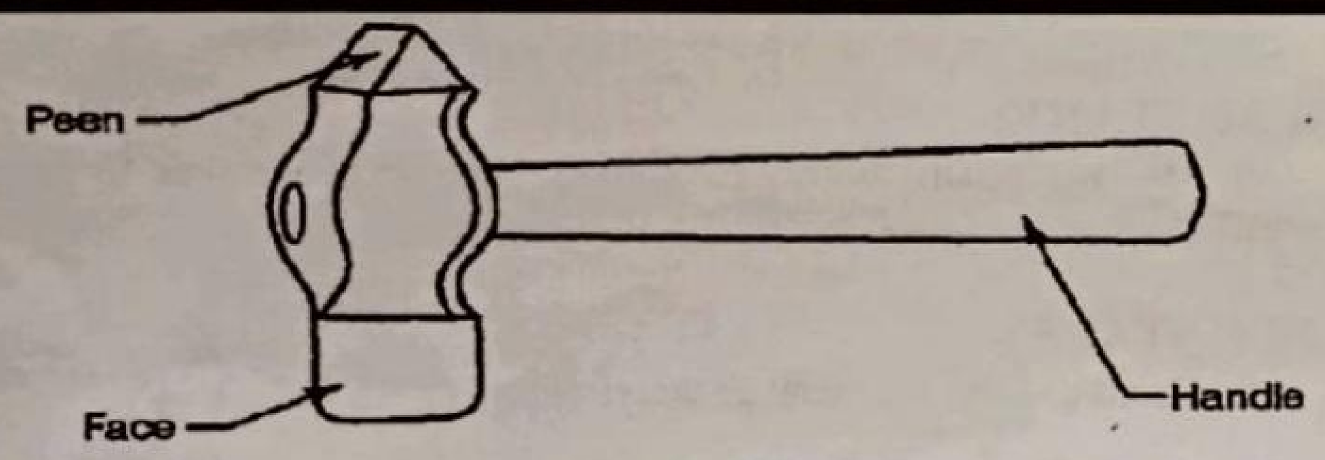
It is similar to the seam joint to get the locked joint.

### WIRED EDGE:

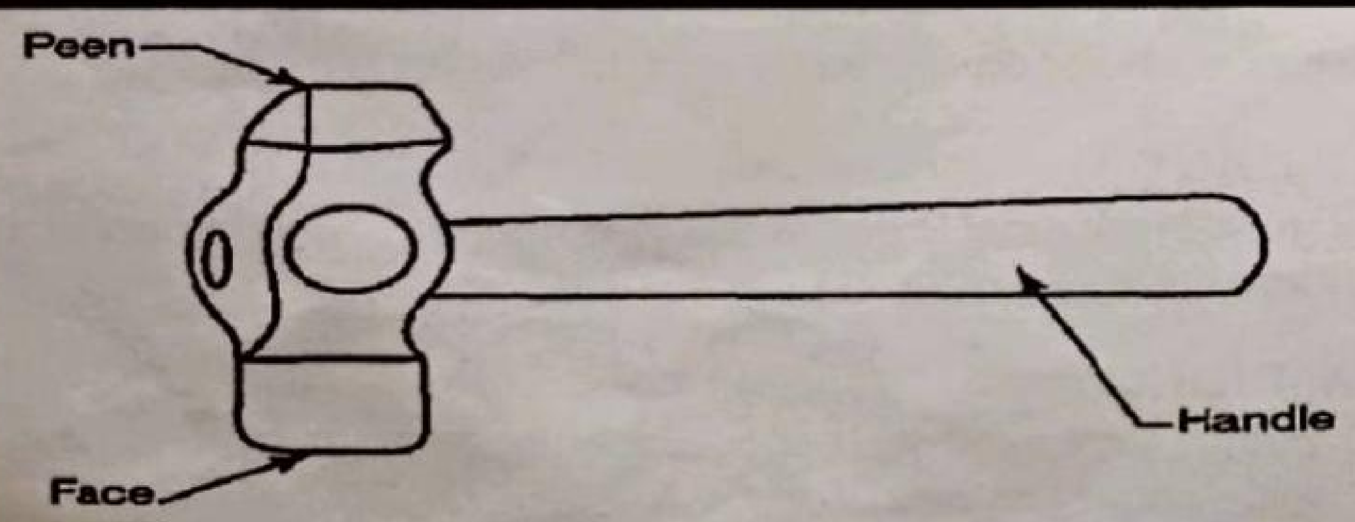
In this a wire is used to join the sheets along them and to add strength to the edges of the sheets



Ball peen hammer



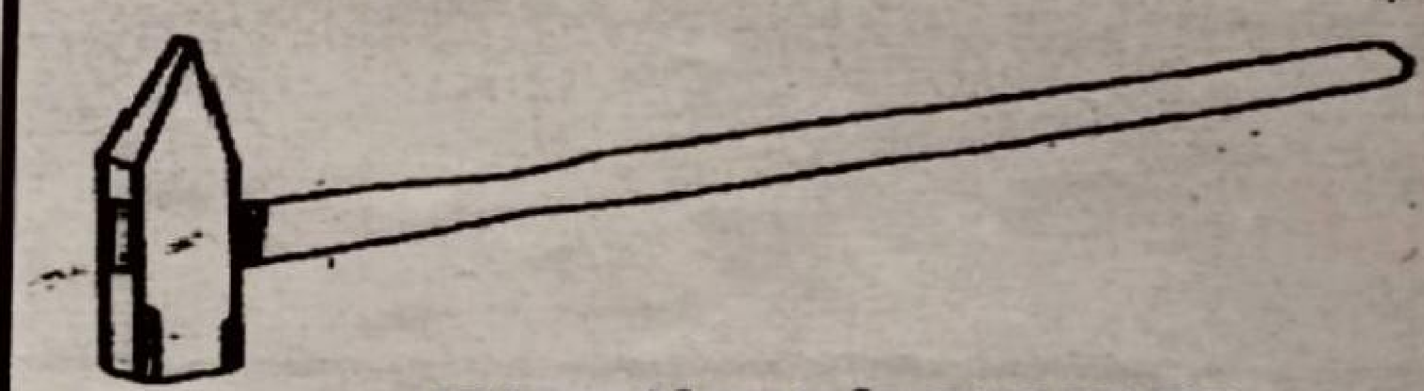
Cross peen hammer

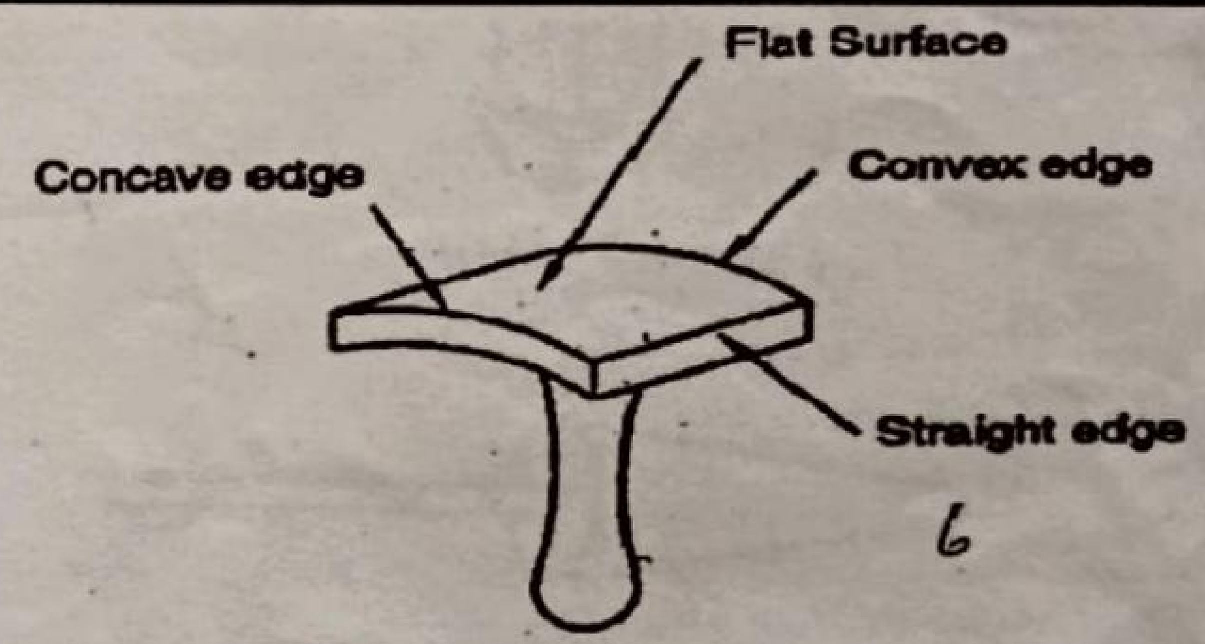


Straight peen hammer



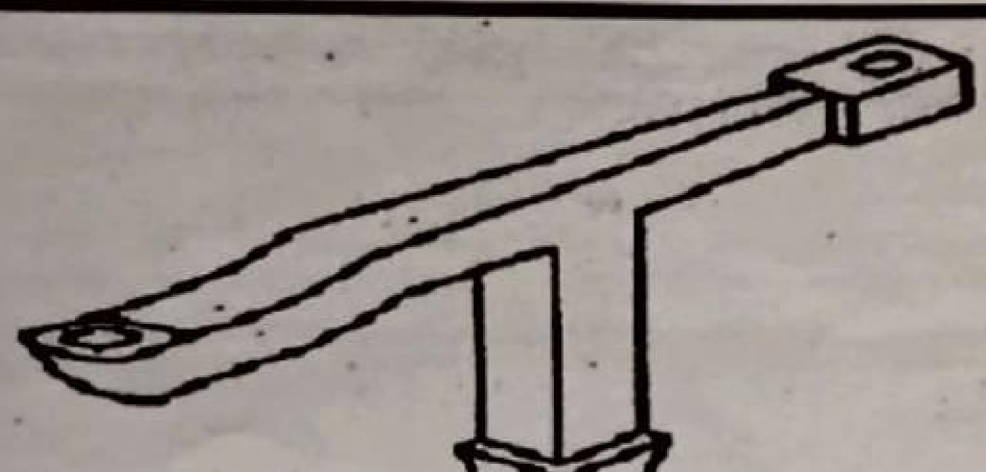



Riveting hammer



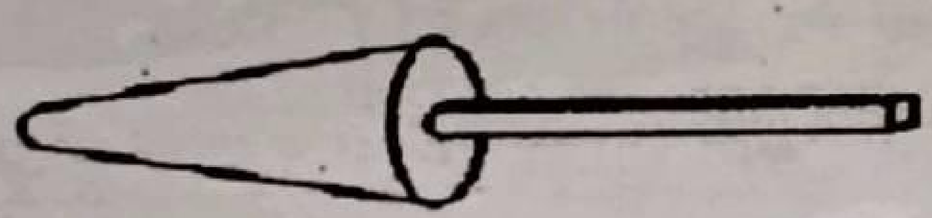
Hand stake

Half moon stake



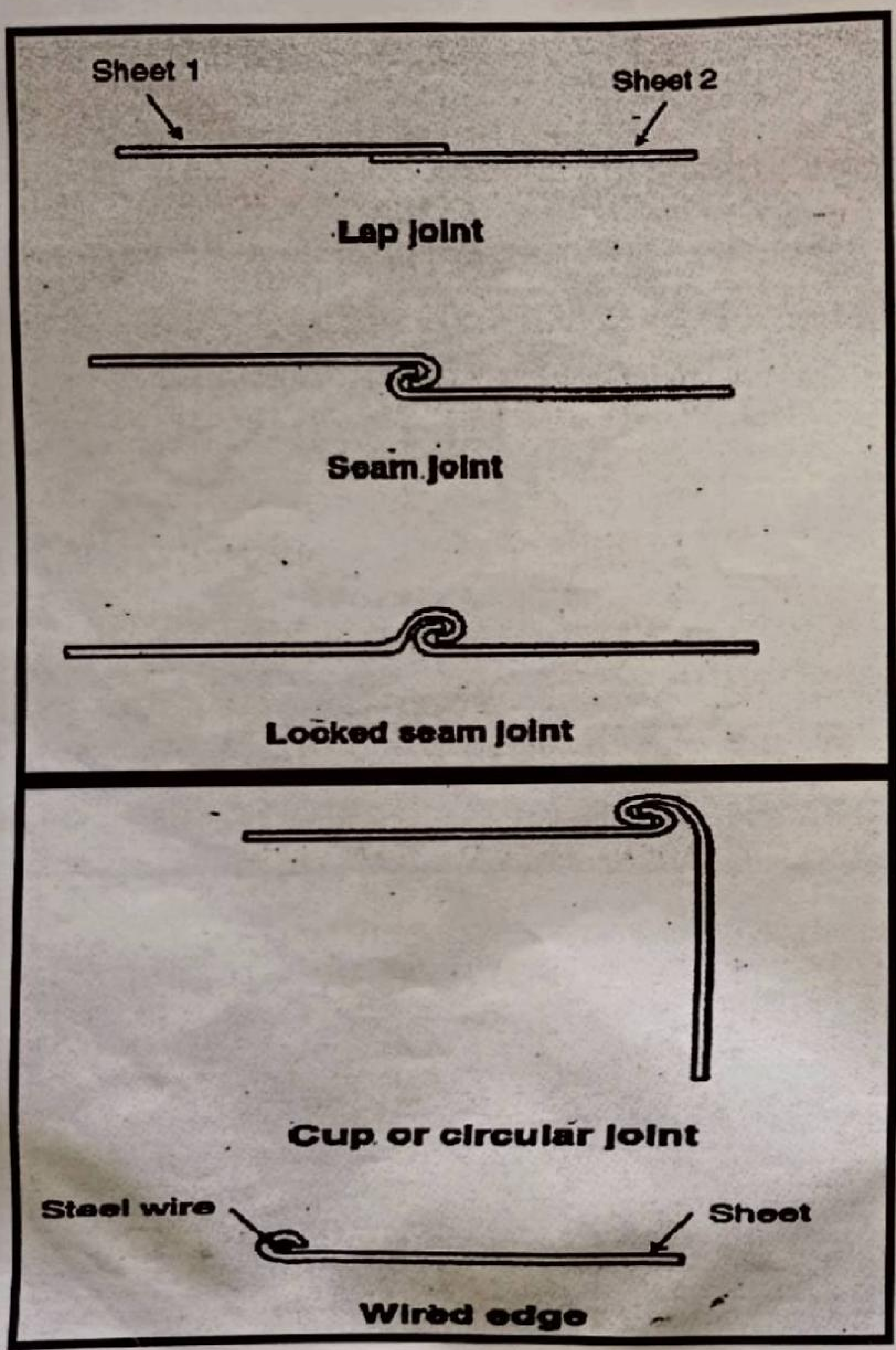
6

**Horse stake**



6

**Taper stake**








### CUP OR CIRCULAR JOINT:

It is also used to join the metal sheets along them.

### SHEET METAL OPERATIONS

Various operations like measuring, marking, laying out, cutting, shearing, edge forming, bending, wiring, joint making, piercing, blanking, nibbling etc. are carried out in a job of metal sheet. Nibbling is the process of continuous cutting along the edges of straight or curved line. Piercing is the process of forming a hole. Blanking is the process of cutting a blank to proceed for any further sheet metal operation.

Laying out is an important operation of scribing the developed shape of the object going to be made. Provide allowances for joining process like overlapping, bending, etc. While scribing the development of the object on the metal sheet.

### RESULT:-

Thus sheet metal process and different types of sheet metal joints have been studied.

# LIST OF EXERCISES:-

**FITTING:-**

1. FILING PRACTICE.
2. VEE-FITTING.
3. ACUTE ANGLE FITTUNG.

**CARPENTARY:-**

1. PLANING.
2. HALF LAP JOINT.
3. CORNER MORTSE JOINT.

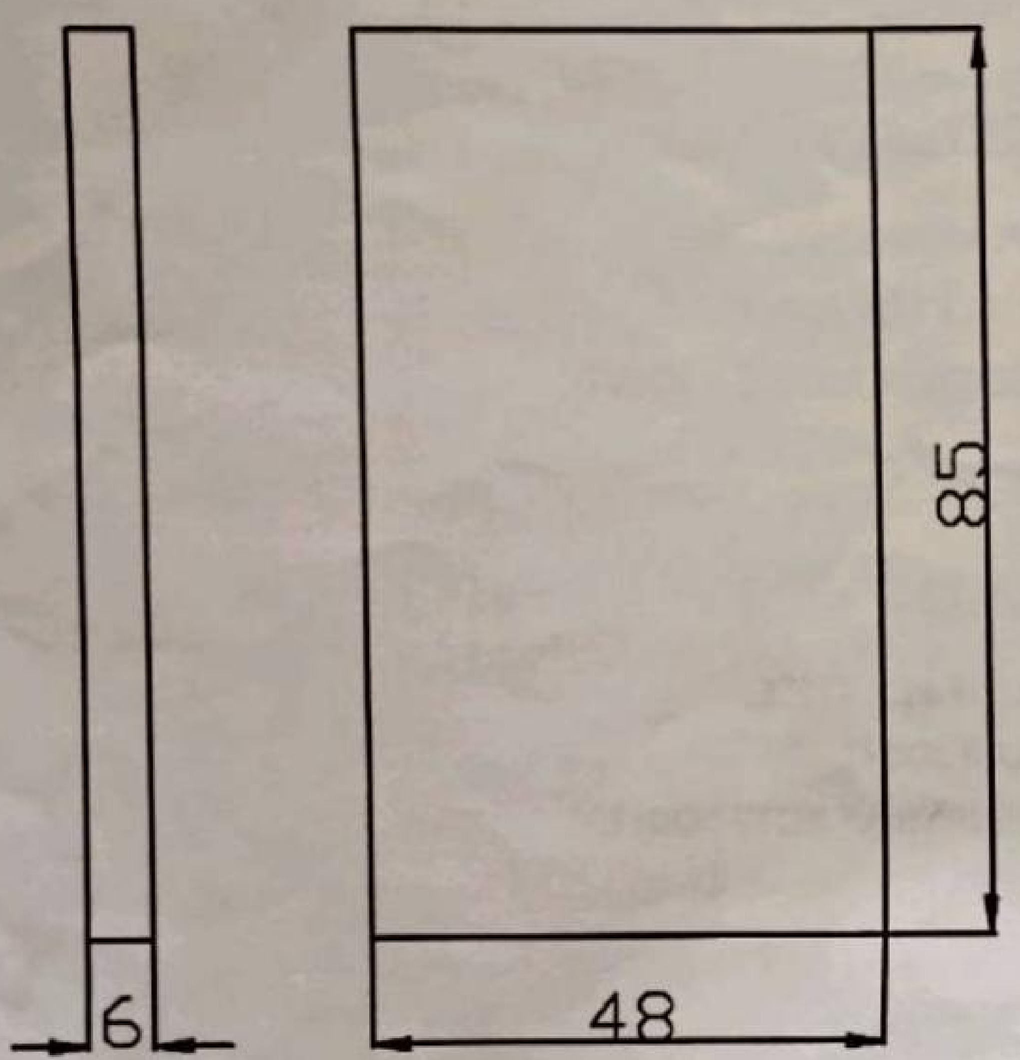
**WELDING:-**

1. LINE PRACTICE.
2. LAP JOINT.
3. SINGLE 'V' BUTT JOINT.

**SHEET METAL:-**

1. RECTANGULAR TRAY.
2. DUST PAN.

**FILING PRACTICE.**




**ALL DIMENSIONS ARE IN MM.**





# FILING PRACTICE

**AIM:**

To file the given work piece according to the given dimensions.

**MATERIALS REQURIED:**

1. Mild Steel. (MS) 85mm*48mm

**TOOLS REQUIRED:**

1. 8'' bench vice
2. Try square
3. Steel rule
4. Dot punch
5. Hacksaw frame with Blade
6. 12''Flat rough file
7. 12''Flat smooth file
8. Surface plate
9. Vernier Height Guage
10. Angle plate
11. 200g Ball Peen Hammer

**PROCEDURE:**

1. The original dimensions of the work piece is checked using try square.
2. The work piece is clamped properly in the bench vice and using a flat file ,file any two sides of the work piece. Check whether the two sides are at right angle using try square.
3. The given dimensions are marked using the surface plate and the vernier height gauge by referring sides as bases.
4. Using Prick punch make impressions on the marked lines.
5. Now the remaining two faces are filed.
6. Filing is continued until the remaining dimensions and smooth surface of the work piece is obtained.

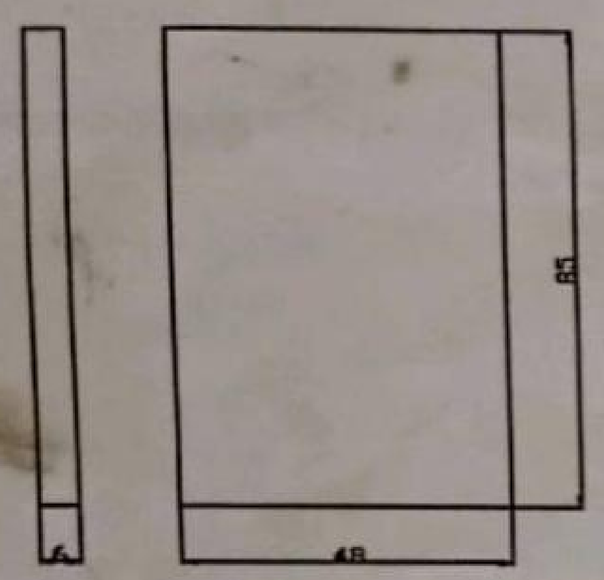
**SAFETY PRECAUTIONS:**

1. Tools should be placed in their respective places after proper cleaning.
2. Sharp edges of the tools must be covered.

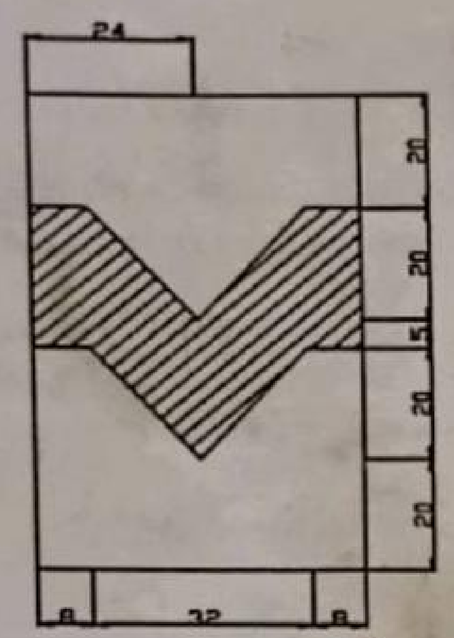
**RESULTS:**

Thus the filing with the required dimensions is obtained.

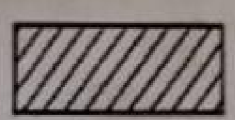
## VEE-FITTING.

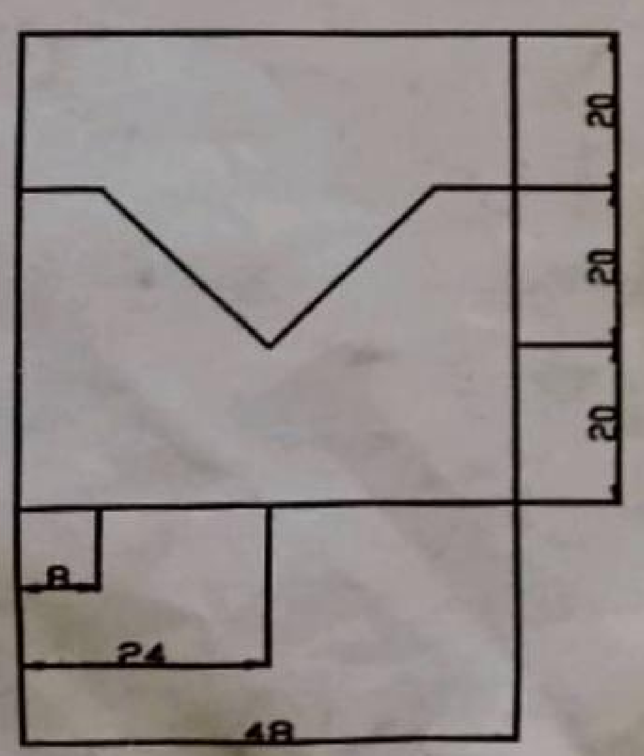



GIVEN JOB.




MARKED JOB.


PORTION TO BE REMOVED.




FINISHED JOB.

ALL DIMENSIONS ARE IN MM.

# VEE FITTING

**AIM :**

To file the given work piece according to the given dimensions.

**MATERIALS REQUIRED:**

1. Mild Steel Flat. (M.S) 85mm*48mm

**TOOLS REQUIRED:**

1. 8'' bench vice
2. Try square
3. Steel rule
4. Dot punch
5. Hacksaw frame with Blade
6. 12''Flat rough file
7. 12''Flat smooth file
8. Surface plate
9. Vernier Height Gauge
10. Angle plate
11. 200g Ball Peen Hammer

**PROCEDURE:**

1. The original dimensions of the work piece is checked using try square.
2. The work piece is clamped properly in the bench vice and, file any two sides of the work piece using a flat file . Check whether the two sides are at right angles using try square.
3. The given dimensions are marked using the surface plate and the vernier height gauge by referring sides as bases.
4. Using Prick punch make impressions on the marked lines.
5. Marking and filing are also done in the same way on the other work piece.
6. Cut and remove the excess material using the hacksaw blade.
7. Filing is continued until the remaining dimensions and smooth surface of the Work piece is obtained.
8. Finally the accuracy and the dimensions of the work piece are verified.

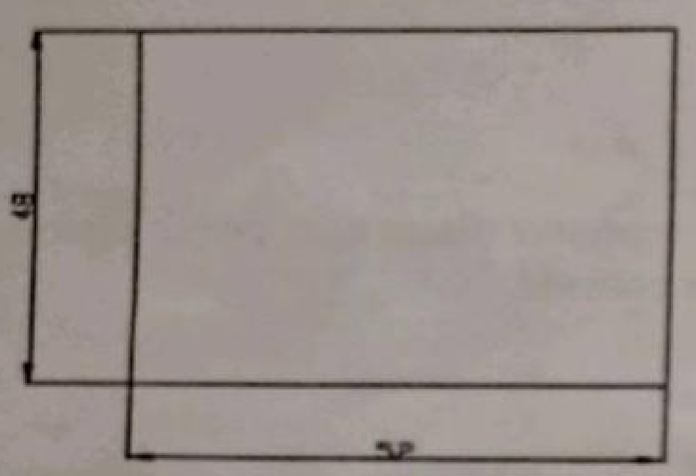
**SAFETY PRECAUTIONS:**

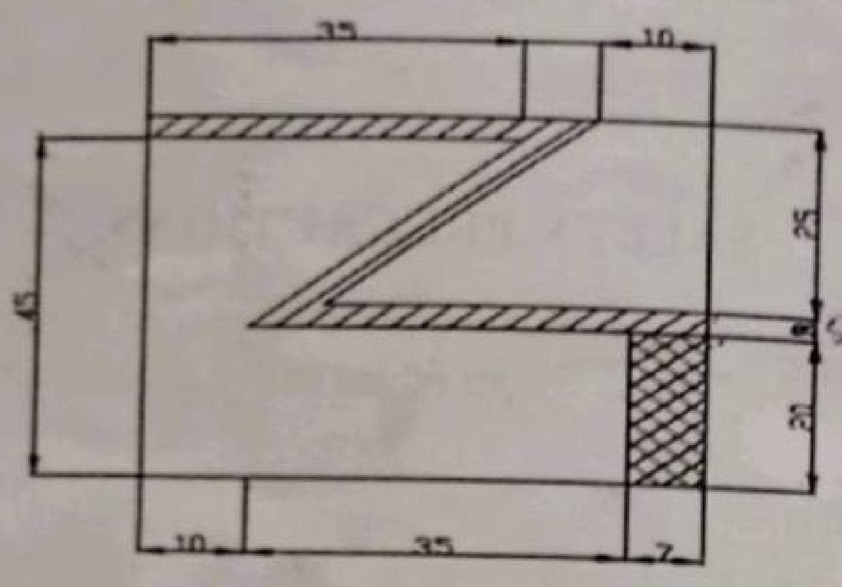
1. Tools should be placed in their respective places after proper cleaning.
2. Sharp edges of the tools must be covered.

**RESULT:**

Thus the required 'VEE' fitting is obtained from the given work pieces.

## ACUTE ANGLE FITTING

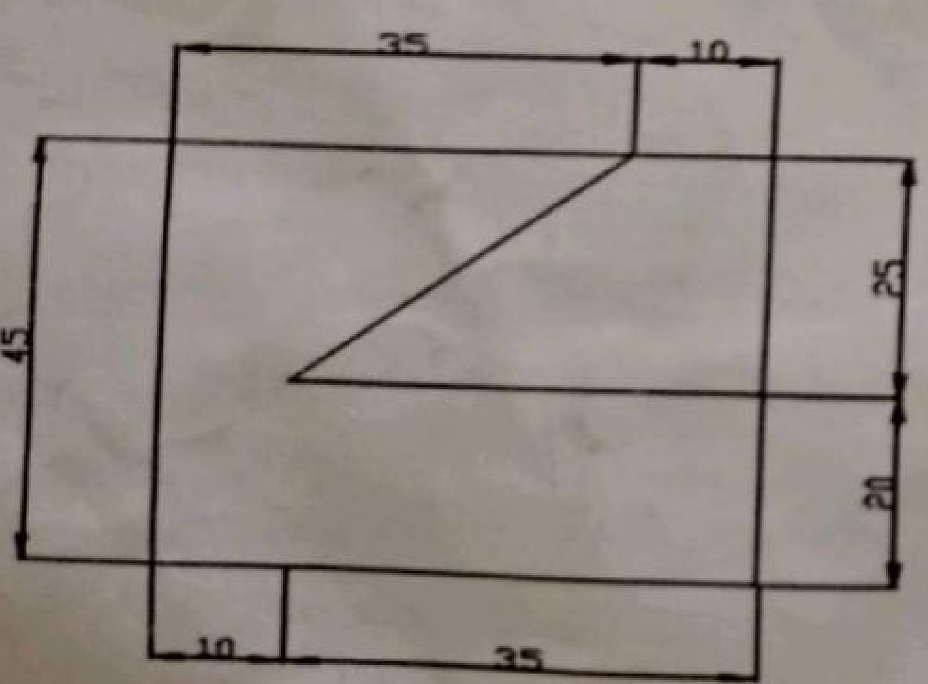



**GIVEN JOB.** **MARKED JOB.**

PORTION TO BE REMOVED.

PORTION TO BE FILED.




**FINISHED JOB.**

**ALL DIMENSIONS ARE IN MM.**

## ACUTE ANGLE FITTING

**AIM :**

To file the given work piece according to the given measurements.

**MATERIALS REQUIRED:**

1. Mild Steel Flat. (M.S) ~~85~~ 52mm*48mm

**TOOLS REQUIRED:**

1. 8'' bench vice
2. Try square
3. Steel rule
4. Dot punch
5. Hacksaw Blade
6. 12''Flat rough file
7. 12''Flat smooth file
8. Surface plate
9. Vernier Height Gauge
10. Angle plate
11. 200g Ball Peen Hammer

**PROCEDURE:**

1. The original dimensions of the work piece is checked using try square.
2. The work piece is clamped properly in the bench vice and using a flat file ,file any two sides of the work piece. Check whether the two sides are at right angles using try square.
3. The given dimensions are marked using the surface plate and the vernier height gauge by referring sides as bases.
4. Using Prick punch make impressions on the marked lines.
5. Marking and filing are also done in the same way on the other work piece.
6. Cut and remove the excess material using the hacksaw blade.

7. Filing is continued until the remaining dimensions and smooth surface of the work piece is obtained.

8. Finally the accuracy and the dimensions of the work piece are verified.

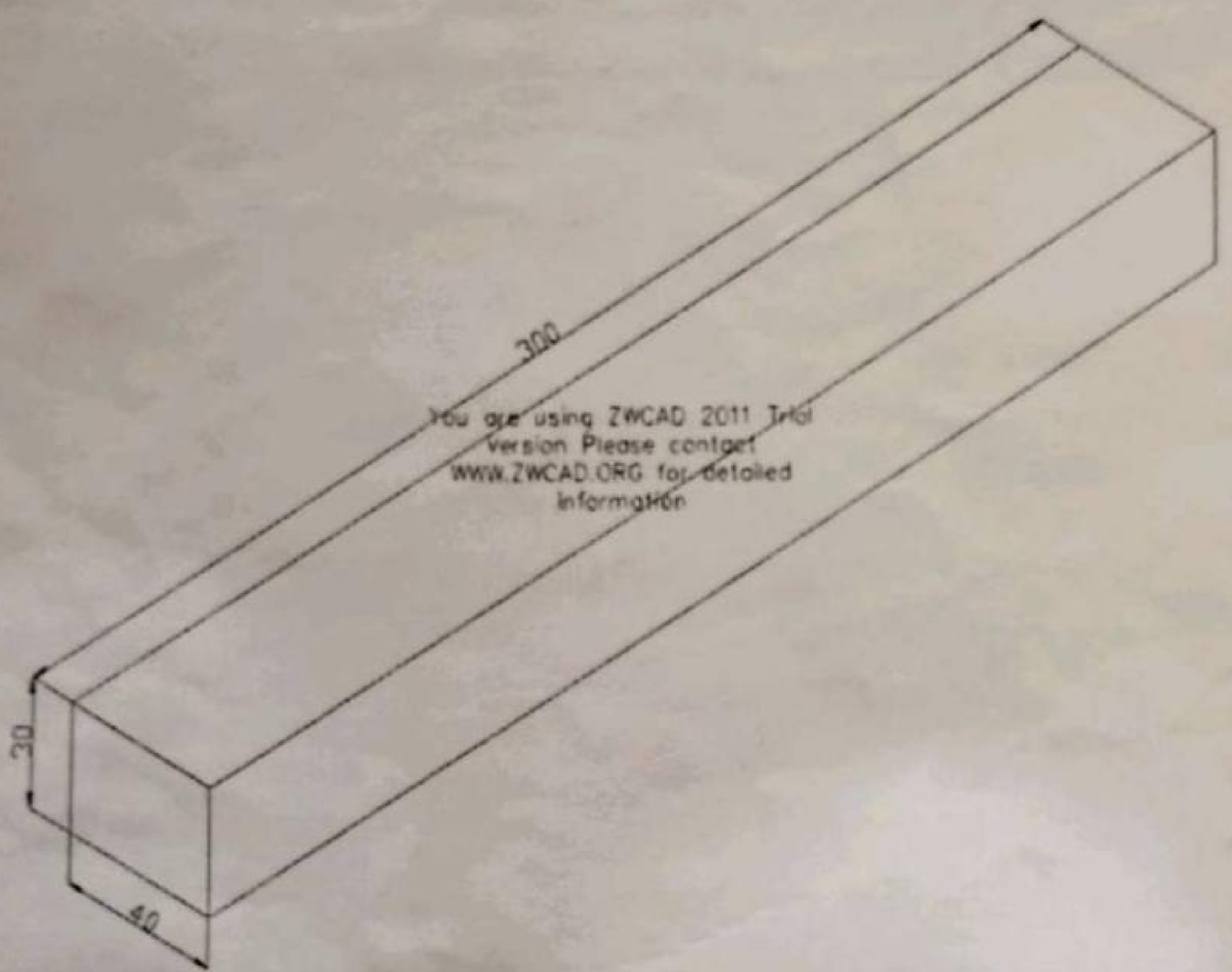
**SAFETY PRECAUTIONS:**

1. Tools should be placed in their respective places after proper cleaning.
2. Sharp edges of the tools must be covered.

**RESULT:**

Thus the Acute angle fitting ~~is~~ was obtained from the given work piece.

## PLANNING.




ALL DIMENSIONS ARE IN MM.



## PLANING

### AIM:

To make planing from the given work piece.

### TOOLS REQUIRED:

1. 8" Carpentry vice
2. Metal Jack plane
3. Try square
4. Mallet
5. Panel saw
6. G-clamp
7. Steel rule
8. Marking gauge etc.,

### PROCEDURE:

1. The job is clamped properly in the vice and any two adjacent surfaces are planned to get right angle using metal jack plane. The accuracy of right angle is checked by using try square.
2. Proper marking is done on the two portions based on the planned surfaces using marking gages.
3. The other two faces of the work piece are planned to the marked dimensions.
4. Finally, the accuracy and proper dimensions are verified by using steel rule and try square.

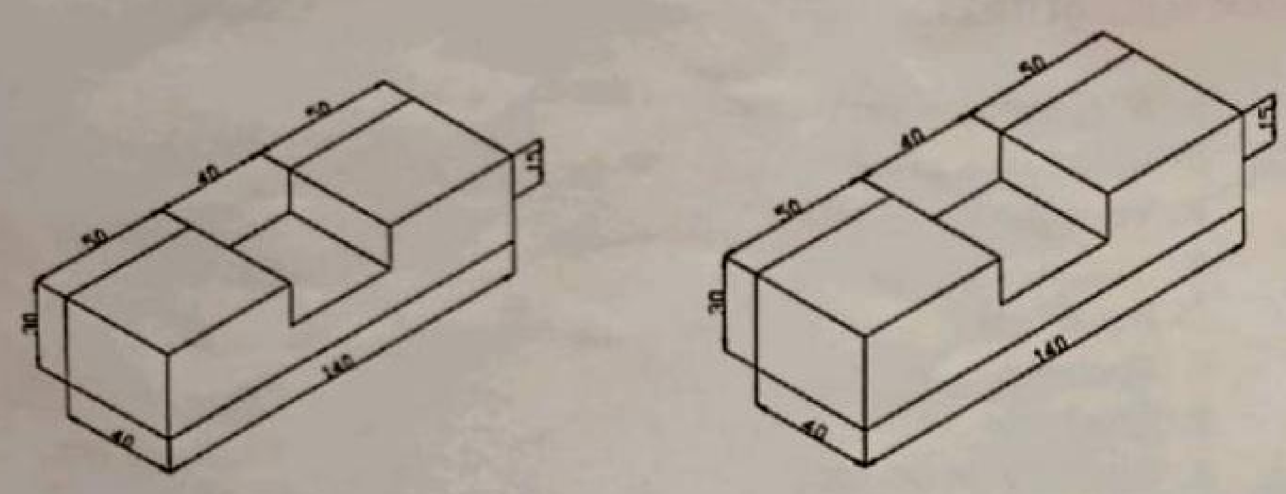
### SAFETY PRECAUTIONS:

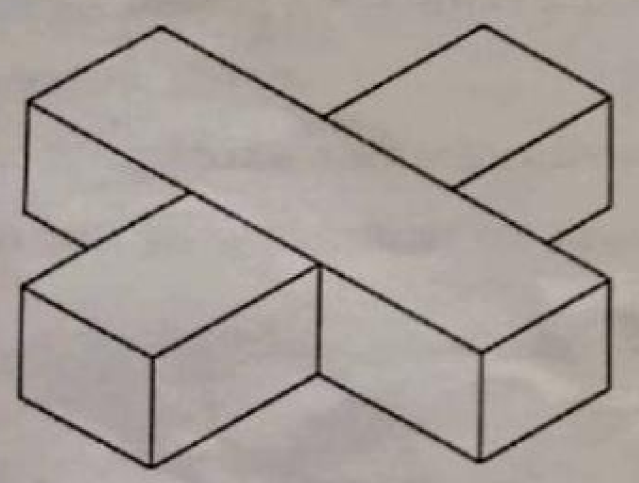
1. Loose clothing should be avoided
2. Wear shoes while working in the shop
3. Tools should always be kept in good working condition

### RESULT:

Thus, the given work piece is planned to the required dimensions.

# HALF LAP JOINT.



FINISHED JOB.

ALL DIMENSIONS ARE IN MM.



# HALF LAP JOINT

## AIM:

To make a half lap joint from the given work piece as per the required dimensions.

## TOOLS REQUIRED:

1. 8" Carpentry vice
2. Metal Jack plane
3. Try square
4. Mallet
5. 1''Firmer Chisel
6. Mortise chisel
7. Panel saw
8. G-clamp
9. Steel rule
10. Marking gauge etc.,

## PROCEDURE:

1. The job is clamped properly in the vice and any two adjacent surfaces are planned to get right angle using metal jack plane. Using try square checks the accuracy of right angle.
2. The job is cut into two halves using panel saw.
3. Proper marking is done on the two portions based on the planned surfaces.
4. One half is cut according to the given dimensions using panel saw and firmer chisel.
5. The other half is also cut in the similar way so as to make correct fit on one another and the extra portions are removed using firmer chisel.
6. Finally, the accuracy and proper dimensions are verified by using steel rule and try square.

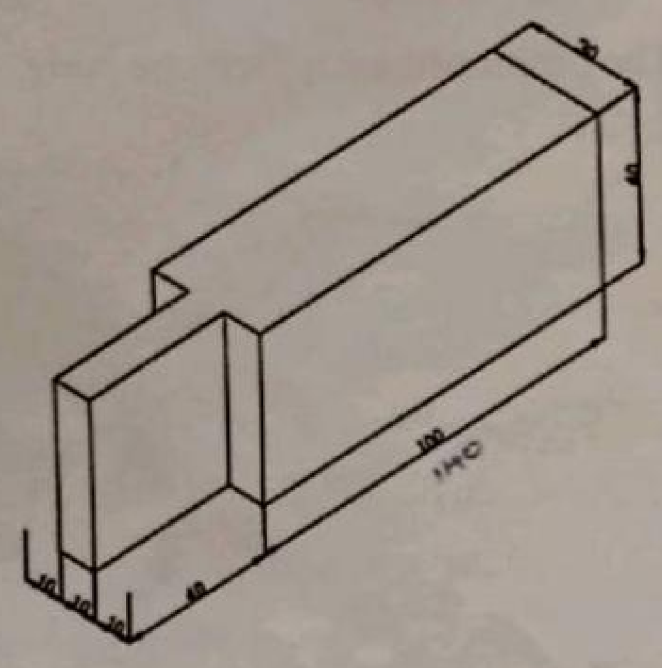
## SAFETY PRECAUTIONS:

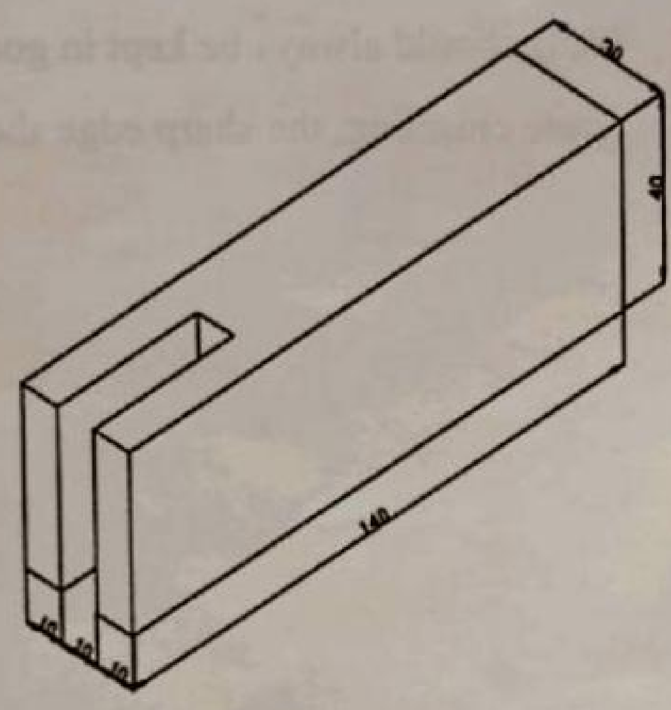
1. Tools should always be kept in good working condition
2. While chiseling, the sharp edge should be positioned away from the body.

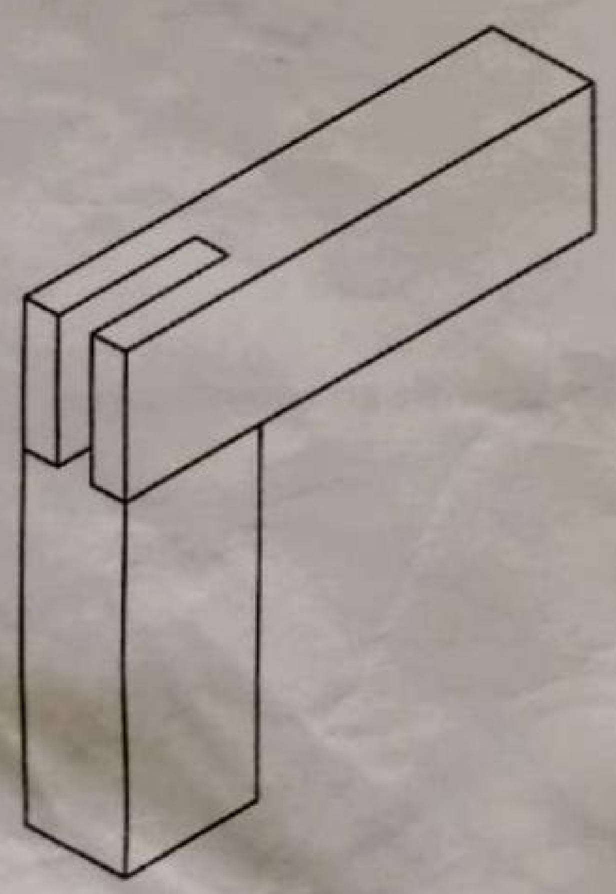
## RESULTS:-

Thus the required half lap joint ~~is~~ was made from the given work piece.

## CORNER MORTSE JOINT.







FINISHED JOB.

ALL DIMENSIONS ARE IN MM.



## CORNER MORTISE JOINT

### AIM:

To make a Corner Mortise joint from the given work piece as per the given dimensions.

### TOOLS REQUIRED:

1. 8" Carpentry vice
2. Metal Jack plane
3. Try square
4. Mallet
5. 1''Firmer Chisel
6. Mortise chisel
7. Panel saw
8. G-clamp
9. Steel rule
10. Marking gauge etc.,

**PROCEDURE:**

1. The job is clamped properly in the vice and any two adjacent surfaces are planned to get right angle using metal jack plane. The accuracy of right angle is checked by using try square.
2. The job is cut into two halves using rip saw.
3. Proper marking is made on the two portions based on the planned surfaces.
4. One half is cut into a 'T' shaped lock having the required dimensions using firmer and mortise chisels.
5. In the other portion, a rectangular hole of the given dimensions is made using firmer and mortise chisels to get mortise
6. Finally, the accuracy and proper dimensions are verified by using steel rule and try square.

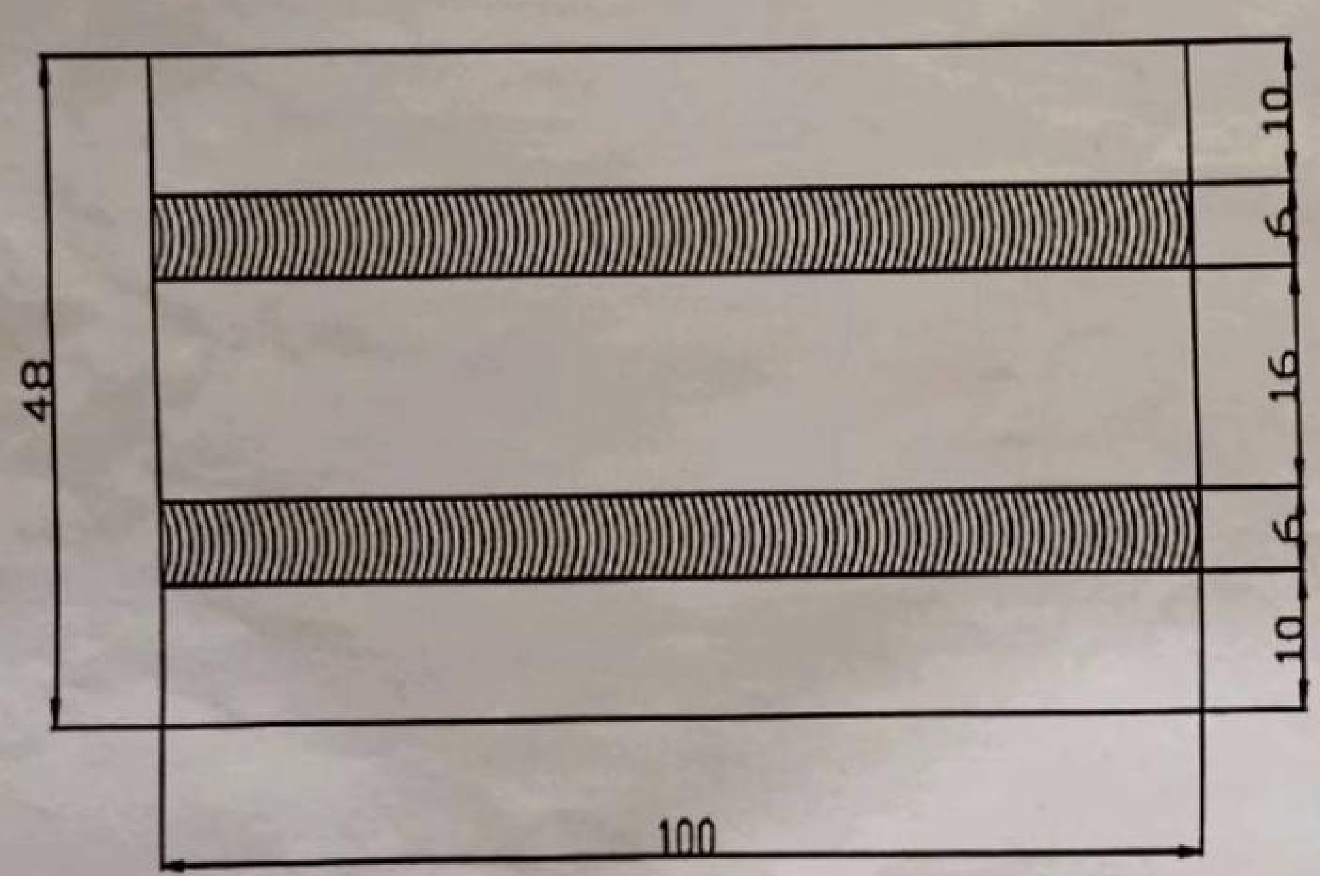
## SAFETY PRECAUTIONS:

1. Tools should always be kept in good working condition
2. While chiseling, the sharp edge should be positioned away from the body

**RESULT:**

Thus the required Corner Mortise joint ~~is~~ was made from the given work piece.

WELDING LINE PRACTICE.




ALL DIMENSIONS ARE IN MM.



## WELDING LINE PRACTICE

**AIM :**

To make weld lines on the given work piece.

**MATERIALS REQURIED:**

1. Mild Steel. (MS) 100mm*48mm   75 x 48 mm

**TOOLS REQUIRED:**

1. A.C.Welding transformer
2. Plain goggles
3. 3.15mm Welding electrode
4. Chipping hammer
5. Wire brush
6. Hand gloves
7. 12'' Flat Rough File
8. Tongs

**PROCEDURE:**

1. The surface to be welded is cleaned and filed for perfect joint and more strength.
2. The welding electrode is held in the electrode holder and the ground clamp is clamped to the plate to be welded.
3. The plate to be welded is positioned and starts welding from one end of the plate.
4. The electric produced melts the welding electrode and it deposits over the metal plate.
5. Maintain the gap of 3mm between the plate and welding electrode.

6. Complete the welding process by removing slag using chipping hammer and wire brush.

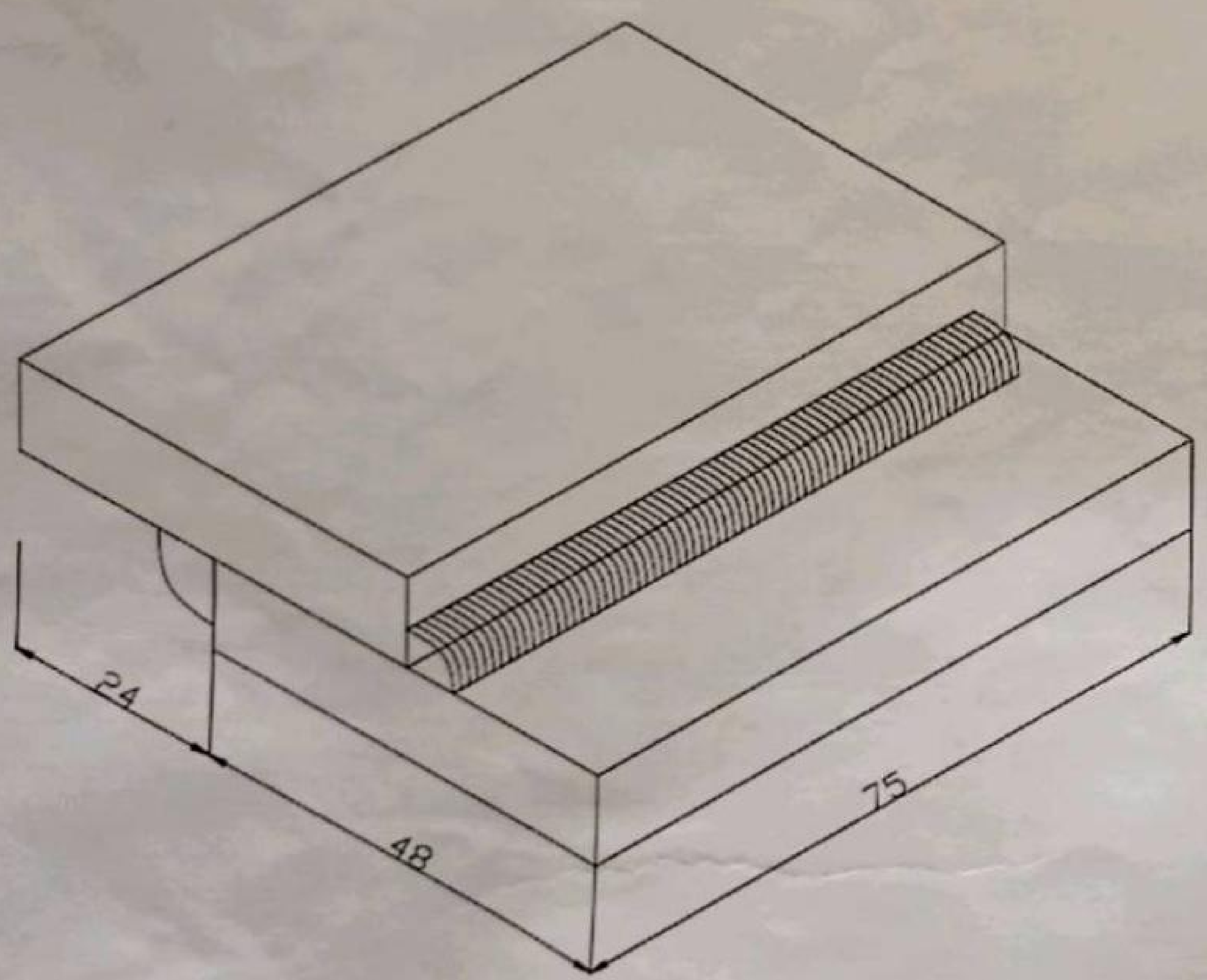
**SAFETY PRECAUTIONS:**

1. Always wear hand gloves and goggles to avoid injuries.
2. Tongs should be used to carry the welding piece.

**RESULT:**

Thus the given metal piece was welded as required.

## LAP JOINT.




ALL DIMENSIONS ARE IN MM.



## LAP JOINT

**AIM :**

To join the given metal plates by a lap joint using arc welding process.

**MATERIALS REQURIED:**

1. Mild Steel. (MS) 75mm*48mm

**TOOLS REQUIRED :**

1. A.C.Welding transformer
2. Plain goggles
3. 3.15mm Welding Electrode
4. Chipping hammer
5. Wire brush
6. Hand gloves
7. 12'' Flat Rough File
8. Tongs

**PROCEDURE :**

1. The surface to be welded is cleaned and filed for perfect joint and more strength.
2. The welding electrode is held in the electrode holder and the ground clamp is clamped to the plate to be welded.
3. The plate to be welded is positioned overlapping and tag welding is done on the ends to avoid the movement of the plate during welding.
4. Now start welding from one end of the plates.
5. The electric produced melts the welding electrode and it deposits over the metal plate.
6. Maintain the gap of 3mm between the plate and welding electrode.
7. Complete the welding process by removing slag using chipping hammer and wire brush.

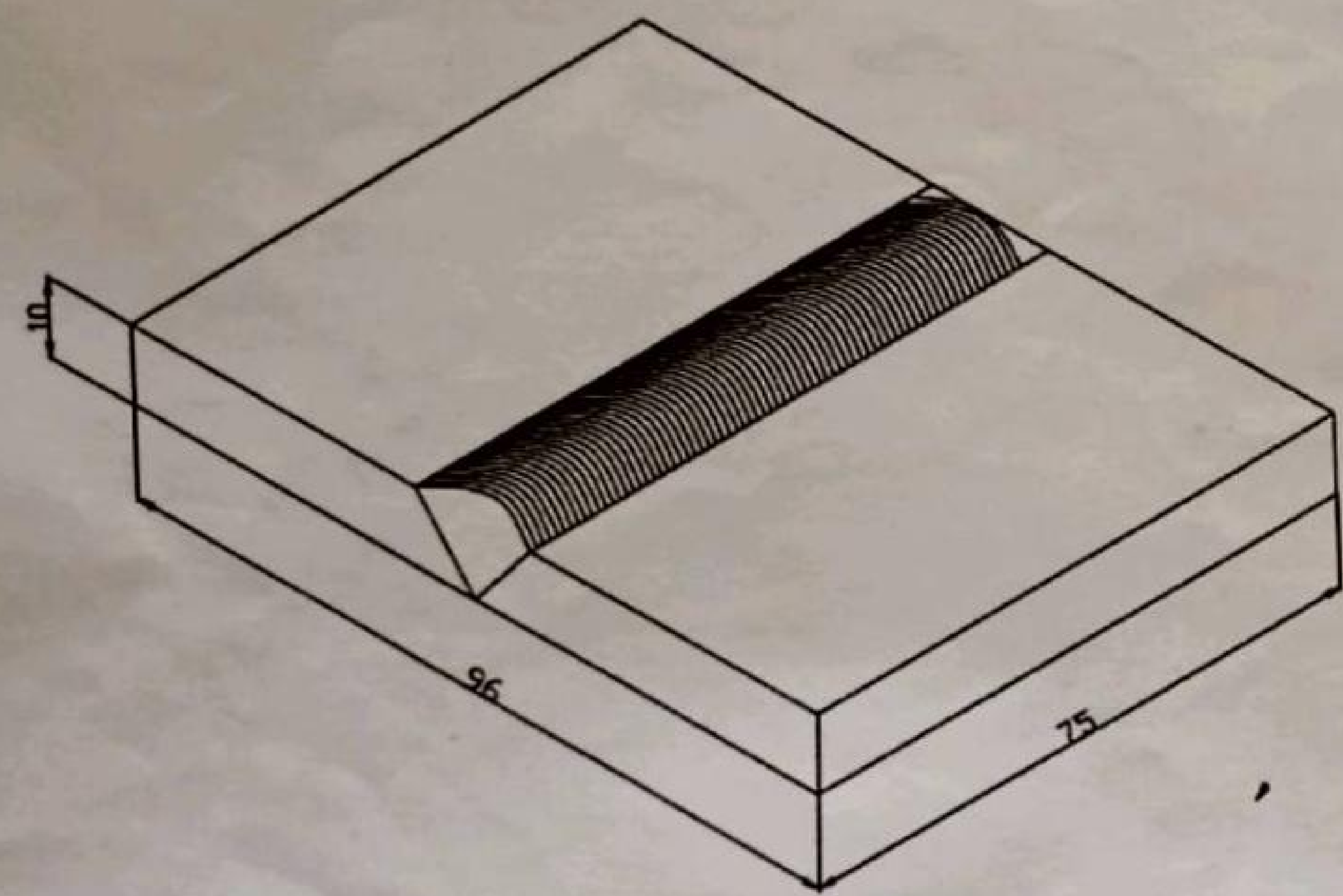
**SAFETY PRECAUTIONS:**

1. Always wear hand gloves and goggles to avoid injuries.
2. Tongs should be used to carry the welding piece.

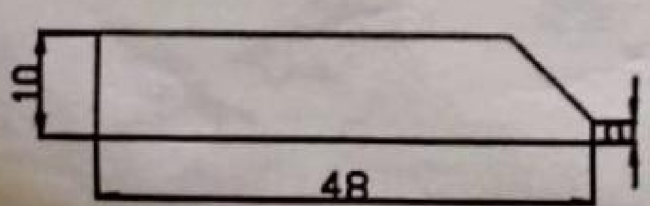
**RESULT:**

Thus the given metal piece was welded as required.

SINGLE " V " BUTT JOINT.




BEVEL.




ALL DIMENSIONS ARE IN MM.



## SINGLE 'V' BUTT JOINT

### AIM:

To join the given metal plates by a single Vee butt joint using arc welding process.

### MATERIALS REQURIED:

1. Mild Steel. (MS) 85mm*48mm

### TOOLS REQUIRED:

1. A.C.Welding transformer
2. Plain goggles
3. 3.15mm Welding electrode
4. Chipping hammer
5. Wire brush
6. Hand gloves
7. 12'' Flat Rough File
8. Tongs

### PROCEDURE :

1. The surface to be welded is cleaned and filed for perfect joint and more strength.
2. The welding electrode is held in the electrode holder and the ground clamp is clamped to the plate to be welded.
3. The plate to be welded is positioned overlapping and tag welding is done on the ends to avoid the movement of the plate during welding.
4. Now start welding from one end of the plates.
5. The electric produced melts the welding electrode and it deposits over the metal plate.
6. Maintain the gap of 3mm between the plate and welding electrode.

7. Complete the welding process by removing slag using chipping hammer and wire brush.

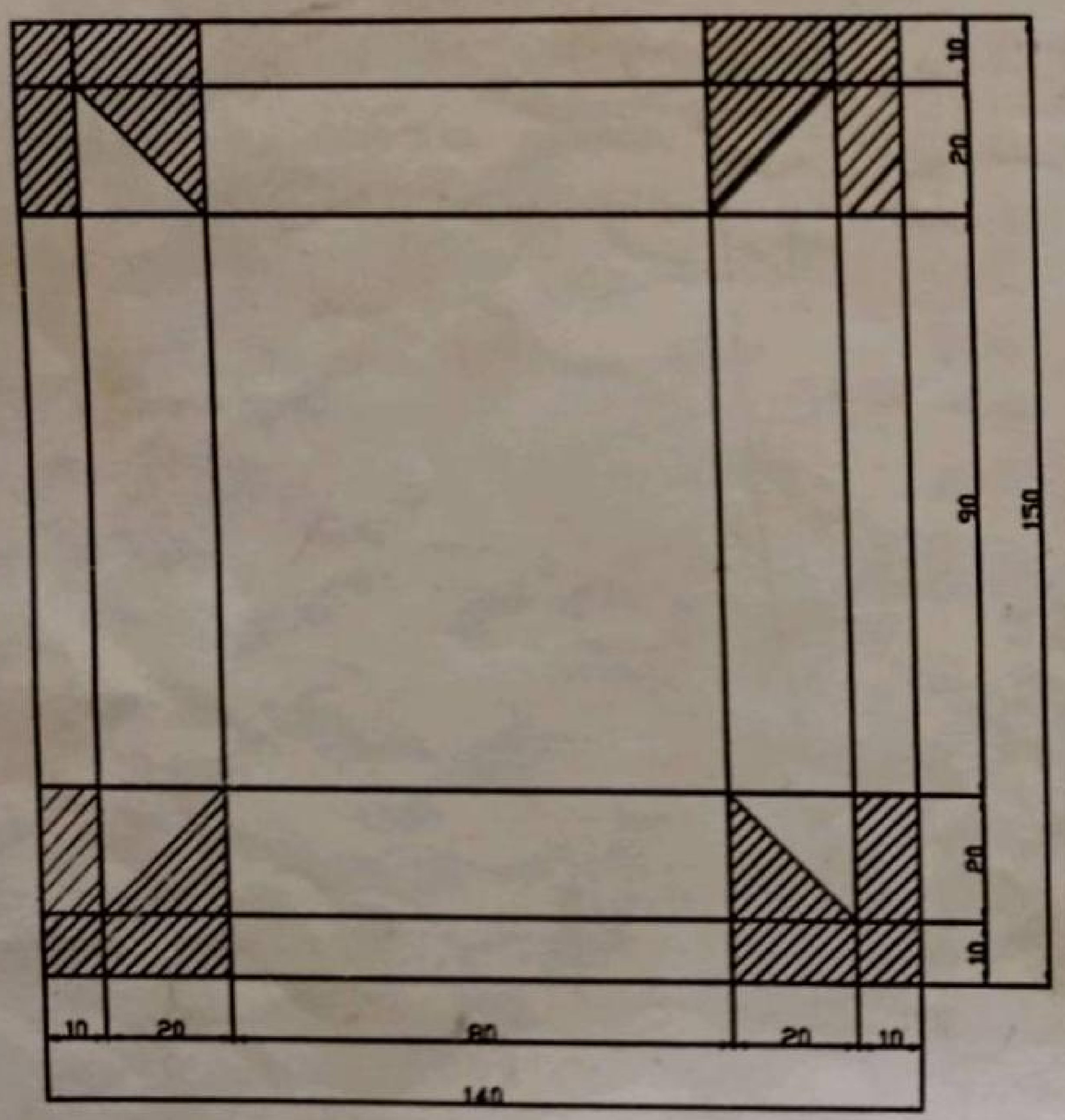
**SAFETY PRECAUTIONS**:

1. Always wear hand gloves and goggles to avoid injuries.
2. During welding, it is not advisable to look at the rays directly.

**RESULT:**

Thus the given metal piece was welded as required dimensions.

## RECTANGULAR TRAY.




MARKED JOB.

PORTION TO BE CUT.

ALL DIMENSIONS ARE IN MM.



## RECTANGULAR TRAY

**AIM:**

To make a rectangular tray from a given metal sheet as required dimensions.

**MATERIALS REQURIED:**

1. Galvanisied Iron Sheet (G.I Sheet).

**TOOLS REQUIRED :**

1. Wooden Mallet
2. Ball Peen Hammer
3. Straight snip
4. Bent Snip
5. Half moon stake
6. Steel rule
7. Cutting plier
8. Scriber

**PROCEDURE :**

1. The given sheet is smoothened using mallet.
2. The development of the rectangular tray is drawn on the sheet with given dimensions using scriber.
3. The unmarked and excess portions in the sheet are removed using sleeper. snipper
4. Folding is done as per the given dimensions using the stake and mallet.
5. Bending is done on the anvil as per the given dimensions.

**SAFETY PRECAUTIONS:**

1. Wearing hand gloves is compulsorily.

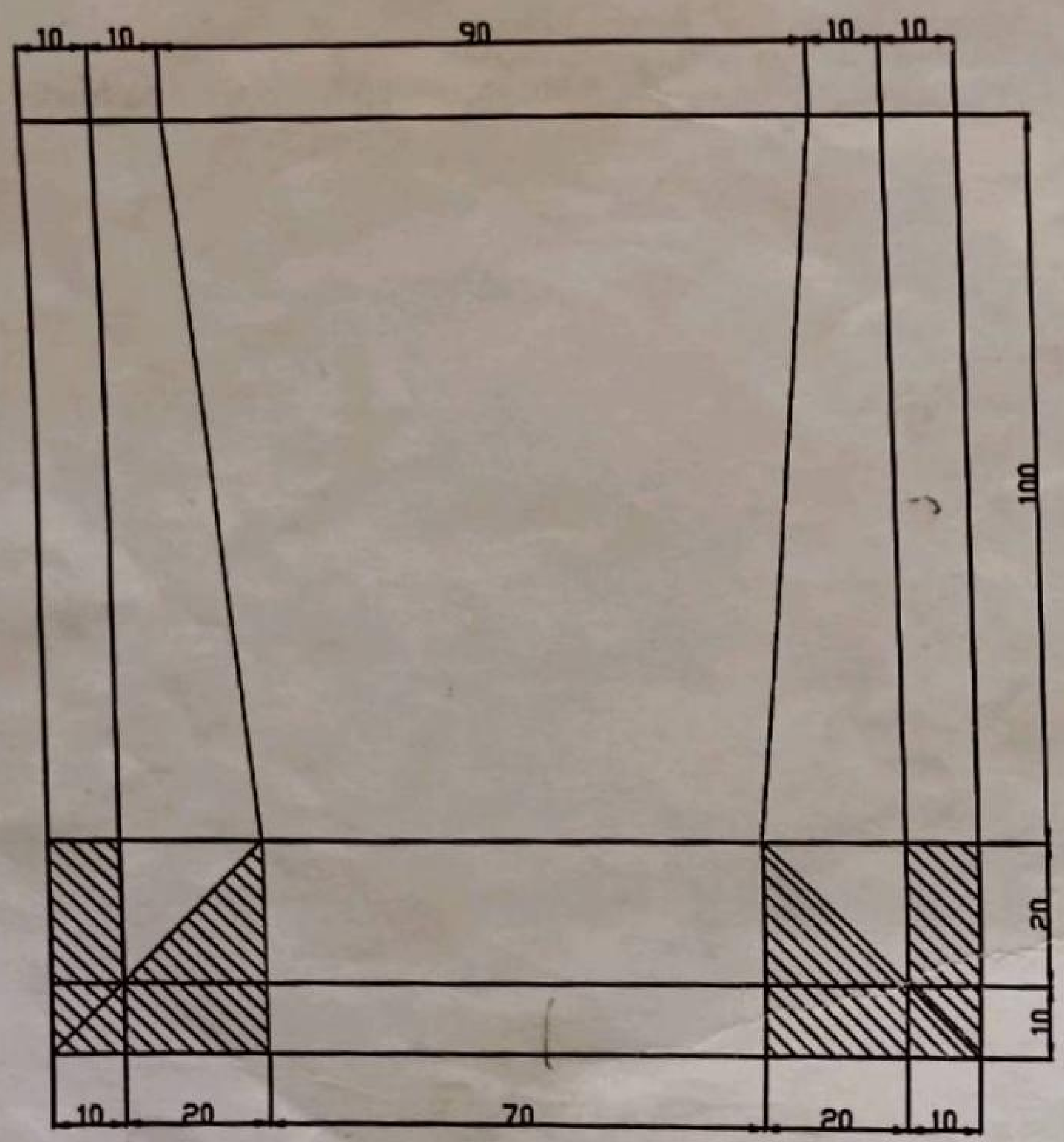
2. While doing bending and folding care should be taken to handle the tools.

**RESULT:**

Thus the rectangular tray with the required dimensions is obtained.

52

## DUST PAN.




**MARKED JOB.**

PORTION TO BE CUT.

**ALL DIMENSIONS ARE IN MM.**

## DUST PAN

**AIM :**

To make a dust pan from a given metal sheet as required dimensions.

**MATERIALS REQURIED:**

1. Galvanisied Iron Sheet (G.I Sheet).

**TOOLS REQUIRED :**

1. Wooden Mallet
2. Ball Peen Hammer
3. Straight snip
4. Bent Snip
5. Half moon stake
6. Steel rule
7. Cutting plier
8. Scriber

**PROCEDURE :**

1. The given sheet is smoothened using mallet.
2. The development of the rectangular tray is drawn on the sheet with given dimensions using scriber.
3. The unmarked and excess portions in the sheet are removed using sleeper.
4. Folding is done as per the given dimensions using the stake and mallet.
5. Bending is done on the anvil as per the given dimensions.

53

**SAFETY PRECAUTIONS:**

1. Wearing hand gloves is compulsorily.

2. While doing bending and folding care should be taken to handle the tools.

**RESULT:**

Thus the dust pan with the required dimensions ~~is~~ was obtained.

# FITTING

1) **What is meant by fitting?**

It is the assembling of parts together and removing metal to secure the necessary fit.

2) **Name the material out of which the vice is made.**

| | | |
|---|---|---|
| Body | - | cast iron |
| Fixed and movable jaw | - | cast steel |
| Handle, square threaded screw and nut | - | mild steel. |

3) **Name the tools used for marking.**

Scriber inside caliper outside caliper compass divider.

4) **What are the types of punches used in fitting shop?**

Prick punch: It is used to make small punch and tapered point has an angle of 40.

Centre punch: It is used for making dots along a line before cutting. This also used for making hole centers that are to be drilled. The tapered point has an angle of 60.

5) **What is the use of V block & clamp and surface gauge?**

V-block & clamp: It is used as a supporting block for round bar stock for marking out.

Surface gauge: it is used for drawing vertical, horizontal and parallel lines.

6) **What materials are the files made up of?**

High grade steel.

7) **What is "hand file" and why is it called "safe edge file"?**

This is rectangular in section and tapered in thickness but parallel in width. The edges are single cut and the faces are double cut.

The other edge does not have any teeth and hence this file is known as safe edge file.

8) **What is a "drill"?**

A drill is a fluted cutting tool used to originate or enlarge a hole in a metal piece.

9) **What material the hacksaw blades are made up of?**

Special steels are generally used for cutting hard metals so hard blades of high speed steel are used.

10) **What is the material out of which the hammers are made?**

Hammers are made of forged steel.

11) **How a tap drill size is determined?**

The size of the tap drill can be derived from the relation

$$D = T - 2d$$

Where, D = diameter of the tap drill size.
T = diameter of tap
D = depth of thread

So tap drill size = outside diameter 0.8

# CARPENTRY

1. **What is carpentry?**
   It is any kind of work with wood.

2. **What is the basic material used for wood working?**
   Timber is the basic material used for any class of wood working. The term "timber" is applied to the trees which provide us with wood.

3. **State the main differences between soft wood and hard wood.**

| Soft wood: | hard wood: |
|---|---|
| ➢ Straight fibres and fine texture | fibres are quite & compact |
| ➢ Light in weight | heavier |
| ➢ Light in colour | dark in colour |
| ➢ Easy to work | difficult to work |

4. **What are the uses of inside, outside calipers and try square?**
   Inside caliper: it is used for measuring diameter of holes length of recesses.
   Outside caliper: it is used for measuring diameter of rods and small lengths.
   Try square: it is used for testing flatness of surfaces.

5. **What is "setting" of saw teeth?**
   The teeth are bent alternately, one to the right, the next to the left. Bending of teeth in this manner is called "setting".

6. **What is TPI?**
   Teeth per inch or point per inch. The number of points (teeth) is 5,6,7 and 8 per 25 mm (inch).

7. **Explain the different types of chisels used in carpentry shop.**
   The types of chisels are used in carpentry shop are:
   Firmer chisel, paring chisel and mortise chisel
   Firmer chisel: the word "firmer" means stronger. This form enables the worker to get into close corners and angles when trimming off and parting off thin shaving.

   Paring chisel: this is used for paring plane surfaces both in the directions of the grain and on the end grain of the wood.

   Mortise chisel: this is much stronger than the firmer chisel. It is used for cutting rectangular mortises.

8. **What is the material out of which the chisels are made up of?**
   The chisels are made up of forged steel.

9. **What is chiseling?**
   Chiseling is the process of cutting a small stock of wood to produce the desired shape.

**10. What is planning?**

Planning is the operation of turning up a piece of wood by a planner. This is also known "facing and Edging".

**11. What is meant by plane?**

A plane is a hand tool and is generally used for smoothening or shaping pieces of wood.

**12. What are the three types of defects**

Heart shake
Cup shake
Twisted fibres



# WELDING

**1) What is metal joining process?**

Metal joining process is one of the manufacturing process in which metal pieces are joined by the application of heat with or without the application of pressure.

**2) Define welding.**

Welding is the process of joining two similar metals by the application of heat with or without the application of pressure and the addition of filler material.

**3) Name at least any two types of welding.**

The welding processes are classified into the following two classes.
a) Plastic welding or pressure welding
b) Fusion welding or non- pressure welding.

**4) What is meant by gas welding?**

Gas welding is the process of joining two metal pieces by flame obtained by the combustion of fuel gas and oxygen.

**5) What are the different types of flames in gas welding?**

The different types of flame in gas welding are:
a) Neutral flame
b) Carburizing flame or reducing flame and
c) Oxidizing flame.

**6) What are the welding tools used in welding shop?**

Face shield, tongs, chipping hammer, wire brush etc.

**7) Name the parts of a gas welding equipment.**

The parts of gas welding equipment are:
Gas cylinders
Pressure regulators
Pressure gauges
Welding torch and
Hoses.

**8) What is arc welding?**

Arc welding is the most extensively used method of joining metal parts. In arc welding the heat generated by an electric arc used to melt and join the work piece.

9) **Name few types of arc welding.**

The main types of arc welding are:

- Carbon arc welding
- Metal arc welding
- Metal inert gas welding (MIG)
- Submerged arc welding (SMAW)
- Tungsten inert gas welding (TIG)
- Atomic hydrogen welding (AHW)
- Plasma arc welding (PAW)

10) **What are the difference between arc and gas welding?**

| Arc welding | gas welding |
|---|---|
| Welding produced by using electric arc | welding produced by using gas flame. |
| The temperature of arc is about 4000 c | the temperature of flame is about 3200 c |
| Joint has good strength | comparatively low strength |
| Risk of electric shock is more | risk of explosive. |

11) **What is meant by filler rod used in welding?**

Filler rod is metallic rod coated with a flux material. It is also melted during the process and mixed with the molten used for this process is similar to that of oxy-acetylene welding except the torch.

12) **What is a welding rod?**

It is an electrode metal, used in arc welding process. During the process, it is also melted and mixed with molten metal. Welding rod can also be called as electrode.

There are two types of welding rods (electrode) namely: (a) consumable electrode and (b) non-consumable electrode.

13) **In what welding used to join non-ferrous metal?**

Arc welding is used for joining the non-ferrous.

## P106 WORKSHOP PRACTICE

***OBJECTIVES***

- *To convey the basics of mechanical tools used in engineering*
- *To establish hands on experience on the working tools*
- *To develop basic joints and fittings using the hand tools*
- *To establish the importance of joints and fitting in engineering applications*
- *To explain the role of basic workshop in engineering*
- *To develop an intuitive understanding of underlying physical mechanism used in mechanical machines.*

| S.No. | Trade | List of Exercises |
|---|---|---|
| 1 | Fitting | Study of tools and Machineries. Exercises on symmetric joints and joints with acute angle. |
| 2 | Welding | Study of arc and gas welding equipment and tools – Edge preparation – Exercise on lap joint and V Butt joints – Demonstration of gas welding |
| 3 | Sheet metal work | Study of tools and Machineries – Exercise on simple products like Office tray and waste collection tray. |
| 4 | Carpentry | Study of tools and Machineries – Exercises on Lap joints and Mortise joints |

**LIST OF EXERCISES**

**I-FITTING**

1. Study of tools and Machineries
2. Symmetric fitting
3. Acute angle fitting

**II-WELDING**

1. Study of arc and gas welding equipment and tools
2. Simple lap welding (Arc)
3. Single V butt welding (Arc)

**III-SHEET METAL WORK**

1. Study of tools and machineries
2. Frustum
3. Waste collection tray

**IV-CARPENTRY**

1. Study of tools and machineries
2. Half lap joint
3. Corner mortise joint.